Transmitter source model (transmitter)

Contract, version 0.13.0

The transmitter module describes Alice's side of a protocol: what she prepares, the system she retains, and whether her source is trusted. Its input is a source specification, with an ordered alphabet, a retained system carrying a sector structure, her measurement and the signal ensemble, declared as binary64 matrices or through a submodule's source model and connected by source links to the virtual ensemble on which the certificate is written. In trusted-source mode a source-replacement certificate, declared as a pure projector or as positive sector blocks or defined from the virtual ensemble, describes the preparation as a measurement of Alice's retained system. The module's main results are these: Theorem 4.1 shows that the certificate's algebraic conditions are, in either declared form, finitely many exact decisions and that her measurement commutes with every attack that leaves her retained system and outcome alone, which retention and the sequential round structure make every attack of a round; Lemma 3.4 constructs from the virtual ensemble, given as sector blocks, a certificate with pure blocks whose defining identities hold with no error term; and Lemma 4.5 states what a block certificate implies for the states an attack reaches. Every condition the module decides is decided exactly on the declared entries, while the retention of Alice's system remains an attestation that the module records and never verifies. The module emits an artefact: the declared data with the source segment, a licence record for a verified certificate, which lets a consumer fix Alice's marginal, and an assumption ledger naming the assumptions the artefact rests on. Its consumers are the protocol compiler, the reductions module and the key-rate engine of meat-core; the module itself states no key rate, builds no optimisation problem and restricts no feasible set. Its decoy submodule declares a weak-coherent-pulse source and the tagging link that makes it finite.

Grey text, here and throughout, is non-normative: it explains, and nothing in it is assumed or proved.

The design notes (non-normative) explain why the module is built as it is and draw its processing flowchart. The dependencies between the sections are drawn in the section dependency matrix, and the graph index opens every section's dependency graph. The statements digest holds every definition, assumption, result and requirement without the proofs. How to read a contract explains this layout and its links.

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Main results

The module's most important results, those its overview names as what it establishes, with every assumption each rests on, directly or through the claims it uses. Each links to its statement and proof; the statements digest holds them in full.

Lemma 3.4 Certificate defined by an ensemble presentation

A virtual ensemble in sector blocks defines, by canonical purification and a square-root measurement, a block certificate with pure blocks that exactly reproduces Alice's marginal and, under that measurement, the ensemble. Only her marginal, a transposed sum of declared blocks, is formed as numbers.

Rests on no assumption

Theorem 4.1 Checkability and sufficiency of the source-replacement certificate

In any form (pure projector, sector blocks, ensemble presentation) the certificate's algebraic conditions are finitely many exact decisions making Alice's measurement reproduce the virtual ensemble. Under retention and sequential rounds she may measure before or after any attack: source replacement.

Rests on A-SEQ, A-SRC

Lemma 4.5 Matched pinching of a block certificate

A block certificate is unchanged by pinching Alice's sectors or the virtual signal's sectors paired with them, so every state an admitted attack reaches from it is block diagonal with Alice's declared marginal. An operator merely constrained to have that marginal need not be block diagonal.

Rests on A-SRC

Proposition 8.1 Tagging channel for weights at or below the Poisson weights

One channel, independent of the label, maps each tagged virtual state to its phase-randomised weak coherent state whenever the declared sector weights lie at or below the Poisson weights. The weights therefore need only a certified inequality, never an exact Poisson weight.

Rests on no assumption

Corollary 9.2 The emitted decoy declaration

The emitted decoy declaration has an exact certificate against the tagged ensemble, its source identity by the tagging channel, and block-diagonal attacked states. Exactness is decided on products of declared weights and formed dyad entries; lowering a weight gives another tagged declaration.

Rests on A-SEQ, A-SRC, A-WCP

Assumptions at a glance

Every assumption this contract states or its claims assume, with its category and the claims of this contract that assume it directly. An assumption of another contract links into that contract's PDF.

Assumption	Category	Title	Stated as	Assumed directly by
A-ENS	source	Emission of the specified ensemble	Assumption 5.1	Fact 3.1

Assumption	Category	Title	Stated as	Assumed directly by
A-SRC	source	Retention of the system A in trusted-source mode	Assumption 5.3	Thm. 4.1, Lem. 4.5, Cor. 9.2
A-WCP	source	Emission of the declared weak-coherent-pulse states	Assumption 10.1	Cor. 9.2
A-SEQ	protocol structure	Sequential round structure	Assumption 5.2	Fact 3.1, Thm. 4.1, Rem. 4.2, Cor. 9.2

Notation

The symbols of the repository's notation catalogue (spec/notation.md) that this contract uses, in the catalogue's order, with their meaning.

Symbol	Meaning
$\mathcal{H}(A)$	Hilbert space of the system label A
$\mathcal{L}(A)$	Bounded linear operators on $\mathcal{H}(A)$
$\mathrm{Pos}(A)$	Positive semidefinite operators
$S_=(A)$	Normalized states
$\mathbb{1}$, $\mathbb{1}_C$	Identity operator (contextual / named)
id	Identity channel
$\mathcal{X}$	Outcome alphabet of the measurement on A — Alice's under the default partition
$\mathfrak{L}$	Declared constraint list $\{(\hat{O}_k, \bowtie_k, \Theta_k)\}$
$\{H, V, +, -\}$	Polarization / detector labels
on, off, dc, sq, mc, nc, sc, vac	Mode / outcome abbreviations
test, gen	Round-type values
T	Transpose, X^{T}

Terms

The phrases this contract defines, and the terms of other contracts it uses, each with a one-line gloss and the definition that fixes it. In the text, a linked term opens its definition.

Term	Meaning	Defined in
binary64 numbers	the finite values of the IEEE 754 binary64 format, each a dyadic rational	foundations/def: binary64
block form	The certificate form declared as positive blocks, one per sector, on matched sector structures of Alice's system and the virtual system.	Definition 3.3
declared descriptor family	The family of source descriptors for which a source link's channel identity is asserted.	Definition 2.1
declared order	a total order on a finite alphabet, fixed together with the alphabet	foundations/def: declared-order
ensemble presentation	The certificate form that declares only the weighted virtual ensemble in sector blocks and defines the certificate from it.	Definition 3.3
entropy cutoff	The largest photon number whose sector of the virtual source stays quantum	Definition 7.2
intensities	The declared mean photon numbers of the weak coherent pulses, non-negative reals	Definition 7.1
interface conditions	The conditions (S1) to (S4) a source segment satisfies: descriptor match, index preservation, the source identity, and the slots each link carries.	Definition 2.2
matched subspace	The direct sum, over the sectors, of the tensor products of matching sectors of Alice's system and the virtual system.	Definition 3.3
obligation record	What a theorem-supplied source link carries in place of a decided identity: its kind, the soundness claim, the assumptions the obligation carries, and the family member its descriptors are.	Definition 2.1
one-sector structure	The default sector structure, whose single sector is the whole retained system.	Definition 1.1
Poisson weight	The probability that a phase-randomised coherent pulse of a given intensity carries a given number of photons	Definition 7.1

Term	Meaning	Defined in
polarisation flag	A one-dimensional sector for one photon number between the two cutoffs and one polarisation, shared by all intensities	Definition 7.2
polarisations	The declared polarisation labels, each standing for a unit vector of the one-photon space that is an object of the model	Definition 7.1
pure form	The certificate form declared as a rank-one projector, on the one-sector structure.	Definition 3.3
retained-register sector map	An optional matched pair of sector structures, on Alice's retained system and on the descriptor the certificate is written against.	Definition 2.1
round type	The third component of a letter, marking a test round or a generation round	Definition 7.1
rounds nowhere	said of an accumulation every one of whose binary64 additions is exact	foundations/def: declared-order
sector index set	The ordered sectors of the virtual source, those that some letter puts positive weight on	Definition 7.2
sector structure	A decomposition of Alice's retained system into orthogonal sectors, each spanned by declared basis vectors.	Definition 1.1
sound	Said of a source link whose channel maps every source member to its target member at every member of its descriptor family.	Definition 2.1
soundness registry	The closed list, emitted with every artefact, of admitted source-link kinds with their soundness claims and carried assumptions.	Definition 2.2
source cutoff	The largest photon number that keeps a sector of its own in the virtual source	Definition 7.2
source descriptor	The transmitted system and signal ensemble a source link starts from.	Definition 2.1
source link	A link of a reduction chain's source stage that maps one Alice-side ensemble to another with the same labels and weights.	Definition 2.1

Term	Meaning	Defined in
source model	A submodule's named definition of the signal ensemble as mathematical objects, possibly infinite dimensional, rather than as matrices.	Definition 1.1
source segment	The ordered, possibly empty list of source links leading from the virtual ensemble to the specified one.	Definition 2.2
source specification	The declared description of Alice's side of a protocol: alphabet, retained system with sectors, measurement, signal ensemble and source mode.	Definition 1.1
source-replacement certificate	An operator on Alice's retained system and the virtual transmitted system whose marginal on her side is her declared marginal and which, measured on her side by her POVM, leaves the weighted virtual members, with retention attested.	Definition 3.3
specified signal ensemble	The weights and signal states, one per letter, that Alice's source is specified to emit.	Definition 1.1
tagged source declaration	A finite virtual source over a weak-coherent-pulse model: two cutoffs, sector weights at most the Poisson weights, flags and one tagging link	Definition 7.2
tail flag	A one-dimensional sector for one polarisation and one intensity, carrying the weight the declared sectors leave out	Definition 7.2
tail weight	One minus the sum of an intensity's declared sector weights, the weight its tail flags carry	Definition 7.2
target descriptor	The transmitted system and signal ensemble a source link produces.	Definition 2.1
theorem constructor	A named claim of a submodule that supplies a source link's channel for each member of a declared family of descriptors.	Definition 2.1
transmitted system	The system Alice sends into the channel, carrying the specified signal ensemble.	Definition 1.1

Term	Meaning	Defined in
transported constraint entries	Entries of a consumer's constraint list carried across a link, each with the certificate case that discharges it.	Definition 2.1
trusted-source mode	The source mode in which Alice's retained system never reaches the adversary, with a declared marginal and, above dimension one, a certificate.	Definition 1.1
untrusted-source mode	The source mode with no certificate and no marginal, in which the adversary may have prepared Alice's retained system.	Definition 1.1
virtual ensemble	The finite-dimensional ensemble the source segment starts from and the certificate is written against.	Definition 2.2
virtual transmitted system	The transmitted system of the virtual ensemble, where the source segment starts.	Definition 2.2
weak-coherent-pulse model	A source model that declares phase-randomised weak coherent states on a two-mode Fock space as mathematical objects, not as matrices	Definition 7.1
weighted members	The positive operators, weight times state, one per letter, through which an ensemble is declared and read.	Definition 1.1

Part I
Mathematics

1 Source specifications and source modes

Alice's side of a protocol is fixed by what she declares and by the mode in which she declares it. Definition 1.1 lists the declared data and the number model in which every condition is decided exactly. In trusted-source mode her retained system never reaches the adversary and a certificate ties it to the signal ensemble; in untrusted-source mode nothing does. Remark 1.2 says what each mode asserts, why a declared marginal needs an attestation, and why untrusted-source mode is always the safe choice.

Alice's alphabet, retained system with sectors, measurement and signal ensemble are declared, with binary64 numbers and exact decisions, or defined from declared data or a source model. Trusted-source mode adds a marginal and, above dimension one, a certificate; untrusted-source mode has neither.

Alice's side of a protocol is described by what she prepares and what she keeps. The definition lists the data a specification declares, the conditions on them, and the objects that are defined from declared data rather than declared themselves; its closing number model says how declared data are given and how every condition is decided. In trusted-source mode the specification asserts that Alice's retained system A never reaches the adversary, and a source-replacement certificate (Definition 3.3) describes her preparation as her measurement $\{M_x\}$ of A. In untrusted-source mode no certificate ties A or $\{M_x\}$ to the signal ensemble: $\{M_x\}$ is Alice's measurement of a system the adversary may have prepared, and the mode is the conservative relaxation of trusted-source mode (Remark 1.2). The sector structure on A is what lets a certificate be declared in block form (Definition 3.3), which a source that first draws a sector and then prepares a state in it needs (Remark 4.4). At $d_A = 1$ the two modes coincide (Remark 4.7).

Definition 1.1 (Source specification)**.** A *source specification* $\mathfrak{S}_{\mathrm{src}}$ consists of the following data. A finite non-empty alphabet $\mathcal{X}$ with a declared order. A finite-dimensional system A with a declared computational basis, $d_A \coloneqq \dim \mathcal{H}(A)$, and a declared *sector structure* on it: an ordered finite index set $\mathbb{T}$ and a decomposition $\mathcal{H}(A) = \bigoplus_{t\in\mathbb{T}} \mathcal{H}(A_t)$ into subspaces each spanned by vectors of the declared basis, $d_{A_t} \coloneqq \dim \mathcal{H}(A_t) \geq 1$, with projectors Π_t^A; the *one-sector structure*, $\mathbb{T}$ a single index and $A_t = A$, is the default. A POVM $\{M_x\}_{x\in\mathcal{X}}$ on A presented as a block list against that structure, $M_x = \bigoplus_t M_x^{(t)}$ with $M_x^{(t)} \in \mathcal{L}(A_t)$, each block Hermitian and positive semidefinite, and $\sum_{x\in\mathcal{X}} M_x^{(t)} = \mathbb{1}_{A_t}$ for every t with the sum taken in the declared order. The *specified signal ensemble* $\{p_x, \rho_x\}_{x\in\mathcal{X}}$ on the *transmitted system* A', with

$p_x \geq 0$, $\sum_x p_x = 1$ and ρ_x a state on A'. With one sector the block list M_x is the matrix M_x, and every condition on its blocks is the same condition on that matrix.

How the specified ensemble is declared. In one of two ways. (a) *As matrices*, for a finite-dimensional A': by its *weighted members* $\sigma_x \in \mathrm{Pos}(A')$, with $p_x := \operatorname{tr} \sigma_x$, $\sum_x p_x = 1$ in the declared order, and $\rho_x := \sigma_x / p_x$ where $p_x > 0$. A declaration by the pair (p_x, ρ_x), $\rho_x \in S_=(A')$, is the declaration of the weighted member $\sigma_x = p_x \rho_x$ with the product formed exactly, and it requires $\rho_x \in S_=(A')$ of *every* letter, a letter with $p_x = 0$ included. Every condition of this module that reads an admitted ensemble reads it through its weighted members only. (b) *By a source model*, for a transmitted system A' whose Hilbert space is separable, possibly infinite dimensional: by a named definition of a submodule of this module that fixes the weights p_x and the states ρ_x as mathematical objects and not as matrices, together with the physical-source assumption which that submodule declares in the place of Assumption 5.1. A specification declared by a source model is presented with a non-empty source segment (Definition 2.2): what a consumer receives is never the model but the virtual ensemble, which is finite dimensional.

The two modes. The specification is declared in one of two modes. In *trusted-source mode* it additionally carries a marginal $\tau_A \in S_=(A)$, presented as a block list $\tau_A = \bigoplus_t \tau_A^{(t)}$ with $\tau_A^{(t)} \in \mathrm{Pos}(A_t)$ and $\sum_t \operatorname{tr} \tau_A^{(t)} = 1$, and, at $d_A \geq 2$, a source-replacement certificate in the sense of Definition 3.3, written against the *virtual* ensemble $\{p_x, \rho_x^{\mathrm{v}}\}$ on the virtual transmitted system A'': the finite-dimensional ensemble the source segment starts from, whose channels produce the specified ensemble from it (Definition 2.2 (S1), (S3)); when the source segment is empty, $A'' = A'$ and $\rho_x^{\mathrm{v}} = \rho_x$. The certificate takes one of the three forms of Definition 3.3: a rank-one projector (the pure form), positive sector blocks (the block form), or an ensemble presentation, which declares only the weighted virtual ensemble, in blocks along a matched sector structure of A'', and defines the certificate from it. A sector structure with more than one sector is declared only in trusted-source mode and only together with a certificate in block form or an ensemble presentation. *With an ensemble presentation* the specification declares neither A, its basis and its sector structure, nor $\{M_x\}$, nor τ_A: they are constituents of the specification that (3.2) defines from the declared virtual ensemble, the conditions this definition places on them hold by Lemma 3.4 (i), and of them only τ_A is formed as numbers, that is, computed entry by entry from the declared blocks (Lemma 3.4 (iii)); $\{M_x\}$ and the certificate are defined and never computed. At $d_A = 1$ no certificate is demanded and $\tau_A = [1]$ is the only marginal, so that there the two modes impose the same condition. In *untrusted-source mode* the specification carries no certificate and no marginal, written $\tau_A = \mathsf{none}$; the specification with $d_A = 1$ and a one-letter $\mathcal{X}$ is its degenerate case, in which Alice prepares nothing that this module

describes.

What the sector structure is not. It is a declaration about $\{M_x\}$ and τ_A, which are block diagonal by their presentation. It is no statement about any operator on A together with a further system, and no consumer may read it as one: that a joint state is block diagonal along $\mathbb{T}$ is either proved of the states a certificate reaches (Lemma 4.5) or licensed by a theorem of the consuming module that restricts its own feasible set, under that module's own preconditions, and is never inferred from the presentation of τ_A or $\{M_x\}$.

The number model. Every declared entry is a binary64 number, hence a dyadic rational, and is admissible exactly as declared. A source model (case (b)) declares no entries: its weights and states are mathematical objects, and what is decided of such a specification is decided on the entries it does declare, those of its virtual ensemble among them. Every condition above is decided *exactly* on the declared entries: a quantity that a condition compares — a sum in the declared order, a product, a partial trace — is *formed* in exact arithmetic on the declared entries as well as compared exactly, so that the identity decided is the identity written and rounding enters nowhere. That the operands are binary64 is not what makes a decision exact; a product of two binary64 numbers need not be one. Positive semidefiniteness is decided by an exact decision on the declared entries and never by a floating-point eigenvalue.

Used by Rem. 1.2, Def. 2.2, Fact 3.1, Def. 3.3, Lem. 3.4, Thm. 4.1, Cor. 4.3, Lem. 4.5, Rem. 4.7, Ass. 5.1, Ass. 5.3, Def. 7.1, Cor. 9.2, meat-core/lem:product-marginal, protocol-compiler/def:compile, protocol-compiler/def:spec, protocol-compiler/thm:gates, reductions/lem:source-transfer · newly derived · implemented · *Code* certiqs_key.transmitter.artifact:SourceSpecification

Only trusted-source mode asserts that Alice's retained system never reaches the adversary. A declared marginal deletes attacks at $d_A \geq 2$ and needs an attestation there; zero never means no marginal, and the maximally mixed marginal means none only at $d_A = 1$, where it is [1].

Remark 1.2 (The two source modes). *(a) What each mode asserts.* In *trusted-source mode* ($\tau_A \in S_=(A)$) the specification asserts, through Assumption 5.3, that Alice's register A never passes through the adversary's hands, so that in every round the joint state of A, the transmitted system and the adversary's registers has A-marginal τ_A. That is the content of the source-replacement step, and the published analyses state it in those words: [3, Eqs. (36)–(39)] adds marginal constraints to the entanglement-based statistics constraints *for prepare-and-measure protocols*, on the stated ground that "system A is inaccessible to Eve, and hence $[\rho_A]$ is fixed, independent of Eve's attack"; [1, Rem. 1] imposes $\mathrm{tr}_{B^n}[\sigma_{A^nB^n}] = \tau_A^{\otimes n}$ "because we assume Eve has no access to Alice's lab". That n-round product form of the condition is the consuming engine's Lemma "Product marginal of Alice's retained systems" (meat-core/lem:product-marginal), which derives it from Assumption 5.3, the independence of the source's rounds (Assumption 5.1 or the premise a source model declares in its place) and Assumption 5.2, together with that engine's premises on private local randomness, a sealed laboratory and the finite dimension

of the adversary's registers. In *untrusted-source mode* ($\tau_A = \mathsf{none}$) the specification asserts nothing about A: the adversary may prepare A jointly with the transmitted system, and no marginal condition is asserted at all. $\{M_x\}$ is then Alice's measurement of whatever system A reaches her, and no condition relates A or $\{M_x\}$ to the signal ensemble, the certificate that would do so being absent; the consuming engine's round model accordingly lets the attack supply A (Definition "Round model and licensed attack sets" (meat-core/def:roundmodel)). A protocol with no trusted source can also be declared at $d_A = 1$, with Alice's measured system placed on the receiver side (Remark 4.7).

(b) The mode is not a free choice. A declared τ_A *deletes* from a consumer's feasible set every state whose A-marginal differs, hence every attack that would have produced one. That is a different *kind* of input from a constraint on a register the adversary holds, and a more demanding one to justify: such a constraint restricts what the adversary's channel may do to a register she holds, whereas τ_A asserts that a register never reached her at all. Declaring one at $d_A \geq 2$ without the attestation of Assumption 5.3 overstates the key in the unsafe direction; at $d_A = 1$ it deletes nothing (Remark 4.7). Untrusted-source mode is always available and always more conservative: every state feasible with the marginal declared is feasible without it, so dropping the marginal loses rate, never gains any, and can never turn an unsound declaration into a sound-looking one. That inclusion is proved where the feasible sets are defined, in Remark "The two source modes on the program" (meat-core/rem:modes-da1) (a) and Proposition "Dropping a trusted marginal enlarges the feasible set" (protocol-compiler/prop:conservative).

(c) How "no marginal" is expressed. The way to say "no marginal" is $\tau_A = \mathsf{none}$. The value $\tau_A = 0$ is inadmissible, for two independent reasons. First, trusted-source mode requires $\tau_A \in S_=(A)$, a normalized state, which $\tau_A = 0$ fails. Second, even with normalization dropped it would express the opposite of leaving the marginal free: a state ω on A and the transmitted system with $\omega_A = 0$ has $\operatorname{tr}\omega = \operatorname{tr}\omega_A = 0$ and hence $\omega = 0$, so the condition admits no state at all. Nor, at $d_A \geq 2$, is $\tau_A = \mathbb{1}_A/d_A$ a neutral way to say that nothing is known about the source: it fixes the marginal to the maximally mixed state and excludes every adversarial source that sends anything else. At $d_A = 1$ it is $[1]$, the only state on A, and imposes the same condition as none (Remark 4.7).

Uses Def. 1.1 · newly derived · accepted

2 Source links and the source segment

A security proof may analyse a replacement of the protocol, reached by a reduction chain whose source stage, the source segment, changes what Alice emits and whose receiver stage changes the measurement. A source link maps one Alice-side ensemble to another with the same labels and weights, and a segment of links runs from the virtual ensemble, on which the certificate is written, to the specified one; each channel is declared data or a named theorem of a submodule, and in the second case the link carries an obligation record. Because each link applies one channel to the transmitted system for every label, the adversary could apply it herself, so, in trusted-source mode, security of the virtual protocol carries over to

the specified one (Lemma "Exactness of the source map in trusted-source mode" (reductions/lem:source-transfer)). Definition 2.2 states which link kinds are admitted and what a consumer decides of a presented segment; the one admitted kind is the decoy submodule's tagging link (Proposition 8.1).

A source link maps one Alice-side signal ensemble to another with the same labels and weights, through a channel given as data or by a named theorem. A theorem-supplied channel carries an obligation record, whose registry entry a consumer decides and whose family membership it does not.

A source link replaces Alice's signal ensemble by another one, carrying the labels and the weights through unchanged; along a source segment the links run from the virtual ensemble a certificate is written against to the specified ensemble (Definition 2.2 (S1), (S3)). Besides its channel, the tuple records the family of descriptors the channel's identity is asserted for, which is what makes a link sound rather than merely well typed. The one kind admitted is the decoy submodule's tagging kind, whose channel is supplied by a theorem (Proposition 8.1); the kind source-map that this module declares has no soundness claim and is refused. How source links compose with the receiver-stage links of a chain is stated by the module that composes chains; which links are admitted at all is Definition 2.2.

Definition 2.1 (Source link)**.** A *source link* Ξ is a typed link of a reduction chain (Definition "Reduction chain" (reductions/def:chain)) at the stage source: a tuple

$$\Xi = \Big(\mathsf{kind},\ \big(A^{\mathrm{in}}, \{\sigma_x^{\mathrm{in}}\}_{x\in\mathcal{X}}\big),\ \big(A^{\mathrm{out}}, \{\sigma_x^{\mathrm{out}}\}_{x\in\mathcal{X}}\big),\ \Gamma_\Xi,\ \mathcal{D}_\Xi,\ \mathfrak{s}_\Xi,\ \mathfrak{L}_\Xi,\ \mathsf{licence},\ \mathsf{status}\Big) \tag{2.1}$$

whose slots are the following.

(a) The kind. kind is a source-stage link kind. The vocabulary is the kinds this module declares, of which there is exactly one, source-map, together with the kinds a submodule of this module declares in its own interface. A submodule that declares a kind supplies its soundness claim, and a kind for which no claim of this repository proves the link's identity is declared and refused (Definition 2.2, "Which kinds are admitted"), as source-map is. An admitted kind has an entry in the soundness registry of that definition, and a kind without one is not admitted whatever a link declares.

(b) The two descriptors. The *source descriptor* $(A^{\mathrm{in}}, \{\sigma_x^{\mathrm{in}}\})$ is the transmitted system and signal ensemble the link starts from and the *target descriptor* $(A^{\mathrm{out}}, \{\sigma_x^{\mathrm{out}}\})$ the one it produces, both indexed by the same alphabet $\mathcal{X}$ in the same declared order, and both read through their weighted members $\sigma_x = p_x\rho_x$, as every ensemble of this module is (Definition 1.1): σ_x^{in} and σ_x^{out} are positive semidefinite with $\operatorname{tr}\sigma_x^{\mathrm{in}} = \operatorname{tr}\sigma_x^{\mathrm{out}} = p_x$, which is what carrying the weights untouched means.

(c) The channel witness, in one of two ways. Γ_Ξ is a channel from A^{in} to A^{out}, supplied either *as data* or *as a theorem constructor*.

(c1) *As data*: both systems are finite dimensional and the channel is declared, so that the identity $\Gamma_\Xi(\sigma_x^{\text{in}}) = \sigma_x^{\text{out}}$ for every $x \in \mathcal{X}$ is an exact matrix identity on declared numbers and is decided as such (Definition 2.2 (S3)).

(c2) *As a theorem constructor*: the witness is supplied by a named claim of a submodule of this module, which constructs one channel for each member of a declared family of descriptors and proves the identity of (c1) for it. The identity is then *not* decided on numbers: the link carries an *obligation record* of four components, the triple

$$\big(\mathsf{kind},\ \text{the id of the soundness claim},\ \mathcal{A}(\mathsf{kind})\big) \tag{2.2}$$

and, as the fourth, the member of the declared family the link's descriptors are. The third component is the assumptions the obligation carries, which Definition 2.2 (A1) defines as the soundness claim's assumption closure together with the physical-source assumption the submodule declares for the source model the claim is stated over, an assumption the soundness claim itself does *not* assume. The obligation record travels with the link and, where the link carries a licence record, in it. *Its first three components are decided*, against the soundness registry of the artefact that carries the link (Definition 2.2 (A1), (A2)), by a comparison of declared strings. The fourth is not: which member of the family the descriptors are is asserted by the module that emits the link and re-proved by no consumer, and no verdict on a number is carried, there being none to decide.

A channel supplied as a theorem constructor is how a link whose source or target is not a matrix is expressed at all, and it is the only way an obligation enters this module: no claim here discharges one.

(d) The declared descriptor family. $\mathcal{D}_\Xi$, the *declared descriptor family*, is the family of source descriptors the witness's identity is asserted for. It contains the declared source descriptor; for a link whose channel is supplied as data it is that descriptor alone; for a constructor link it is the family the named claim proves the identity for. The link is *sound* when $\Gamma_\Xi(\sigma_x^{\text{in}}) = \sigma_x^{\text{out}}$ holds for every $x \in \mathcal{X}$ at every member of $\mathcal{D}_\Xi$; for a data link this is what (S3) decides, and for a constructor link it is what the named claim concludes and the obligation record names. A link whose family does not contain its own declared descriptor asserts nothing about the descriptor it carries, and is refused. For a constructor link that membership is not a decidable condition: the family is the extension of a hypothesis of a claim, the emitting module asserts that its descriptors satisfy it, and what a consumer decides instead

is the triple (2.2) against the registry and the provenance of the artefact (Definition 2.2 (A1) to (A3)). The refusal of a family that excludes its own descriptor therefore applies where membership is decidable, to a data link, whose family is by definition its own descriptor, so that it refuses a data link whose family slot excludes that descriptor; elsewhere membership is an obligation of the emitting module, and what the registry bounds is which claims a link may name for it.

(e) The retained-register sector map. $\mathfrak{s}_\Xi$, the *retained-register sector map*, is optional: a link may emit a matched pair of sector structures over one ordered index set, one on Alice's retained system A and one on the Alice-side descriptor *the certificate is written against*, which is the *source* descriptor of the segment's first link, the virtual pair (Definition 2.2 (S1)), in the sense the block form of Definition 3.3 is written against. It is not the descriptor the link produces: a link's target may be a system that carries no sector structure at all, and for a constructor link whose target is infinite dimensional it never carries one. The sector map is a presentation datum: it says how the certificate, the marginal and Alice's measurement are presented, and it is no statement about any joint state (Definition 1.1, "What the sector structure is not"). Where the slot is empty the one-sector structure is meant.

(f) The transported constraint entries and the licence. $\mathfrak{L}_\Xi$ is the list of *transported constraint entries*, entries of a consumer's constraint list carried across the link, each carrying, *per entry*, the certificate case that discharges it — the statement, for that entry, that what the entry asserts of the link's target is implied by what holds of its source — and $\mathsf{licence}$ is the link's licence record, the slot every link of a chain has for its transported entries (Definition "Typed link" (reductions/def:link)), present exactly when $\mathfrak{L}_\Xi \neq \varnothing$. *A non-empty list is refused*: a consumer's constraint operators live on the measured system, and no claim of this repository transports an Alice-side entry. An admitted link therefore carries an empty list and no licence record of its own.

(g) The status. $\mathsf{status} \in \{\mathsf{exact}, \mathsf{certified_outer}, \mathsf{exploratory_numeric}\}$, the link statuses of Definition "Typed link" (reductions/def:link), with exact meaning that the identity of (c1) holds with no error term — for a constructor link, that the named claim concludes it with none.

Alice-side descriptor discipline. Both descriptors of a source link are Alice-side: each names a transmitted system and an ensemble on it, and no condition on a source link mentions the receiver's systems, its outcome alphabets or its measurements. A source link changes what Alice emits and nothing else.

The degenerate link. A link that supplies its channel as data, whose declared descriptor family is that one descriptor, whose sector-map slot is empty and whose entry list is empty gains nothing from its slots (d) and (e): they carry exactly what the identity of (c1) asserts and the one-sector

structure.

Used by Def. 2.2, Def. 7.2, Prop. 8.1, Cor. 9.2, reductions/def:chain, reductions/def:link · newly derived · implemented · *Code* certiqs_key.transmitter.links:SourceLink

A source segment chains source links from the virtual ensemble to the specified one, preserving labels and weights, with the source identity decided exactly for data links. A link kind is admitted only with a soundness claim, and a consumer decides its registry entry, ledger and provenance.

Where the source of a protocol is not declared directly as the ensemble the security analysis is about, a chain of source links connects the two. The definition fixes what a consumer decides of such a chain and which link kinds may appear in it at all; Remark 2.3 says what these conditions leave to other claims. The conditions are read by the modules that consume the artefact this module emits: the declared data with the segment, a soundness registry of admitted link kinds, and an assumption ledger listing the assumptions the artefact rests on.

Definition 2.2 (Source segment and its interface conditions)**.** The *source segment* of a reduction chain is an ordered list $\mathfrak{R}_{\mathrm{src}} = (\Xi_1, \dots, \Xi_{L_\mathrm{s}})$ of source links (Definition 2.1), $L_\mathrm{s} \geq 0$, whose links act on the transmitted system and replace Alice's signal ensemble; it is presented together with a source specification $\mathfrak{S}_{\mathrm{src}}$ (Definition 1.1). The segment satisfies the *interface conditions* when the following hold.

(S1) *(descriptor match)* the target descriptor of Ξ_i equals the source descriptor of Ξ_{i+1} for $1 \leq i < L_\mathrm{s}$; the target descriptor of Ξ_{L_s} is the specified pair, the transmitted system A' named by $\mathfrak{S}_{\mathrm{src}}$ with the specified ensemble $\{p_x, \rho_x\}$ on it; and the source descriptor of Ξ_1 is the virtual pair, the *virtual transmitted system* A'' with the *virtual ensemble* $\{p_x, \rho_x^{\mathrm{v}}\}_{x \in \mathcal{X}}$ on it, on which the source-replacement certificate is written (Definition 3.3). Every descriptor of this segment is an Alice-side descriptor: no condition of the segment mentions the receiver's systems, its outcome alphabets or its measurements;

(S2) *(index preservation)* each link carries the signal index x through unchanged: it maps a family indexed by $\mathcal{X}$ to a family indexed by $\mathcal{X}$, with the weights p_x untouched;

(S3) *(source identity, where checkable)* writing $\Xi := \Xi_{L_\mathrm{s}} \circ \cdots \circ \Xi_1$ for the composite when every link supplies its channel as data,

$$\Xi(\rho_x^{\mathrm{v}}) = \rho_x \qquad \text{for every } x \in \mathcal{X}, \tag{2.3}$$

an *exact* matrix identity, decided entry by entry. The composite Ξ is *formed* in exact arithmetic as well, so that the identity decided

is the identity written; that the operands are binary64 is not what makes the decision exact (the number model of Definition 1.1). The channel runs from the *virtual* states to the *specified* ones, and the object a consumer receives is the virtual protocol, the protocol with the virtual ensemble in the place of the specified one. *Where a link supplies its channel as a theorem constructor* (Definition 2.1 (c2)) the identity of *that* link is not decided on numbers: it is the conclusion of the claim the link's obligation record names, at the member of that claim's declared descriptor family which the link's descriptors are, and the identity of the composite is the composition of the per-link identities — each one either decided, where its link supplies data, or concluded by its named claim, where its link supplies a constructor. In a segment that mixes the two kinds of witness, each data link's identity is therefore decided link by link. So (2.3) is what the composite satisfies in every case, and what changes with the kind of witness is *by what* it is established and therefore what a consumer may re-decide: nothing of a constructor link's identity, which is why the obligation travels;

(S4) *(slots)* each link carries its constraint entries and its licence slot, both possibly empty, and its status tag; a link whose channel is a theorem constructor carries in addition its obligation record, the claim and the assumptions it names, which are the strings (A2) below decides.

Which kinds are admitted, and what a consumer decides. No source link kind is admitted unless the submodule that declares it supplies its soundness claim: a link of kind k may appear in a segment only where a submodule of this module declares k and supplies, as a claim of this repository, the statement that the link's witness satisfies the identity of Definition 2.1 (c1) at every member of its declared descriptor family — decided there, for a data link, or proved there, for a constructor link. A link of a kind for which no such claim exists is *refused*, and so is a link with a non-empty list of transported constraint entries, whatever its kind: no claim of this repository transports an Alice-side entry.

That much is a condition on the *kind*, and a consumer cannot decide it of a claim it does not read. What a consumer decides is the following, and it is decided on data.

(A1) *(the soundness registry)* this module emits, with every artefact, a finite list — the *soundness registry* — with one entry per admitted source-link kind, *closed* in that a kind without an entry is not admitted (Definition 2.1 (a)), each entry the triple

$$\big(\mathsf{kind},\ \text{the id of its soundness claim},\ \mathcal{A}(\mathsf{kind})\big), \tag{2.4}$$

written out in the module's interface and emitted verbatim. *The third component $\mathcal{A}(\mathsf{kind})$ is the assumptions the obligation carries*, and it has two sources, both of which a registry entry lists:

(a) the *assumption closure of the soundness claim*: the assumptions in the transitive closure of that claim under its *uses* and *assumes* edges, the claims and assumptions each claim lists as its dependencies; it may be empty, and is empty if and only if no claim of that closure, the soundness claim included, lists an assumption among its dependencies;

(b) the *physical-source assumption the submodule declares for the source model the claim is stated over*, where the claim is stated over one. That assumption is not a premise of the soundness claim and must not be made one: what it attests is that the model the claim is about describes the physical device, which is what an obligation on a physical source owes and a mathematical identity does not.

The registry of an artefact is the registry of the version of this module that emitted it, not a datum a request supplies.

(A2) *(the triple is decided)* a presented chain's link is *refused* unless the triple (its kind, the claim id of its obligation record, the assumption short ids of that record) appears in the registry of the artefact that carries it — a comparison of finitely many declared strings, taken entry by entry — and unless every assumption of that entry is in the artefact's assumption ledger, the list of assumptions the artefact rests on. *Both components of $\mathcal{A}(\mathsf{kind})$ are compared*, so a link that names the soundness claim's closure and omits the model's attestation is refused exactly as one that names neither. A link supplying its channel as data carries no obligation record and is decided by (S3) instead.

(A3) *(provenance)* a non-empty source segment is accepted only inside an artefact emitted by the module that admits its links. A segment assembled from links a consumer received in any other way is refused, whatever its triples: the registry says which kinds this repository has proved, not that a particular link was built by the module that proved them.

The admission rule on the kind is applied where a link is admitted, which is this module and the submodule whose kind it is; it is not among the conditions (S1) to (S4), which are the conditions a consumer decides on a presented chain, and (A1) to (A3) are conditions on the artefact and its registry rather than on the segment's descriptors.

The empty segment. When $L_{\mathrm{s}} = 0$ the segment is empty, Ξ is the identity, $A'' = A'$ and $\rho_x^{\mathrm{v}} = \rho_x$, so (2.3) holds by inspection. That is the case of every

specification whose ensemble is declared as matrices and that carries no link. A specification whose ensemble is declared by a source model is presented with a non-empty segment (Definition 1.1, case (b)), which is the other case and the one the conditions above are written for.

Uses Def. 1.1, Def. 2.1 · *Used by* Rem. 2.3, Def. 3.3, Rem. 4.6, Def. 7.2, Cor. 9.2, meat-core/lem:product-marginal, protocol-compiler/ass:link, protocol-compiler/def:banda, protocol-compiler/def:chaincond, protocol-compiler/def:spec, protocol-compiler/req:da1-source-identity, protocol-compiler/thm:band-a, protocol-compiler/thm:chain-consumption, reductions/def:chain, reductions/def:link, reductions/lem:source-transfer, reductions/thm:two-sided · newly derived · implemented · *Code* certiqs_key.transmitter.links:check_source_link

The registry conditions decide that a link names a claim proved for its kind, carries that claim's assumptions and comes from the admitting module; they do not decide that the claim applies to these descriptors. Composing the segment with the receiver segment is stated elsewhere.

Remark 2.3 (Scope of the source-segment conditions). *What (A1) to (A3) decide.* The conditions (A1) to (A3) of Definition 2.2 decide that a link's obligation names a claim of this repository that was proved for its kind, that the assumptions that obligation carries — the claim's closure and the attestation for its model — travel in the ledger, and that the segment came from the module that admits it. They do not decide that the named claim's conclusion applies to these descriptors: family membership is asserted by the emitting module and recorded as the fourth component of the obligation record (Definition 2.1 (c2), (d)), and no consumer re-proves a theorem. A link's obligation thus has two halves, what a consumer decides and what it takes from the emitting module. The second is exactly what rests on the emitting module for every artefact this repository passes between modules, and it is the smaller half: the registry closes the case of a link naming a claim that proves nothing about its kind, which is the case no other condition reaches.

What is decided elsewhere. The definition admits a *kind* as a type, on the strength of a claim and a registry entry, and it decides of a presented segment the conditions (S1) to (S4) and (A1) to (A3). It asserts of no specification that it declares a non-empty segment: whether one does is *data* of the presented chain, read off the segment itself. Two things a non-empty segment needs are not the definition's and are named rather than restated: the rule by which the segment composes with a receiver segment, which is Lemma "Licensed admissibility survives precomposition with a source channel" (reductions/lem:precomposition) and Theorem "The two-sided reduction in trusted-source mode" (reductions/thm:two-sided) with the admission rule of Definition "Reduction chain" (reductions/def:chain), in trusted-source mode; and the consumer that decides that rule on an assembled declaration, which is Definition "Interface conditions on a chain" (protocol-compiler/def:chaincond). That consumer also refuses a non-empty segment in untrusted-source mode.

Uses Def. 2.2 · newly derived · accepted

3 The source-replacement certificate

Source replacement describes a preparation as the measurement of a system Alice retains, so that a security proof can work with an entangled state. The imported construction is canonical, and Definition 3.3 abstracts the property that makes declared data such a description, as an operator on Alice's system and the virtual transmitted system, the system of the virtual ensemble from which the source segment produces the specified one. It is declared as a pure projector or as positive sector blocks, or defined from the virtual ensemble alone. The section's main result, Lemma 3.4, constructs the certificate an ensemble presentation defines, so that no square root or quotient of a declared number is ever declared. Every decided condition is decided exactly on the declared binary64 entries, in the sense of the number model of Definition 1.1; the retention condition is an attestation.

Preparing labelled signals is equivalent to preparing purifications with a shield system and measuring a retained label register. This is the entanglement-based description on which the trusted-source constraints of the module rest.

Fact 3.1 (Source replacement for prepare-and-measure protocols)**.** *([2, Lemma 8.3, Eqs. (83)–(84)], with the protocol framing of its Def. 8.1 and Def. 8.2.)* Consider a protocol in which Alice first prepares the global state $\rho_{X_1^n (A')_1^n} = \bigotimes_{j=1}^n \rho^{(j)}_{X_j A'_j}$ with $\rho^{(j)}_{X_j A'_j} = \sum_{k=1}^{d_A} p_k^{(j)} \, |k\rangle\langle k|_{X_j} \otimes (\rho_k^{(j)})_{A'_j}$, where $\{|k\rangle\}_{X_j}$ is the classical basis of X_j; she keeps X_1^n and sends the $(A')_1^n$ systems to the adversary. Then the state $\rho_{X_1^n (A')_1^n}$ can equivalently be obtained by *(1)* preparing the global state $\rho_{\bar{A}_1^n \hat{A}_1^n (A')_1^n} = \bigotimes_j \rho^{(j)}_{\bar{A}_j \hat{A}_j A'_j}$ with $\rho^{(j)}_{\bar{A}_j \hat{A}_j A'_j} = \sum_{k,i=1}^{d_A} \sqrt{p_i^{(j)} p_k^{(j)}} \; |i\rangle\langle k|_{\bar{A}} \otimes \left|\psi_i^{(j)}\right\rangle\!\left\langle\psi_k^{(j)}\right|_{\hat{A}_j A'_j}$, where $\left|\psi_k^{(j)}\right\rangle_{\hat{A}_j A'_j}$ is a purification of $(\rho_k^{(j)})_{A'_j}$, $\{|k\rangle\}$ is an orthonormal basis of $\bar{A}$ and $\hat{A}_j$ is the *shield system*; *(2)* sending $(A')_1^n$ to the adversary; and *(3)* for each j, measuring $\bar{A}_j \hat{A}_j$ with the POVM $\{|k\rangle\langle k|_{\bar{A}_j} \otimes \mathbb{1}_{\hat{A}_j}\}_{k \in \{1,\ldots,d_A\}}$ and storing the result in X_j.

Symbol map and hypothesis check. The source's X_j is this module's outcome register with alphabet $\mathcal{X}$ and $k \leftrightarrow x$; its A'_j is the transmitted system carrying the ensemble the lemma is applied to, and its $\{p_k^{(j)}, \rho_k^{(j)}\}$ is that ensemble. That ensemble is the virtual pair $(A'', \{p_x, \rho_x^{\mathrm{v}}\})$ on which the source-replacement certificate is stated, which is the specified pair $(A', \{p_x, \rho_x\})$ of Definition 1.1 exactly when the source segment is empty. This module describes one round; that every round is described by the same data is the consumer's round-homogeneity assumption and is not asserted here, and it is under that assumption that the superscript j is dropped in what follows. The source's $\bar{A}_j \hat{A}_j$ is this module's retained system A, and its measurement POVM is $\{M_x\}$ with $M_x = |x\rangle\langle x|_{\bar{A}} \otimes \mathbb{1}_{\hat{A}}$. The d_A of the cited lemma and the d_A of this module are *not* the same number, and the identification does not claim they are: the published index runs $k \in \{1, \ldots, d_A\}$ over an or-

thonormal basis of $\bar{A}$, so there $d_A = \dim \mathcal{H}(\bar{A}) = |\mathcal{X}|$, whereas throughout this module $d_A = \dim \mathcal{H}(A) = \dim \mathcal{H}(\bar{A}) \cdot \dim \mathcal{H}(\hat{A})$; the two agree exactly when the shield system is trivial. The lemma's one hypothesis is that Alice prepares the classical-quantum source state of its Eq. (83) — this is Assumption 5.1 for the physical device, and the declared source data for the specified object. Its only timing element is that the global state is prepared first; it places no condition on the sequential structure of later operations, being an identity between the state of its Eq. (83) and the measured state of its Eq. (84). The cited work applies it within the protocol framing of its Def. 8.1 and Def. 8.2, and this module applies it to protocols with the sequential structure of Assumption 5.2, which supplies that structure: the assumption is the hypothesis under which this module applies the fact, and not a hypothesis of the cited lemma. The published statement quantifies over an arbitrary purification $|\psi_k\rangle$, so the shield system is trivial exactly when every state of that ensemble is pure — the case, for a pure *specified* ensemble, of [3, Eqs. (37)–(39)], which writes the source-replaced vector as $|\psi\rangle_{\bar{A}A'} = \sum_i \sqrt{p_i}\,|i\rangle_{\bar{A}}\,|\varphi_i\rangle_{A'}$ and records that the resulting marginal $\rho_{\bar{A}} = \mathrm{tr}_{A'}(|\psi\rangle\langle\psi|)$ is fixed independently of the adversary's attack.

Uses Def. 1.1 · *Assumes* A-ENS, A-SEQ · *Used by* Rem. 3.2, Cor. 4.3 · published in [2, Lemma 8.3, Eqs. (83)-(84)]; [2, Def. 8.1, Def. 8.2]; [3, Eqs. (37)-(39)] · accepted · *Oracles* registry_transmitter.py

The imported construction gives Alice's retained label register one dimension per letter, with a singular marginal whenever the signal states are linearly dependent, as in BB84. Theorem 4.1 removes that constraint; the import says nothing about granting the adversary a purification.

Remark 3.2 (Scope of the imported source replacement). Fact 3.1 supplies the *canonical* source replacement: one particular purification and one particular measurement, with $\bar{A}$ of dimension $|\mathcal{X}|$. A consumer that admitted only that form would be forced to declare $d_A = |\mathcal{X}|$ with a singular marginal in every protocol whose signal states are not linearly independent — including BB84, where the canonical $\tau_{\bar{A}}$ has rank two in a four-dimensional space. Theorem 4.1 removes that constraint: it states the property that makes a declared triple $(\Psi, \tau_A, \{M_x\})$ a source replacement (Definition 3.3), proves that any triple with that property describes the same protocol, and thereby admits the qubit form that Corollary 4.3 exhibits for BB84. The import supplies no statement about the adversary being granted a purification of Alice's retained system; that step belongs to the consuming engine's round model and is not claimed here.

Uses Fact 3.1 · newly derived · accepted · *Oracles* registry_transmitter.py

An operator on Alice's system and the virtual signal whose partial trace over the signal is her marginal and whose partial traces against her measurement elements are the weighted virtual members. It is declared in pure or block form or defined by an ensemble presentation; retention is attested.

The imported source replacement is one canonical way of describing a preparation as the measurement of a system Alice retains. The certificate abstracts the property that makes declared data such a description. Its algebraic conditions are either decided exactly on declared numbers, in the sense of the number model of Definition 1.1, or, in an ensemble presentation, theorems of the declared data; its retention condition is an attestation. It comes in three forms because what is exactly declarable depends on the source. It is written against the virtual ensemble on A'', from which the links of the source segment, of length L_{s}, produce the specified ensemble on A' (Definition 2.2); with no links the two coincide.

Definition 3.3 (Source-replacement certificate)**.** Let $\{p_x, \rho_x^{\mathrm{v}}\}_{x\in\mathcal{X}}$ be the *virtual* signal ensemble on the virtual transmitted system A'' — the pair the source segment starts from, Definition 2.2 (S1), (S3) — with weighted members $\sigma_x^{\mathrm{v}} \coloneqq p_x\rho_x^{\mathrm{v}} \in \mathrm{Pos}(A'')$, $p_x = \operatorname{tr}\sigma_x^{\mathrm{v}} \geq 0$ and $\sum_x p_x = 1$. These conditions are decided *exactly*, as every declared condition is, the weight sum in the declared order and rounding nowhere. A *source-replacement certificate* for the declared data $(A, \mathbb{T}, \tau_A, \{M_x\})$ of a source specification (Definition 1.1) is an operator $\Psi \in \mathcal{L}(AA'')$ such that

(SR1) $\operatorname{tr}_{A''}\Psi = \tau_A$;

(SR2) $\operatorname{tr}_A\big[(M_x \otimes \mathbb{1}_{A''})\,\Psi\big] = p_x\,\rho_x^{\mathrm{v}}$ for every $x \in \mathcal{X}$; and

(SR3) *(retention, an attestation)* A is retained in Alice's laboratory and never transmitted; no operation of the protocol and no operation of the adversary acts on A, except Alice's declared measurement $\{M_x\}$.

It is given in one of the three forms below. In the pure form and in the block form Ψ is declared, together with the declared computational basis of A. In an ensemble presentation neither Ψ nor a basis of A is declared: Ψ is defined from a declared ensemble by (3.2).

Pure form. In the *pure form* the sector structure of A is the one-sector structure, and Ψ is a *rank-one projector* — Ψ Hermitian, $\Psi^2 = \Psi$ and $\operatorname{tr}\Psi = 1$, each decided exactly and the trace summed without rounding, so that $\Psi = |\psi\rangle\langle\psi|$ for a unit vector $|\psi\rangle \in \mathcal{H}(AA'')$.

Block form. Let $\mathcal{H}(A) = \bigoplus_{t\in\mathbb{T}}\mathcal{H}(A_t)$ be the sector structure of A declared in Definition 1.1, along which $\tau_A = \bigoplus_t \tau_A^{(t)}$ and $M_x = \bigoplus_t M_x^{(t)}$ are presented as block lists. In the *block form* the virtual transmitted system carries a declared sector structure *matched* to that of A: the same ordered index set $\mathbb{T}$ and a decomposition $\mathcal{H}(A'') = \bigoplus_{t\in\mathbb{T}}\mathcal{H}(S_t)$ into subspaces each

spanned by vectors of the declared basis of A'', with projectors Π_t^{v}. The certificate is a block list $(\Psi_t)_{t\in\mathbb{T}}$ with $\Psi_t \in \mathcal{L}(A_tS_t)$, each block Hermitian and positive semidefinite, decided exactly, and $\sum_t \operatorname{tr}\Psi_t = 1$, the traces summed in the declared order without rounding. It denotes the operator $\Psi = \bigoplus_t \Psi_t$ on the *matched subspace* $\bigoplus_t \mathcal{H}(A_t)\otimes\mathcal{H}(S_t) \subseteq \mathcal{H}(AA'')$, extended by zero to its orthogonal complement. No purity condition is imposed, on Ψ or on any block. The weighted virtual members are required to be *block diagonal* along $(S_t)_t$: $\Pi_t^{\mathrm{v}}\sigma_x^{\mathrm{v}}\Pi_{t'}^{\mathrm{v}} = 0$ for all $t \neq t'$ and every x, that is, every entry of σ_x^{v} whose row lies in one sector and whose column lies in another is zero, decided entry by entry on the declared entries of σ_x^{v} (a virtual ensemble declared as block lists along $(S_t)_t$ meets it by its presentation). Write $\sigma_x^{\mathrm{v},(t)} \coloneqq \Pi_t^{\mathrm{v}}\sigma_x^{\mathrm{v}}\Pi_t^{\mathrm{v}}$, read as an operator on S_t, for its diagonal blocks. Against a block list the two identities read block by block,

$$\operatorname{tr}_{S_t}\Psi_t = \tau_A^{(t)}, \qquad \operatorname{tr}_{A_t}\left[(M_x^{(t)}\otimes\mathbb{1}_{S_t})\,\Psi_t\right] = \sigma_x^{\mathrm{v},(t)} \qquad \text{for every } t\in\mathbb{T},\ x\in\mathcal{X}, \tag{3.1}$$

and (3.1) together with the block diagonality of every σ_x^{v} is equivalent to (SR1) and (SR2) of the operator Ψ. Indeed Ψ vanishes off the matched subspace and M_x is block diagonal, so the partial traces of Ψ are the direct sums of those of its blocks: $\operatorname{tr}_{A''}\Psi = \bigoplus_t \operatorname{tr}_{S_t}\Psi_t$, which is $\tau_A = \bigoplus_t \tau_A^{(t)}$ exactly when the first identity of (3.1) holds; and $\operatorname{tr}_A[(M_x\otimes\mathbb{1}_{A''})\Psi] = \bigoplus_t \operatorname{tr}_{A_t}[(M_x^{(t)}\otimes\mathbb{1}_{S_t})\Psi_t]$, which is block diagonal along $(S_t)_t$ and so equals σ_x^{v} exactly when σ_x^{v} is block diagonal and its diagonal blocks satisfy the second identity. Without the block diagonality, (3.1) is (SR2) against the pinched members $\bigoplus_t \sigma_x^{\mathrm{v},(t)}$ and not against σ_x^{v}.

Ensemble presentation. A block-form certificate need not be declared at all. In an *ensemble presentation* the declared data are the matched sector structure $(S_t)_{t\in\mathbb{T}}$ on A'' and the weighted virtual ensemble as block lists, $\sigma_x^{\mathrm{v}} = \bigoplus_t \sigma_x^{\mathrm{v},(t)}$ with $\sigma_x^{\mathrm{v},(t)} \in \mathcal{L}(S_t)$, and *nothing else*: no Ψ, no $\{M_x\}$, no τ_A and no basis of A. The conditions are

(E1) every block $\sigma_x^{\mathrm{v},(t)}$ is Hermitian and positive semidefinite, decided exactly;

(E2) $p_x = \sum_t \operatorname{tr}\sigma_x^{\mathrm{v},(t)}$ for every x and $\sum_x p_x = 1$, the sums formed in the declared orders and rounding nowhere; and

(E3) for every t the sum $\sum_{x\in\mathcal{X}} \sigma_x^{\mathrm{v},(t)}$, accumulated entry by entry in the declared order of $\mathcal{X}$, rounds nowhere, so that its entries are binary64 numbers.

The system A, its sector structure, the marginal τ_A, the POVM $\{M_x\}$ and the blocks Ψ_t are then *defined* by (3.2): A_t is a copy of S_t, $\tau_A^{(t)}$ is the transpose of the sum in (E3), and Ψ_t and $M_x^{(t)}$ are the canonical purification and

its square-root measurement. By Lemma 3.4 they are a block-form certificate with (SR1) and (SR2) holding as identities, with no error term, against the declared ensemble itself, and with every non-zero block a multiple of a rank-one projector. (E1) to (E3) are all that is decided. (SR1), (SR2), positivity of the blocks, the POVM condition and the purity of the blocks are *theorems* of the presentation and are decided by nothing, there being no second declared datum they could be compared with; (SR3) is the attestation it is in every form. Of the defined objects only τ_A is ever formed as numbers, and by (E3) it is admissible declared data. Ψ_t and $M_x^{(t)}$ carry square roots and quotients of the declared numbers, are formed by no one, and are never rounded.

What a consumer reads of a presentation. What a consumer can read of Alice's measurement without Ψ_t and $M_x^{(t)}$ is the family $N_x^{(t)} = (\sigma_x^{\mathrm{v},(t)})^{\mathsf{T}}$ of Lemma 3.4 (v). *A consumer whose declaration carries Alice's measurement as the POVM $\{M_x\}$ cannot receive an ensemble presentation and refuses it*; the form of declaration that carries $\{N_x\}$ in its place is that consumer's to define, and the consuming modules define it: the protocol compiler and the key-rate engine receive a presentation through the ensemble form of their declarations, which carries $\{N_x\}$ in the place of $\{M_x\}$ (Definition "Protocol declaration" (protocol-compiler/def:declaration), Definition "The protocol declaration as the MEAT core reads it" (meat-core/def:declaration)), admitted beside a presentation by Definition "Key-length eligibility" (protocol-compiler/def:eligible). The mode, the alphabet, the source segment and the attestations of a specification are what they are in the other forms.

What is decided. (SR1) and (SR2) are *exact matrix identities* on declared data, decided entry by entry. The conditions a consumer decides — in pure form the three on Ψ, (SR1) and (SR2); in block form Hermiticity and positive semidefiniteness of every block, the trace, the block diagonality of every σ_x^{v} along $(S_t)_t$, and (3.1), which together are (SR1) and (SR2); in an ensemble presentation (E1) to (E3) and no others — are stated in these terms and in no others: none of them is stated on a vector, which is not a declared datum. (SR3) carries no number and is decided by nothing; it is the attestation of Assumption 5.3. A rank-one projector of unit trace is positive semidefinite of unit trace, so a pure-form certificate satisfies every condition of the block form at the one-sector structure, and one condition more.

One reading at every L_{s}. The pair $(A'', \{p_x, \rho_x^{\mathrm{v}}\})$ is the only pair a certificate is ever written against, at every length of the source segment, because the object a consumer receives is the virtual protocol (Definition 2.2 (S3)): the specification's τ_A and $\{M_x\}$ describe the preparation of $\{p_x, \rho_x^{\mathrm{v}}\}$, and it is the source identity (2.3), not the certificate, that carries that description back to the specified ensemble $\{p_x, \rho_x\}$ on A'. At $L_{\mathrm{s}} = 0$ the two pairs are literally equal, $A'' = A'$ and $\rho_x^{\mathrm{v}} = \rho_x$. So (SR1)–(SR2) have exactly one reading whatever L_{s} is.

Uses Def. 1.1, Def. 2.2 · *Used by* Lem. 3.4, Thm. 4.1, Cor. 4.3, Rem. 4.4, Lem. 4.5, Rem. 4.6, Rem. 4.7, Ass. 5.3, Req. 6.1, Req. 6.3, Prop. 9.1, Cor. 9.2, Req. 11.7, Req. 11.8, meat-core/lem:product-marginal, protocol-compiler/def:eligible, protocol-compiler/def:spec, protocol-compiler/req:da1-source-identity, reductions/lem:source-transfer · newly derived · implemented · *Code* certiqs_key.transmitter.certificate:SourceReplacementCertificate · *Oracles* registry_instances_bb84.py, registry_protocol_compiler.py, registry_transmitter.py

A virtual ensemble in sector blocks defines, by canonical purification and a square-root measurement, a block certificate with pure blocks that exactly reproduces Alice's marginal and, under that measurement, the ensemble. Only her marginal, a transposed sum of declared blocks, is formed as numbers.

The ensemble presentation of Definition 3.3 declares only the weighted virtual ensemble, in blocks along a sector structure of the virtual transmitted system. The lemma supplies the objects it defines and shows that they form a block-form certificate, so that no square root or quotient of a declared number ever has to be declared; of the defined objects only Alice's marginal is formed as numbers, that is, computed entry by entry.

Lemma 3.4 (Certificate defined by an ensemble presentation). Let the virtual transmitted system carry a sector structure $\mathcal{H}(A'') \supseteq \bigoplus_{t\in\mathbb{T}} \mathcal{H}(S_t)$, each S_t spanned by vectors $(|i\rangle_{S_t})_i$ of the declared basis, and let $(\sigma_x^{\mathrm{v},(t)})_{t\in\mathbb{T}}$, $x \in \mathcal{X}$, be block lists with $\sigma_x^{\mathrm{v},(t)} \in \mathrm{Pos}(S_t)$ and $\sum_x \sum_t \operatorname{tr} \sigma_x^{\mathrm{v},(t)} = 1$. For every t let A_t be a copy of S_t with basis $(|i\rangle_{A_t})_i$ in the same order, let transposition $X \mapsto X^{\mathsf{T}}$ be taken in these bases, and put

$$
\begin{aligned}
\tau_S^{(t)} &\coloneqq \sum_{x\in\mathcal{X}} \sigma_x^{\mathrm{v},(t)}, \qquad \tau_A^{(t)} \coloneqq \big(\tau_S^{(t)}\big)^{\mathsf{T}}, \\
|\psi_t\rangle &\coloneqq \sum_i |i\rangle_{A_t} \otimes \big(\tau_S^{(t)}\big)^{1/2} |i\rangle_{S_t}, \qquad \Psi_t \coloneqq |\psi_t\rangle\langle\psi_t|, \\
M_x^{(t)} &\coloneqq \big(\tau_A^{(t)}\big)^{-1/2} \big(\sigma_x^{\mathrm{v},(t)}\big)^{\mathsf{T}} \big(\tau_A^{(t)}\big)^{-1/2} + \delta_{x,x_0} \big(\mathbb{1}_{A_t} - P_t\big),
\end{aligned}
\tag{3.2}
$$

where P_t is the projector onto the support of $\tau_A^{(t)}$, the inverse square root is taken on that support and is zero off it, and x_0 is the first letter of $\mathcal{X}$ in its declared order. Then, with $A \coloneqq \bigoplus_t A_t$:

(i) *(a certificate in block form)* every Ψ_t is positive semidefinite, $\sum_t \operatorname{tr} \Psi_t = 1$, $\{M_x\}_x$ with $M_x = \bigoplus_t M_x^{(t)}$ is a POVM presented as a block list, $\tau_A = \bigoplus_t \tau_A^{(t)}$ is a state presented as a block list, and both identities of (3.1) hold: $(\Psi_t)_t$ satisfies every condition Definition 3.3 places on a block-form certificate for $(A, \mathbb{T}, \tau_A, \{M_x\})$ against the virtual ensemble with weighted members $\sigma_x^{\mathrm{v}} = \bigoplus_t \sigma_x^{\mathrm{v},(t)}$, (SR3) apart, which is an attestation;

(ii) *(pure blocks)* every non-zero Ψ_t is $g_t \coloneqq \operatorname{tr} \tau_S^{(t)}$ times a rank-one projector, so the converse of Theorem 4.1 (ii) holds of this certificate;

(iii) *(what is formed as numbers)* the entry (i,j) of $\tau_A^{(t)}$ is the entry (j,i) of the sum $\tau_S^{(t)}$ of the declared blocks. No other object of (3.2) enters (i) or (ii) through its entries: Ψ_t and $M_x^{(t)}$ contain square roots and quotients of the declared numbers and are, in general, not declarable;

(iv) *(the completion is immaterial)* the letter x_0 enters the POVM condition and nothing else: (SR1), (SR2), (ii) and (v) hold with the term $\mathbb{1}_{A_t} - P_t$ assigned to any letter, or split among the letters by any positive semidefinite operators on the kernel of $\tau_A^{(t)}$ that sum to $\mathbb{1}_{A_t} - P_t$;

(v) *(Alice's side without a square root)* $\big(\tau_A^{(t)}\big)^{1/2} M_x^{(t)} \big(\tau_A^{(t)}\big)^{1/2} = \big(\sigma_x^{\mathrm{v},(t)}\big)^{\mathsf{T}}$ for every x and t: the operators $N_x^{(t)} := (\sigma_x^{\mathrm{v},(t)})^{\mathsf{T}}$ are entrywise the declared numbers, and $\sum_x N_x^{(t)} = \tau_A^{(t)}$.

Uses Def. 1.1, Def. 3.3 · *Used by* Thm. 4.1, Prop. 9.1, Cor. 9.2 · newly derived · implemented · *Code* certiqs_key.transmitter.presentation:alice_marginal_blocks

Proof. Idea. Each block is the canonical purification of the summed ensemble of its sector, written through the maximally entangled vector; three facts about transposition reduce every claim to an identity between the declared blocks and their sum.

Preliminaries. Fix t and drop it from the notation; write $K := \tau_S^{1/2}$, $|\Omega\rangle := \sum_i |i\rangle_A |i\rangle_S$, so that $|\psi\rangle = (\mathbb{1}_A \otimes K)\,|\Omega\rangle$, and let Q be the projector onto the support of τ_S. Three facts are used. (a) $\mathrm{tr}_A\big[(Y \otimes \mathbb{1}_S)\,|\Omega\rangle\langle\Omega|\,\big] = Y^{\mathsf{T}}$ for $Y \in \mathcal{L}(A)$: with $|\Omega\rangle\langle\Omega| = \sum_{ij} |i\rangle\langle j| \otimes |i\rangle\langle j|$ the left-hand side is $\sum_{ij} \langle j|\,Y\,|i\rangle\; |i\rangle\langle j|$, whose (i,j) entry is Y_{ji}. (b) $(\mathbb{1} \otimes Z)\,|\Omega\rangle = (Z^{\mathsf{T}} \otimes \mathbb{1})\,|\Omega\rangle$, by comparing coefficients. (c) Transposition is linear, reverses products, commutes with the adjoint and maps positive semidefinite operators to positive semidefinite operators with the same spectrum; hence $f(X^{\mathsf{T}}) = f(X)^{\mathsf{T}}$ for every real function f on the spectrum of a Hermitian X. In particular $(\tau_A)^{1/2} = K^{\mathsf{T}}$, $(\tau_A)^{-1/2} = (\tau_S^{-1/2})^{\mathsf{T}}$ and $P = Q^{\mathsf{T}}$.

(SR1). By (b), $\Psi = (K^{\mathsf{T}} \otimes \mathbb{1})\,|\Omega\rangle\langle\Omega|\,(K^{\mathsf{T}} \otimes \mathbb{1})^{\dagger}$, and $\mathrm{tr}_S\,|\Omega\rangle\langle\Omega| = \mathbb{1}_A$, so $\mathrm{tr}_S\,\Psi = K^{\mathsf{T}}(K^{\mathsf{T}})^{\dagger} = (K^{\mathsf{T}})^2 = (K^2)^{\mathsf{T}} = \tau_S^{\mathsf{T}} = \tau_A$, the operator K^{T} being Hermitian by (c).

(SR2). $\mathrm{tr}_A\big[(M_x \otimes \mathbb{1})\Psi\big] = K\,\mathrm{tr}_A\big[(M_x \otimes \mathbb{1})\,|\Omega\rangle\langle\Omega|\,\big]K = K\,M_x^{\mathsf{T}}K$ by (a), K acting on S alone. By (c), $M_x^{\mathsf{T}} = \tau_S^{-1/2}\sigma_x^{\mathrm{v}}\tau_S^{-1/2} + \delta_{x,x_0}(\mathbb{1}_S - Q)$, and $K\tau_S^{-1/2} = Q$, $K(\mathbb{1} - Q) = 0$, so $K\,M_x^{\mathsf{T}}K = Q\,\sigma_x^{\mathrm{v}}Q$. Finally $0 \preceq \sigma_x^{\mathrm{v}} \preceq \tau_S$ puts the support of σ_x^{v} inside that of τ_S, so $Q\,\sigma_x^{\mathrm{v}}Q = \sigma_x^{\mathrm{v}}$.

The remaining conditions of (i). $\Psi = |\psi\rangle\langle\psi| \succeq 0$ and $\mathrm{tr}\,\Psi = \mathrm{tr}\,\mathrm{tr}_S\,\Psi = \mathrm{tr}\,\tau_A = \mathrm{tr}\,\tau_S$, whose sum over t is $\sum_x \sum_t \mathrm{tr}\,\sigma_x^{\mathrm{v},(t)} = 1$. Each M_x is positive semidefinite: $(\sigma_x^{\mathrm{v}})^{\mathsf{T}} \succeq 0$ by (c), a conjugation by the Hermitian operator $\tau_A^{-1/2}$ preserves positivity, and $\mathbb{1} - P \succeq 0$. Their sum is $\tau_A^{-1/2}\,\tau_S^{\mathsf{T}}\,\tau_A^{-1/2} + (\mathbb{1} - P) = P + (\mathbb{1} - P) = \mathbb{1}_A$, by linearity of transposition. τ_A is posi-

tive semidefinite by (c), and $\sum_t \operatorname{tr} \tau_A^{(t)} = 1$ was just shown. The weighted members $\bigoplus_t \sigma_x^{\mathrm{v},(t)}$ of the virtual ensemble are block diagonal along $(S_t)_t$ by their presentation as block lists, with the declared blocks as their diagonal blocks, which is the condition the block form places on the virtual ensemble; together with (3.1), shown above sector by sector, this is (SR1) and (SR2) of $\Psi = \bigoplus_t \Psi_t$ by Definition 3.3.

(ii) $\Psi = |\psi\rangle\langle\psi|$ with $\langle\psi|\psi\rangle = \operatorname{tr} \Psi = g$; where $g > 0$, Ψ/g is a rank-one projector. Theorem 4.1 (ii) states its converse for exactly such block lists.

(iii) is the definition of the transpose; (i) and (ii) were proved from the operator identities above and read no entry of Ψ or of M_x. That these are not declarable in general is seen on S a qubit: $\tau_S = \operatorname{diag}(\frac{1}{2}, \frac{1}{2})$ gives Ψ with every non-zero entry equal to $\frac{1}{2}$, while $\tau_S = \operatorname{diag}(\frac{3}{4}, \frac{1}{4})$ gives Ψ the entry $\sqrt{3/16}$, which is irrational.

(iv) In the proof of (SR2) the completion term contributed $K(\mathbb{1} - Q)K = 0$, and the same holds of any operator supported on the kernel of τ_S; in (SR1) and (ii) M_x does not occur; in (v) below it contributes $\tau_A^{1/2}(\mathbb{1} - P)\tau_A^{1/2} = 0$.

(v) $\tau_A^{1/2}\tau_A^{-1/2} = P$, so $\tau_A^{1/2} M_x \tau_A^{1/2} = P(\sigma_x^{\mathrm{v}})^{\mathsf{T}} P = \big(Q\,\sigma_x^{\mathrm{v}} Q\big)^{\mathsf{T}} = (\sigma_x^{\mathrm{v}})^{\mathsf{T}}$, using (c) and the support inclusion above. The sum is $\tau_S^{\mathsf{T}} = \tau_A$. □

4 Sufficiency of the certificate and its scope

The section's first main result, Theorem 4.1, shows that the certificate's algebraic conditions are, in either declared form, finitely many exact decisions and that Alice's measurement may be moved across every attack that acts only on the transmitted system and the adversary's registers, starting from a register in a product state with the source; retention and the sequential round structure make the attack of a round such a channel. Remark 4.2 shows that the hypothesis on the attack cannot be dropped, Corollary 4.3 that BB84 admits a certificate on a qubit, and Remark 4.4 what purity adds. Lemma 4.5, the second main result, states what a block certificate implies for the states an attack reaches; the closing remarks bound what a verified certificate buys, down to a one-dimensional retained system.

In any form (pure projector, sector blocks, ensemble presentation) the certificate's algebraic conditions are finitely many exact decisions making Alice's measurement reproduce the virtual ensemble. Under retention and sequential rounds she may measure before or after any attack: source replacement.

The theorem licenses source replacement at the boundary of this module. Recall the conditions of Definition 3.3: (SR1) $\operatorname{tr}_{A''} \Psi = \tau_A$, (SR2) $\operatorname{tr}_A[(M_x \otimes \mathbb{1}_{A''})\Psi] = p_x \rho_x^{\mathrm{v}}$, and the attestation (SR3) that nothing but Alice's measurement acts on A; an ensemble presentation has instead the decided conditions (E1) to (E3), positive declared blocks, normalised weights and per-sector sums that round nowhere. Part (i) is about declared data before

its conditions are known to hold, and says that the conditions other than the attestation (SR3) can be decided. Part (ii) relates a certificate to Alice's measurement and is stated for any Hermitian family, since its converse concerns a family that fails the POVM condition on the support of τ_A, that is, $\Pi(\sum_x M_x)\Pi \neq \Pi$ for the projector Π onto that support, which no source specification declares, a mismatch the pure form detects (Remark 4.4). Part (iii) shows that her measurement may be moved across any attack that leaves her retained system and her outcome register alone and starts from an adversary register in a product state with the source. Its identities hold for every channel of that form; the attestation (SR3) of Assumption 5.3 and Assumption 5.2 are what make the attack of a round such a channel, and Remark 4.2 shows that the restriction on the attack is necessary and not a weakening of the intended reading.

Theorem 4.1 (Checkability and sufficiency of the source-replacement certificate). Let a source specification (Definition 1.1) carry a source-replacement certificate Ψ for $(A, \mathbb{T}, \tau_A, \{M_x\})$ against the virtual ensemble $\{p_x, \rho_x^{\mathrm{v}}\}$, in any form of Definition 3.3. In (ii) and (iii) and their proofs an ensemble presentation is read as the block-form certificate it defines (Lemma 3.4 (i)), so that "either form" below covers it.

(i) *(checkability)* the conditions on Ψ of its declared form — in pure form Hermiticity, $\Psi^2 = \Psi$ and $\operatorname{tr}\Psi = 1$; in block form Hermiticity and positive semidefiniteness of every block and $\sum_t \operatorname{tr}\Psi_t = 1$ — together with (SR1) and (SR2), are finitely many exact identities and exact decisions on the declared data, decided by finitely many arithmetic operations. In block form (SR1) and (SR2) are decided as the block identities (3.1) together with the block diagonality of every weighted virtual member σ_x^{v} along $(S_t)_t$, decided entry by entry, which by Definition 3.3 is their conjunction; (3.1) alone decides (SR2) only against the pinched members $\bigoplus_t \Pi_t^{\mathrm{v}} \sigma_x^{\mathrm{v}} \Pi_t^{\mathrm{v}}$. For a certificate given by an ensemble presentation the decided conditions are (E1) to (E3) of Definition 3.3, equally finite and exact, and the conditions of the block form hold of the defined certificate by Lemma 3.4 (i) without being decided. (SR3) is an attestation and is decided by no function of the declared data. (SR1) moreover implies $\tau_A \in S_=(A)$, $\operatorname{tr}_{A''}\Psi$ being positive semidefinite with trace $\operatorname{tr}\Psi = 1$. In an ensemble presentation τ_A is not declared: it is formed as the transposed sum of (E3), and that it is a state is Lemma 3.4 (i).

(ii) *(consistency)* more generally than in a source specification, let $\{M_x\}$ be any family of Hermitian operators on A, presented as block lists against the sector structure, of which $\sum_x M_x = \mathbb{1}_A$ is not assumed; let Ψ meet the conditions on Ψ of its form listed in (i) and (SR1); and let $p_x\rho_x^{\mathrm{v}}$ stand for $\operatorname{tr}_A[(M_x \otimes \mathbb{1}_{A''})\Psi]$, so that (SR2) holds. Writing

$U = \operatorname{supp}\tau_A$ with projector Π_U, the certificate's signal-side marginal and the virtual average signal state agree,

$$\sum_x p_x \rho_x^{\mathrm{v}} = \operatorname{tr}_A \Psi, \tag{4.1}$$

if $\Pi_U\big(\sum_x M_x\big)\Pi_U = \Pi_U$, in either form; in particular they agree for the certificate of a source specification, whose $\{M_x\}$ is a POVM by Definition 1.1. The converse, *only if*, holds in pure form, and in block form whenever every non-zero block Ψ_t is a positive multiple of a rank-one projector; it fails for a general positive block.

(iii) *(sufficiency at the module boundary)* Assume (SR1), (SR2), the attestation (SR3) of Assumption 5.3, and Assumption 5.2; in block form (SR2) is the identity of operators on A'' against the weighted virtual members $p_x\rho_x^{\mathrm{v}}$ themselves, and not against their diagonal blocks alone: for a declared block form (i) decides it with their block diagonality, and for an ensemble presentation it is proved by Lemma 3.4 (i). Let $\mathcal{N}_{\mathsf{A}}$ be Alice's measurement map, from her retained system A to the classical outcome register X,

$$\mathcal{N}_{\mathsf{A}}(\xi) \coloneqq \sum_{x\in\mathcal{X}} \operatorname{tr}[M_x\,\xi]\ |x\rangle\langle x|_X\,, \qquad \xi \in \mathcal{L}(A), \tag{4.2}$$

and let $\mathcal{E}$ be *any* channel from $A''E$ to E', where E is the adversary's initial register and E' collects her output registers including whatever is delivered to the receiver. Thus $\mathcal{E}$ models the adversary's attack and the remainder of the round *on the transmitted system and the adversary's registers*, and it acts on neither A nor X: it is not permitted to depend on Alice's outcome. Write $\rho_{XA''} \coloneqq \sum_{x\in\mathcal{X}} p_x\ |x\rangle\langle x|_X \otimes \rho_x^{\mathrm{v}}$ for the virtual prepare-and-measure source state. Then:

(a) *(the interchange)*

$$(\mathrm{id}_X\otimes\mathcal{E})\circ(\mathcal{N}_{\mathsf{A}}\otimes\mathrm{id}_{A''E}) \;=\; \mathcal{N}_{\mathsf{A}}\otimes\mathcal{E} \;=\; (\mathcal{N}_{\mathsf{A}}\otimes\mathrm{id}_{E'})\circ(\mathrm{id}_A\otimes\mathcal{E}) \tag{4.3}$$

as linear maps on $\mathcal{L}(AA''E)$, hence in particular on $\Psi\otimes\sigma$ for every state σ on E: Alice's measurement may be performed before the attack or after it, indifferently. This uses no property of the certificate, only that the two maps act on complementary registers;

(b) *(the source identification)* for every state σ on E,

$$(\mathcal{N}_{\mathsf{A}}\otimes\mathcal{E})\big(\Psi\otimes\sigma\big) \;=\; (\mathrm{id}_X\otimes\mathcal{E})\big(\rho_{XA''}\otimes\sigma\big), \tag{4.4}$$

and both sides are states on XE'. Together with (a): on every register other than A, the declared entanglement-based description and the virtual prepare-and-measure description produce the same state, against every attack of this form. The states ρ_x^{v} in $\rho_{XA''}$ are the declared virtual states, in block form as in every form, and not their pinching along $(S_t)_t$.

Uses Def. 1.1, Def. 3.3, Lem. 3.4 · *Assumes* A-SEQ, A-SRC · *Used by* Rem. 4.2, Cor. 4.3, Rem. 4.6, Req. 6.4, Req. 6.5, Cor. 9.2, meat-core/lem:product-marginal, protocol-compiler/def:eligible, protocol-compiler/thm:gates, reductions/lem:source-transfer · newly derived · implemented (effective: accepted) · *Code* certiqs_key.transmitter.certificate: verify_certificate · *Oracles* registry_instances_bb84.py, registry_protocol_compiler.py, registry_transmitter.py

Proof. Idea. Part (i) counts the operations each condition needs and observes that they are formed exactly. Part (ii) reduces the consistency identity to the vanishing of the compression $\Pi_U Y \Pi_U$ of the defect $Y = \sum_x M_x - \mathbb{1}_A$, using that Ψ lives on the support of τ_A. Part (iii) is the interchange law for a tensor product of maps on complementary registers, followed by (SR2).

(i) In pure form the conditions on Ψ compare $(d_A \dim \mathcal{H}(A''))^2$ entries each — $\Psi = \Psi^\dagger$ and $\Psi^2 = \Psi$ — and one accumulated trace; in block form they are, per block, one comparison of $(d_{A_t} \dim \mathcal{H}(S_t))^2$ entries for Hermiticity and one exact decision of positive semidefiniteness — a symmetric elimination over the Gaussian rationals in which a negative pivot refuses the block and a zero pivot requires its whole row and column to vanish, the block being refused otherwise, which decides that every principal minor is non-negative — and one accumulated trace over all blocks; in an ensemble presentation the same decision per declared block, the traces of (E2) and the entrywise sums of (E3); in pure form (SR1) compares d_A^2 entries after a partial trace and (SR2) compares $|\mathcal{X}|$ matrices of size $\dim \mathcal{H}(A'')$, each obtained from Ψ by a multiplication and a partial trace; in block form (3.1) compares, per sector, $d_{A_t}^2$ entries and $|\mathcal{X}|$ matrices of size $\dim \mathcal{H}(S_t)$, each obtained from Ψ_t in the same way, and the block diagonality compares with zero the entries of every σ_x^{v} outside its diagonal blocks, at most $|\mathcal{X}|(\dim \mathcal{H}(A''))^2$ of them. By Definition 3.3 the block identities and the block diagonality together are (SR1) and (SR2) of Ψ: the partial traces of Ψ are the direct sums of those of its blocks, and the one of (SR2) is block diagonal along $(S_t)_t$, so it equals σ_x^{v} exactly when σ_x^{v} has no entry outside its diagonal blocks and those blocks satisfy the second identity of (3.1). Each is a finite comparison, and each is *formed as well as taken* in exact arithmetic on the declared dyadic entries (the number model of Definition 1.1): the products entering Ψ^2 and (SR2), and the additions entering the partial traces and the accumulated trace, are exact, so what is compared is the quantity written and not a rounded image of it. That the operands are exactly represented does not on its own deliver that; the exact formation does. For the implication: $\mathrm{tr}_{A''} \Psi$ is positive semidefinite as the partial trace of a positive semidefinite operator — in pure form Ψ is a projector, hence $\Psi = \Psi^\dagger \Psi \succeq 0$; in block form Ψ is a direct sum of positive semidefinite blocks — and its trace is $\mathrm{tr}\, \Psi = 1$; (SR1) is a matrix identity, so $\tau_A \in S_=(A)$. (SR3) is a statement about the apparatus, not about matrices, and no matrix function decides it; it is the attestation of Assumption 5.3.

(ii) Put $Y \coloneqq \sum_x M_x - \mathbb{1}_A$, which is Hermitian, each M_x being Hermitian by hypothesis and a sum of Hermitian matrices less $\mathbb{1}_A$ being Hermitian; Y is block diagonal along the sector structure, every M_x being presented as a block list. Summing (SR2) over x gives

$$\operatorname{tr}_A\Big[\Big(\big(\textstyle\sum_x M_x\big) \otimes \mathbb{1}\Big)\Psi\Big] = \textstyle\sum_x p_x \rho_x^{\mathrm{v}},$$

and the left-hand side is $\operatorname{tr}_A[(Y \otimes \mathbb{1})\Psi] + \operatorname{tr}_A \Psi$ by linearity of the partial trace, so (4.1) holds if and only if $\operatorname{tr}_A[(Y \otimes \mathbb{1})\Psi] = 0$.

If, in either form. $\Psi \succeq 0$ in either form. Let $|\phi\rangle$ be a unit vector of $\mathcal{H}(A)$ orthogonal to U and put $P \coloneqq |\phi\rangle\langle\phi| \otimes \mathbb{1}_{A''}$. By (SR1), $\operatorname{tr}[P\Psi] = \langle\phi|\,\tau_A\,|\phi\rangle = 0$, so $\Psi^{1/2} P \Psi^{1/2}$ is a positive semidefinite operator of trace zero, hence zero, hence $P\Psi^{1/2} = 0$ and $P\Psi = 0$. Summing over an orthonormal basis of the orthogonal complement of U gives $\Psi = (\Pi_U \otimes \mathbb{1})\Psi(\Pi_U \otimes \mathbb{1})$, and then

$$\operatorname{tr}_A\big[(Y \otimes \mathbb{1})\,\Psi\big] = \operatorname{tr}_A\big[(Y\Pi_U \otimes \mathbb{1})\,\Psi\,(\Pi_U \otimes \mathbb{1})\big] = \operatorname{tr}_A\big[(\Pi_U Y \Pi_U \otimes \mathbb{1})\,\Psi\big],$$

the partial trace over A being cyclic under operators that act on A alone. This vanishes when $\Pi_U Y \Pi_U = 0$, i.e. when $\Pi_U(\sum_x M_x)\Pi_U = \Pi_U$. If $\{M_x\}$ is a POVM then $Y = 0$ and the condition holds outright, and Definition 1.1 delivers exactly that: $\sum_x M_x^{(t)} = \mathbb{1}_{A_t}$ for every t as an identity of the declared numbers, its declared-order sum decided exactly.

Only if, in pure form. $\Psi = |\psi\rangle\langle\psi|$ for a unit vector $|\psi\rangle$; take a Schmidt decomposition $|\psi\rangle = \sum_i \sqrt{\mu_i}\,|e_i\rangle_A\,|f_i\rangle_{A''}$ with every $\mu_i > 0$ and $\{|e_i\rangle\}$, $\{|f_i\rangle\}$ orthonormal. By (SR1), $\tau_A = \operatorname{tr}_{A''}\Psi = \sum_i \mu_i\,|e_i\rangle\langle e_i|$, which is a spectral decomposition, so $\{|e_i\rangle\}$ is an orthonormal basis of $U = \operatorname{supp}\tau_A$. Then

$$\operatorname{tr}_A\big[(Y \otimes \mathbb{1})\,\Psi\big] = \sum_{i,j} \sqrt{\mu_i \mu_j}\;\langle e_j|\,Y\,|e_i\rangle\;|f_i\rangle\langle f_j|\,,$$

and since the $|f_i\rangle$ are orthonormal and every $\mu_i > 0$, this operator vanishes if and only if $\langle e_j|\,Y\,|e_i\rangle = 0$ for all i, j, i.e. if and only if $\Pi_U Y \Pi_U = 0$.

Only if, in block form with rank-one blocks. Let $\Psi_t = w_t\,|\psi_t\rangle\langle\psi_t|$ with $w_t > 0$ and $|\psi_t\rangle$ a unit vector of $\mathcal{H}(A_t S_t)$ for every non-zero block. $Y = \bigoplus_t Y_t$ is block diagonal because every M_x is, so $\operatorname{tr}_A[(Y \otimes \mathbb{1})\Psi] = \bigoplus_t w_t\,\operatorname{tr}_{A_t}[(Y_t \otimes \mathbb{1}_{S_t})\,|\psi_t\rangle\langle\psi_t|]$, a direct sum along $(S_t)_t$, which vanishes if and only if every summand does. The pure-form argument, applied to $|\psi_t\rangle$ with $\tau_A^{(t)}/w_t$ in the place of τ_A, shows that the t-th summand vanishes if and only if $\Pi_{U,t} Y_t \Pi_{U,t} = 0$, where $\Pi_{U,t}$ projects onto $\operatorname{supp}\tau_A^{(t)}$; and $\Pi_U = \bigoplus_t \Pi_{U,t}$, so the conjunction over t is $\Pi_U Y \Pi_U = 0$.

The converse fails for a general positive block. Take one sector, $\Psi = \tau_A \otimes \varsigma$ for a state ς on A'', and any Hermitian Y with $\operatorname{tr}[Y\tau_A] = 0$ and $\Pi_U Y \Pi_U \neq 0$; such a Y exists whenever $\dim U \geq 2$, and scaled so that $\mathbb{1}_A + Y \succeq 0$ it is realised by the one-letter family $M = \mathbb{1}_A + Y$, Hermitian and positive semidefinite but not a POVM, which the hypothesis of (ii)

admits, with $p\rho^{\mathrm{v}} = \mathrm{tr}_A[(M \otimes \mathbb{1})\Psi]$ as in the statement. The block Ψ is positive semidefinite of unit trace and satisfies (SR1). Then $\mathrm{tr}_A[(Y \otimes \mathbb{1})\Psi] = \mathrm{tr}[Y\tau_A]\,\varsigma = 0$.

(iii)(a) *The interchange.* The registers do the work, and no numerical hypothesis enters. The input operator lives on $AA''E$. By hypothesis $\mathcal{N}_{\mathsf{A}}$ is a linear map out of A alone and $\mathcal{E}$ a channel out of $A''E$ alone, and A and $A''E$ are complementary tensor factors of the input; the outputs X and E' are likewise distinct registers. Hence both composites equal the tensor product $\mathcal{N}_{\mathsf{A}} \otimes \mathcal{E}$ of the two maps, which is (4.3); this is the elementary interchange law for a tensor product of linear maps and needs nothing about the particular $\mathcal{N}_{\mathsf{A}}$, $\mathcal{E}$ or input operator. That $\mathcal{E}$ does not act on A is (SR3); that it does not act on X is a hypothesis of the statement, and Remark 4.2 shows it cannot be dropped.

(iii)(b) *The source identification.* Applying $\mathcal{N}_{\mathsf{A}} \otimes \mathrm{id}_{A''E}$ to $\Psi \otimes \sigma$ gives, by (4.2) and (SR2),

$$\sum_x |x\rangle\langle x|_X \otimes \mathrm{tr}_A\left[(M_x \otimes \mathbb{1}_{A''})\,\Psi\right] \otimes \sigma = \sum_x p_x\ |x\rangle\langle x|_X \otimes \rho_x^{\mathrm{v}} \otimes \sigma = \rho_{XA''} \otimes \sigma,$$

the right-hand side being exactly the classical-quantum source state from which Fact 3.1 starts, under the symbol map stated with that fact. Applying the linear map $\mathrm{id}_X \otimes \mathcal{E}$ to both sides gives (4.4), the left-hand side being $(\mathcal{N}_{\mathsf{A}} \otimes \mathcal{E})(\Psi \otimes \sigma)$ by (a). That both sides are states: $\{M_x\}$ is a POVM by Definition 1.1, so $\mathcal{N}_{\mathsf{A}}$ is a channel; Ψ is a state, being positive semidefinite of unit trace in either form; and $\mathcal{E}$ is a channel by hypothesis. Nothing in (iii) reads $\Psi^2 = \Psi$. □

Without the hypothesis that the attack leaves Alice's outcome register alone, part (iii) of Theorem 4.1 is ill-typed and false. The restriction costs nothing, because A-SEQ places the announcements after both parties' measurements.

Remark 4.2 (Necessity and cost of the outcome-register hypothesis). The hypothesis of Theorem 4.1 (iii) that $\mathcal{E}$ acts trivially on the outcome register cannot be dropped: without it the statement is both ill-typed and false. *Ill-typed*, because X does not exist until $\mathcal{N}_{\mathsf{A}}$ creates it: in the right-hand composite of (4.3) the channel $\mathcal{E}$ acts first, on an operator on $AA''E$ carrying no such register. *False*, because a channel that did act on X could destroy exactly the information the equality is about: let $\mathcal{E}_{\mathrm{ow}}$ overwrite X with a fixed $|x_0\rangle\langle x_0|$ and act as the identity elsewhere; then $\mathcal{E}_{\mathrm{ow}} \circ (\mathcal{N}_{\mathsf{A}} \otimes \mathrm{id})$ produces the point distribution at x_0, while any composition in which $\mathcal{E}_{\mathrm{ow}}$ precedes $\mathcal{N}_{\mathsf{A}}$ leaves the outcome distribution $(p_x)_x$ intact. One overwriting channel is enough to refute a commutation asserted for *every* channel.

What the restriction excludes is exactly the operations that read Alice's outcome — in a QKD round, the public announcements. Nothing is lost, because Assumption 5.2 places those *after* both parties' measurements: the round is preparation and transmission, then measurement by both parties, then announcements. So the channel that models "the adversary's attack and the rest of the round up to the announcements" is of the admitted form, and the delayed-measurement reading, in which Alice measures after the attack, survives in the form that is used — Alice's

measurement may be moved across the attack, not across the announcements. The same restriction, in the same place, is what the published framework this module imports from relies on: its security reduction postpones Alice's measurement on the ground that her measurements commute with all operations except the public announcements, which by design occur only after the receiver's measurement in each round [2, Thm. 8.3]. That corroboration is recorded, not used: no step of the proof of Theorem 4.1 appeals to it.

Uses Thm. 4.1 · *Assumes* A-SEQ · newly derived · accepted · *Oracles* registry_instances_bb84.py, registry_protocol_compiler.py, registry_transmitter.py

The canonical source replacement satisfies the pure-form certificate conditions, but it is not the only data that do. Four-state BB84 admits a pure certificate with a two-dimensional retained system, where the canonical form needs four dimensions.

The imported construction fixes Alice's label register $\bar{A}$ at dimension $|\mathcal{X}|$. The corollary shows that the certificate is strictly more permissive already in its pure form, the form in which both the canonical data and the witness below are declared.

Corollary 4.3 (Certificates beyond the canonical form)**.** Throughout, the certificate is in the *pure* form of Definition 3.3, against the one-sector structure on A that Definition 1.1 makes the default.

(i) Fact 3.1's canonical data — $A = \bar{A}\hat{A}$ with $\dim \mathcal{H}(\bar{A}) = |\mathcal{X}|$, $M_x = |x\rangle\langle x|_{\bar{A}} \otimes \mathbb{1}_{\hat{A}}$, and $\Psi = |\psi\rangle\langle\psi|$ with $|\psi\rangle = \sum_x \sqrt{p_x}\,|x\rangle_{\bar{A}}\,|\psi_x\rangle_{\hat{A}A''}$ and $|\psi_x\rangle$ a purification of ρ_x^{v} — satisfy (SR1) and (SR2).

(ii) They are not the only such data: Theorem 4.1 (iii) applies to every triple $(\Psi, \tau_A, \{M_x\})$ satisfying (SR1)–(SR3), and there are source specifications with $d_A < |\mathcal{X}|$ carrying a certificate. In particular the four-state BB84 ensemble on a qubit, $\mathcal{X} = \{(b,k) : b,k \in \{0,1\}\}$ with $p_{(b,k)} = \frac{1}{4}$ and $\rho^{\mathrm{v}}_{(b,k)} = |k_b\rangle\langle k_b|$ for the computational basis ($b = 0$) and the Hadamard basis ($b = 1$), admits the certificate $A = A'' = \mathbb{C}^2$, $M_{(b,k)} = \frac{1}{2}\,|k_b\rangle\langle k_b|$, $\tau_A = \frac{1}{2}\mathbb{1}_A$, $\Psi = |\Phi^+\rangle\langle\Phi^+|$ with $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$: $d_A = 2$ where the canonical form has $d_A = 4$.

Uses Def. 1.1, Fact 3.1, Def. 3.3, Thm. 4.1 · newly derived · accepted · *Oracles* registry_instances_bb84.py, registry_protocol_compiler.py, registry_transmitter.py

Proof. (i) *The canonical data.* Ψ is a rank-one projector of unit trace, $|\psi\rangle$ being a unit vector — the $|\psi_x\rangle$ being unit vectors and the p_x summing to one. (SR1) is the definition of τ_A for the canonical data. For (SR2), $(M_x \otimes \mathbb{1}_{A''})\,|\psi\rangle = \sqrt{p_x}\,|x\rangle_{\bar{A}}\,|\psi_x\rangle_{\hat{A}A''}$, since $|x\rangle\langle x|\,|x'\rangle = \delta_{xx'}\,|x\rangle$ and $\mathbb{1}_{\hat{A}}$ acts trivially; M_x is a projector, so $\mathrm{tr}_A[(M_x \otimes \mathbb{1})\,\Psi] = \mathrm{tr}_A[(M_x \otimes \mathbb{1})\,\Psi\,(M_x \otimes \mathbb{1})] = p_x\,\mathrm{tr}_{\hat{A}}\,|\psi_x\rangle\langle\psi_x| = p_x \rho_x^{\mathrm{v}}$, the last step because $|\psi_x\rangle$ purifies ρ_x^{v}.

(ii) *The qubit witness.* $\{M_{(b,k)}\}$ is a POVM on $\mathbb{C}^2$: each element is positive semidefinite and $\sum_{b,k} \frac{1}{2} |k_b\rangle\langle k_b| = \frac{1}{2}\mathbb{1} + \frac{1}{2}\mathbb{1} = \mathbb{1}$, each basis resolving the identity; so the data are a source specification (Definition 1.1). $|\Phi^+\rangle\langle\Phi^+|$ is a rank-one projector of unit trace with every entry in $\{0, \frac{1}{2}\}$. (SR1): $\mathrm{tr}_{A''} |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}\mathbb{1}_A = \tau_A$. (SR2): both bases are real, so $(\langle k_b| \otimes \mathbb{1}) |\Phi^+\rangle = \frac{1}{\sqrt{2}} |k_b\rangle$, whence $\mathrm{tr}_A[(M_{(b,k)} \otimes \mathbb{1}) |\Phi^+\rangle\langle\Phi^+|] = \frac{1}{2} \cdot \frac{1}{2} |k_b\rangle\langle k_b| = p_{(b,k)} \rho^{\mathrm{v}}_{(b,k)}$. Four vectors in a two-dimensional space are not linearly independent, so the canonical form has $\dim \mathcal{H}(\bar{A}) = |\mathcal{X}| = 4$ with a $\tau_{\bar{A}}$ of rank two; the witness has $d_A = 2$. □

Only the converse in part (ii) of Theorem 4.1 depends on purity. A certificate vector, a globally pure operator over sectors and a pure block over non-orthogonal states can all need irrational entries, which is why the certificate is an operator with a block form and an ensemble form.

Remark 4.4 (Purity and exact declarability of the certificate forms). *What purity adds.* A pure-form certificate satisfies every condition of the block form at the one-sector structure and one condition more (Definition 3.3). Every statement this module proves of the block form therefore holds of the pure form, and Theorem 4.1 (ii) is the only statement whose *conclusion* depends on the additional condition: in pure form a source/POVM mismatch of the kind (ii) describes is a detected defect and not a silent one, while in block form it is the POVM condition of Definition 1.1, decided on its own, that excludes it. Part (i) of the theorem lists purity as a condition of the pure form, and a consumer that demands purity of every certificate re-decides it and therefore refuses a block form. Everything Corollary 4.3 exhibits is admitted in block form as well. Not every block-form certificate is pure: already at one sector the block form admits a positive semidefinite operator of unit trace and rank above one, which the pure form refuses.

Why the certificate is an operator and not a vector. No binary64 number a satisfies $a^2 = \frac{1}{2}$, so a declared vector with entries $\mathrm{fl}(1/\sqrt{2})$ yields $|\psi\rangle\langle\psi|$ with entries $\mathrm{fl}(1/\sqrt{2})^2 \neq \frac{1}{2}$, and (SR1) fails exactly *for the certificate as an instance declares it* — $|\Phi^+\rangle$ in that instance's own computational basis, whose amplitudes are $1/\sqrt{2}$. The *operator* $|\Phi^+\rangle\langle\Phi^+|$, by contrast, has every entry equal to $\frac{1}{2}$, which is a binary64 number; and every quantity (SR1), (SR2) and Theorem 4.1 read off the certificate is a function of Ψ and never of a vector. In pure form the purity conditions — $\Psi^2 = \Psi$ with $\mathrm{tr}\,\Psi = 1$ — are what recover the vector where an argument needs one (Theorem 4.1 (ii)), and they are two exact identities on the declared matrix. Whether a given declaration of the canonical data of Corollary 4.3 is admissible is decided, not assumed: the canonical Ψ must be declared as a binary64 rank-one projector, and a declaration whose entries do not satisfy $\Psi^2 = \Psi$ exactly is refused.

Why a block form and an ensemble presentation. Exact purity is not declarable for a source that draws a sector and then prepares a state in it. A globally pure Ψ over sectors of weights g_t carries cross-sector amplitudes $\sqrt{g_t g_{t'}}$, which need not be dyadic rationals, and a pure block whose marginal has the spectrum $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ carries the entry $\sqrt{1/8}$, which is not one. The direct sum of positive blocks is exactly declarable, it is the state of the source that first draws t with probability $\mathrm{tr}\,\Psi_t$ and then prepares $\Psi_t / \mathrm{tr}\,\Psi_t$, and the sufficiency statement, Theorem 4.1 (iii), reads purity nowhere. Where even the blocks are not declarable — a pure block over non-orthogonal signal states carries square roots of their weights — the ensemble

presentation is the form to use: it has the purity (Lemma 3.4 (ii)) and declares none of the numbers, which is what it exists for.

Uses Def. 3.3 · newly derived · accepted

A block certificate is unchanged by pinching Alice's sectors or the virtual signal's sectors paired with them, so every state an admitted attack reaches from it is block diagonal with Alice's declared marginal. An operator merely constrained to have that marginal need not be block diagonal.

A block certificate declares its sectors on Alice's side and on the virtual signal at once. The lemma shows that this matching survives every admitted attack, for the states the attack actually reaches; it is about those states only, and part (iii) shows why it does not restrict any optimisation's feasible set.

Lemma 4.5 (Matched pinching of a block certificate). Let a source specification carry a certificate in block form (Definition 3.3), with sector projectors Π_t^A on A and Π_t^{v} on A'', $t \in \mathbb{T}$, which sum to the identity on A and on A'' respectively (Definitions 1.1 and 3.3), and write

$$\mathrm{Pin}_A(\xi) \coloneqq \sum_{t\in\mathbb{T}} \Pi_t^A\, \xi\, \Pi_t^A, \qquad \mathrm{Pin}_{\mathrm{v}}(\zeta) \coloneqq \sum_{t\in\mathbb{T}} \Pi_t^{\mathrm{v}}\, \zeta\, \Pi_t^{\mathrm{v}}$$

for the two pinching channels.

(i) *(matched structure)* for every $t \in \mathbb{T}$,

$$(\Pi_t^A \otimes \mathbb{1}_{A''})\,\Psi = \Psi\,(\Pi_t^A \otimes \mathbb{1}_{A''}) = (\mathbb{1}_A \otimes \Pi_t^{\mathrm{v}})\,\Psi = \Psi\,(\mathbb{1}_A \otimes \Pi_t^{\mathrm{v}}) = \Psi_t, \tag{4.5}$$

the block Ψ_t read as an operator on $\mathcal{H}(AA'')$ extended by zero. Hence $(\mathrm{Pin}_A \otimes \mathrm{id}_{A''})(\Psi) = \Psi = (\mathrm{id}_A \otimes \mathrm{Pin}_{\mathrm{v}})(\Psi)$.

(ii) *(reached states)* assume (SR3), and let $\mathcal{E}$ be any channel from $A''E$ to E' and σ any state on E, as in Theorem 4.1 (iii). The state $\omega \coloneqq (\mathrm{id}_A \otimes \mathcal{E})(\Psi \otimes \sigma)$ on AE' satisfies

$$\mathrm{tr}_{E'}\,\omega = \tau_A, \qquad (\mathrm{Pin}_A \otimes \mathrm{id}_{E'})(\omega) = \omega, \qquad \omega = (\mathrm{id}_A \otimes \mathcal{E}')(\Psi \otimes \sigma) \;\text{ with }\; \mathcal{E}' \coloneqq \mathcal{E} \circ (\mathrm{Pin}_{\mathrm{v}} \otimes \mathrm{id}_E). \tag{4.6}$$

So $\omega = \bigoplus_t \omega_t$ with $\omega_t \in \mathrm{Pos}(A_t E')$ and $\mathrm{tr}_{E'}\,\omega_t = \tau_A^{(t)}$; and the attack $\mathcal{E}$ and the attack $\mathcal{E}'$ that first measures the sector of the virtual signal reach the same state.

(iii) *(operators that are not reached)* the block diagonality of (ii) does not extend to operators merely constrained to have the marginal: in general a positive operator on AE' with marginal $\tau_A = \bigoplus_t \tau_A^{(t)}$ need not be block diagonal along $\mathbb{T}$, and this happens already for two one-dimensional sectors, $\tau_A = \frac{1}{2}\mathbb{1}_A$ and E' a qubit.

Uses Def. 1.1, Def. 3.3 · *Assumes* A-SRC · *Used by* Rem. 4.6, Cor. 9.2 · newly derived · accepted

Proof. (i) By Definition 3.3, $\Psi = \bigoplus_t \Psi_t$ on the matched subspace $\bigoplus_t \mathcal{H}(A_t) \otimes \mathcal{H}(S_t)$ and zero on its orthogonal complement. The projector $\Pi_t^A \otimes \mathbb{1}_{A''}$ has range $\bigoplus_{t'} \mathcal{H}(A_t) \otimes \mathcal{H}(S_{t'})$, whose intersection with the matched subspace is $\mathcal{H}(A_t) \otimes \mathcal{H}(S_t)$, and it commutes with the projector onto the matched subspace, both being diagonal in the product of the declared bases. So $(\Pi_t^A \otimes \mathbb{1})\Psi$ is the part of Ψ with range in $\mathcal{H}(A_t) \otimes \mathcal{H}(S_t)$, which is Ψ_t; taking adjoints, Ψ and Ψ_t being Hermitian, gives the second equality, and the same argument with $\mathbb{1}_A \otimes \Pi_t^{\mathrm{v}}$, whose range meets the matched subspace in the same summand, gives the third and fourth. Summing $(\Pi_t^A \otimes \mathbb{1})\Psi(\Pi_t^A \otimes \mathbb{1}) = \Psi_t(\Pi_t^A \otimes \mathbb{1}) = \Psi_t$ over t gives $(\mathrm{Pin}_A \otimes \mathrm{id})(\Psi) = \Psi$, and likewise on the other factor.

(ii) $\mathcal{E}$ is trace preserving and does not act on A, the latter being (SR3); so $\mathrm{tr}_{E'}\, \omega = \mathrm{tr}_{A''E}[\Psi \otimes \sigma] = \mathrm{tr}_{A''}\, \Psi = \tau_A$ by (SR1). The maps Pin_A and $\mathcal{E}$ act on complementary tensor factors, so they commute, and by (i)

$$(\mathrm{Pin}_A \otimes \mathrm{id}_{E'})(\omega) = (\mathrm{id}_A \otimes \mathcal{E})\big((\mathrm{Pin}_A \otimes \mathrm{id}_{A''})(\Psi) \otimes \sigma\big) = (\mathrm{id}_A \otimes \mathcal{E})(\Psi \otimes \sigma) = \omega.$$

Replacing $(\mathrm{Pin}_A \otimes \mathrm{id})(\Psi)$ by $(\mathrm{id} \otimes \mathrm{Pin}_{\mathrm{v}})(\Psi)$, equal to it by (i), in the middle expression gives $\omega = (\mathrm{id}_A \otimes \mathcal{E}')(\Psi \otimes \sigma)$. An operator fixed by $\mathrm{Pin}_A \otimes \mathrm{id}_{E'}$ is the direct sum of its compressions $\omega_t = (\Pi_t^A \otimes \mathbb{1})\omega(\Pi_t^A \otimes \mathbb{1})$, each positive semidefinite as a compression of one, and $\mathrm{tr}_{E'}\, \omega_t = \Pi_t^A \tau_A \Pi_t^A = \tau_A^{(t)}$.

(iii) With two one-dimensional sectors spanned by $|0\rangle$ and $|1\rangle$, $\tau_A = \frac{1}{2}\mathbb{1}_A$ and E' a qubit, the projector ω onto $(|0\rangle\,|0\rangle + |1\rangle\,|1\rangle)/\sqrt{2}$ is positive semidefinite with marginal τ_A on A, and its compression $(\Pi_0^A \otimes \mathbb{1})\,\omega\,(\Pi_1^A \otimes \mathbb{1}) = \frac{1}{2}\,|0\rangle\langle 1| \otimes |0\rangle\langle 1|$ is not zero. □

A verified certificate buys the interchange and the source identification of Theorem 4.1 outright, and in block form the reached-state statement of Lemma 4.5. It is no security reduction, restricts no consumer's feasible set and decides nothing about retention.

Remark 4.6 (Scope of a verified certificate). Theorem 4.1 (iii) is a statement about the *source*: it says that Alice's measurement commutes with every attack of the admitted form — every channel on the transmitted system and the adversary's registers, which by Assumption 5.2 is every operation of the round up to the announcements (Remark 4.2) — and that the declared trusted-source data describe the virtual preparation (the specified one exactly when the source segment is empty) against every such attack. It is not a security reduction, and it does not say that the consuming engine's optimization is sound; that argument — granting the adversary a purification of Alice's retained system and the transmitted system, restricting to the declared feasible set, accumulating over rounds — lives in the engine's claims, which import Assumption 5.3 for it. What this module owes depends on the form of the certificate (Theorem 4.1 (i)). In pure and in block form it is the decided conditions of the declared certificate, those on Ψ of its form together with (SR1) and (SR2), and the attestation (SR3). The conditions on Ψ are Hermiticity, $\Psi^2 = \Psi$ and $\mathrm{tr}\,\Psi = 1$ in pure form, and Hermiticity and positive semidefiniteness of every block and $\sum_t \mathrm{tr}\,\Psi_t = 1$ in block form. For an ensemble presentation it is exactly (E1) to (E3), decided on the declared blocks, and the attestation (SR3); (SR1)

and (SR2) then hold of the certificate the presentation defines as theorems and are decided by nothing.

What a verified certificate buys. A verified certificate, one whose decided conditions (Theorem 4.1 (i)) hold, buys (4.3) and (4.4) outright: the commutation, and the identification of the declared trusted-source data with the virtual source, as identities for every channel of the admitted form, with no residue and no qualification. That is what the exact reading of (SR1)–(SR2) is for, and it is what a consumer of the licence record, the record of the verified certificate that the artefact carries (Requirement 6.5), is entitled to read the record as asserting. What the certificate does *not* buy is (SR3), which no function of the declared data decides (Assumption 5.3); it is (SR3), with Assumption 5.2, that makes the attack of a round a channel of the admitted form.

What the block form adds. A certificate declared in block form (Definition 3.3) buys one thing more, and it is a statement about *reached* states: Lemma 4.5, that every state an admitted attack reaches from that certificate is block diagonal along the declared sectors with A-marginal τ_A, and that pinching A after the attack is the same operation as pinching the virtual signal before it. That is what a consumer may read the record as asserting of the declared sector structure, and it is all of it. *It does not buy a restriction of any consumer's feasible set.* A feasible set is a set of operators constrained to have the marginal τ_A, not a set of states reached from a certificate, and a positive operator with that marginal need not be block diagonal (Lemma 4.5 (iii)). A consumer that minimises over block-diagonal operators alone does so under a theorem of its own module, with its own preconditions on declared data, and neither this module's licence record nor the presentation of τ_A and $\{M_x\}$ as block lists is that theorem (Definition 1.1, "What the sector structure is not").

Links supplied by a theorem. The same boundary, in the other direction, governs a source link whose channel is supplied by a named claim of a submodule rather than as data (a theorem constructor, Definition 2.1 (c2)). For such a link, what the artefact carries about its source identity (2.3) is the link's obligation record, which travels with the link in the source segment (Definition 2.2 (S4)) and not in the licence record: a named claim and the assumptions the obligation carries, never a verdict this module decided, and there is nothing of that identity a consumer could re-decide.

Uses Def. 2.2, Def. 3.3, Thm. 4.1, Lem. 4.5 · *Used by* Req. 6.5 · newly derived · accepted · *Oracles* registry_instances_bb84.py, registry_protocol_compiler.py, registry_transmitter.py

For a one-dimensional retained system the two source modes impose the same condition and the licence excludes no attack. A trusted non-empty source segment there for which no certificate exists rests on the supplier obligation A-LINK, a case no artefact reaches.

Remark 4.7 (The one-dimensional retained system). Suppose $d_A = 1$. Then $S_=(A) = \{[1]\}$, so the only marginal a trusted-mode specification can carry is $\tau_A = [1]$; and for any state ω on A and the transmitted system the A-marginal ω_A is the 1×1 matrix $\operatorname{tr}\omega$, so the conditions $\omega_A = \tau_A$ and $\operatorname{tr}\omega = 1$ are the same condition on the same number. The two source modes, trusted-source mode and untrusted-source mode, therefore impose the same condition on every consumer, and what the source-replacement certificate licenses at $d_A = 1$ is the condition $\operatorname{tr}\omega = 1$, which every normalized state satisfies: it excludes nothing, and Assumption 5.3 is *vacuous* —

it deletes no attack. That is why the certificate of Definition 3.3 is demanded only at $d_A \geq 2$ (Requirement 6.1). A specification at $d_A = 1$ is also the shape in which a protocol with no trusted source is declared: the consuming declaration folds Alice's measured system into the receiver-side systems and names, in its partition of the outcome among the parties, which components are hers (Remark "The two source modes on the program" (meat-core/rem:modes-da1)). That shape is the *safe* direction (Remark 1.2): it removes an assumption rather than adding one.

A non-empty source segment at $d_A = 1$. Every such specification that an artefact of this module can carry is covered by theorems, and the one case left to an assumption is reached by no artefact. In detail, the theorem route applies to a non-empty segment whose consumer declares the ensemble form (Definition 3.3, "What a consumer reads of a presentation") and which passes every condition that admits it: the admission rule on the kind, (S1) to (S4) and (A1) to (A3) of Definition 2.2, the refusals of Definition 2.1 (d) and (f), and the admission rule of Definition "Reduction chain" (reductions/def:chain), which admits the segment only in trusted-source mode and only where every entry of the constraint list is licensed, as Definition "Interface conditions on a chain" (protocol-compiler/def:chaincond) decides it. There the security statement is carried across the segment by the source identity (2.3) at the descriptors of the emitted links, by Theorem 4.1 (iii) with (SR1) to (SR3) for the source-replacement recasting, and by Theorem "The two-sided reduction in trusted-source mode" (reductions/thm:two-sided) for the composition with the receiver segment; at $d_A \geq 2$ Assumption 5.3 supplies the attestation. The degenerate declaration of the decoy submodule (Definition 7.2) is covered by the same route: with one polarisation, intensity set $\{0\}$, its entropy cutoff and source cutoff both zero and all weight on the zero-photon sector it has one one-dimensional sector, hence $d_A = 1$; its ensemble presentation supplies (SR1) and (SR2) as theorems (Lemma 3.4 (i)), its defined source state and marginal are both [1], and (SR3) is vacuous, a trace-preserving map being unable to act nontrivially on a one-dimensional retained register. The hypotheses of Lemma "Exactness of the source map in trusted-source mode" (reductions/lem:source-transfer) (ii), (iii) and Theorem "The two-sided reduction in trusted-source mode" (reductions/thm:two-sided) therefore hold there as well.

The case left to an assumption. A trusted specification at $d_A = 1$ that declares no certificate, with a non-empty segment and a virtual ensemble that is not constant over the letters with $p_x > 0$, has none to declare: with a one-dimensional retained register (SR2) would force all positively weighted virtual states to coincide. The theorem route does not cover that case. Its transfer across the segment is the source clause of the protocol compiler's supplier obligation A-LINK (Assumption "Soundness of every reduction link" (protocol-compiler/ass:link)), declared on that module's chain-condition definition, Definition "Interface conditions on a chain" (protocol-compiler/def:chaincond), and on Theorem "Decided identities and named obligations of a consumed chain" (protocol-compiler/thm:chain-consumption) (v′), never discharged, and emitted in its assumption ledger. No artefact reaches the case: (A3) of Definition 2.2 accepts a segment only inside an artefact of the admitting module, and the one admitted constructor kind emits an ensemble presentation, which supplies the certificate described above.

Uses Def. 1.1, Def. 3.3 · *Used by* Req. 6.1, Req. 6.2, Req. 6.3 · newly derived · accepted

5 Assumptions

Three assumptions carry what declared matrices cannot establish, that the source emits the declared ensemble in every round and independently of every other round, that each round runs preparation, measurement and then announcements, and, in trusted-source mode, that Alice's retained system never reaches the adversary. Each is an attestation about the devices and how the protocol is run; the module records each in the artefact's assumption ledger, the list of assumptions the artefact rests on, and never verifies it. A specification declared by the decoy submodule's source model carries that submodule's premise, A-WCP, in the place of A-ENS.

The physical source draws each round's letter independently with the declared weights and emits the product of the declared states, no round depending on another. It ties a source specification declared as matrices to the device it describes.

The assumption is about the source as it runs: in every round it emits exactly the declared ensemble, not an approximation of it, and it does so afresh, independently of every other round. The published prepare-and-measure protocol makes the same premise, Alice preparing each round's state "independently for each round" [2, Protocol 2, step 1(a)]; the product it gives the n rounds is the state from which the source-replacement lemma starts. A specification declared by a source model carries the premise its submodule declares in the place of this one. In untrusted-source mode the protocol compiler's assumption on Alice's measurement governs the measurement by which Alice obtains a letter from a received system.

Assumption 5.1 [A-ENS] (Emission of the specified ensemble)**.** For the declared signal ensemble $\{p_x, \rho_x\}$ of Definition 1.1 on the transmitted system, the letters the physical source draws and the systems it emits satisfy the following; a letter that Alice obtains in untrusted-source mode by measuring a received system is not drawn by the source, and nothing below concerns it: (i) the letters $x_1, \ldots, x_n$ it draws in the n rounds are drawn independently, each with the weights p_x; (ii) given those letters, the systems it emits in the n rounds are in the state $\rho_{x_1} \otimes \cdots \otimes \rho_{x_n}$; (iii) neither the letter it draws in a round nor the state it emits in it depends on what happens in the other rounds, their outcomes and public messages included.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Your source prepares, in every round, one of the declared states with the declared probabilities. Each round's letter is drawn independently with weights p_x, and the states emitted over the n rounds form a product: no round's letter or state depends on another round, its detection results or its public messages. A source with memory or drift between rounds (for example pattern effects in a modulator) violates this. Where your source is a pair source inside your laboratory with one half measured there to herald or fix the letter, those local detectors belong to the declared ensemble: their efficiency mismatch must be carried by the declared weights and states, and their memory across rounds (for example afterpulsing or dead time) violates this like any other dependence between rounds.

Uses Def. 1.1 · *Category* source · *Affects* secrecy · *Assumed by* Fact 3.1, meat-core/lem:product-marginal, meat-core/thm:accumulated-entropy, meat-core/thm:keylength · *Machine key* `source_ensemble` · newly derived · accepted

Each round runs the delivery of its quantum systems, then every party's measurement, then the round's public messages, then sifting; each party measures in round order, post-processing follows all rounds and every message is public. It is the timing premise of the security analysis.

With one measuring party this is the structure of [2, Def. 8.1, Def. 8.2], whose state evolution is its Lemma 8.2, and it is the hypothesis under which this module applies the imported source-replacement lemma, whose own statement places no condition on the sequential structure of the operations after the preparation. The cited definitions are that case and no more: they have one measuring party and one announcement block per round, and the reading for several measuring parties is this contract's own. Of the messages, the assumption fixes only that all of them follow all measurements of the round; a message may depend on earlier messages of the same round, which is a property of the declared messages and not of this assumption. The clauses have two consequences worth naming. Where one party measures, as in a prepare-and-measure protocol, every measurement of the round is that party's. Where two parties measure systems delivered by a source neither of them prepared, neither party's message precedes the other party's measurement. Clause (v) is the measurement part of the timing condition of the published Protocol 2 (Bob's measurement times increasing there), read for each measuring party [2, Sec. 8.1.1, Protocol 2]; the monotone order of preparations and of announcements is not adopted, and the analysis does not need it; with (i) it puts every measurement of a round before every public message of a later round.

Assumption 5.2 [A-SEQ] (Sequential round structure)**.** The physical protocol has the sequential structure the declared protocol describes: (i) each round runs, in this order, state preparation and transmission; the measurements of the round, by every party that measures in it; the public messages of the round, in their declared order, *each of them sent after every measurement of the round is complete*, whichever party sends it and whichever party measured; and announcement-determined sifting; (ii) the classical post-processing is applied after the n rounds; (iii) every message is public; (iv) no operation on a quantum system of round j depends on a public message of round j; (v) every party that measures makes its measurements in round order: its measurement of round j is complete before its measurement of round $j+1$ is.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: In every round the quantum systems come first: where your source is trusted, it prepares and sends that round's signal, and otherwise the systems of the round are delivered to the measuring stations. Then every party that measures completes its measurement, and only then are that round's public messages sent, in the declared order.

Sifting is then determined by the announcements, and no operation on a quantum system of a round depends on a public message of that round. All classical post-processing (for example error correction, verification and privacy amplification) runs after all n rounds, and every classical message is public. A message sent before a measurement of the same round has finished (for example an early basis announcement) violates this. Each party that measures completes its measurement of each round before it completes that of the next round.

Category protocol structure · *Affects* secrecy · *Assumed by* Fact 3.1, Thm. 4.1, Rem. 4.2, Cor. 9.2, meat-core/lem:honest-attacks, meat-core/lem:product-marginal, meat-core/thm:accumulated-entropy, meat-core/thm:keylength, reductions/lem:source-transfer, reductions/thm:factorization · *Machine key* `sequential_round_model` · newly derived · accepted

In trusted-source mode the retained system A stays in the laboratory of the party that holds it and is acted on only by that party's declared measurement, condition (SR3) of the certificate. It is what lets a declared marginal delete attacks.

The attestation below is the premise the published trusted-source analyses state in words: [3, Eqs. (36)–(39)] adds the marginal constraints to the entanglement-based statistics constraints for prepare-and-measure protocols on the stated ground that "system A is inaccessible to Eve", and [1, Rem. 1] imposes the n-round product form "because we assume Eve has no access to Alice's lab". Under it, after the source-replacement recasting, the register A is *inaccessible to the adversary* in every round and its marginal is τ_A whatever the adversary does; together with the independence of the source's rounds, or the premise a source model declares in its place, it licenses that n-round product form. For a prepare-and-measure source the register A exists only in the source-replacement description; what is attested of the apparatus is then what those analyses assume in words, that the adversary has no access to Alice's laboratory. (SR3) is the one part of a source-replacement certificate that no declared number carries, and it is demanded only at $d_A \geq 2$: at $d_A = 1$ the marginal excludes nothing.

Assumption 5.3 [A-SRC] (Retention of the system A in trusted-source mode)**.** For a source specification (Definition 1.1) in trusted-source mode, which fills the marginal slot with $\tau_A \in S_=(A)$, condition (SR3) of Definition 3.3 holds of the apparatus: (i) the system A is retained in the laboratory of the party whose label the party partition of the consuming declaration (Definition "Protocol declaration" (protocol-compiler/def:declaration)) gives A, and is never transmitted; (ii) no operation of the protocol or of the adversary acts on A except the declared measurement $\{M_x\}$ of that party.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: The source's retained half, system A, never leaves the laboratory of the party the declaration assigns it to, and nothing acts on it except that party's own declared measurement: no other step of the protocol and no action of the adversary. In a prepare-and-measure setup A exists only in the source-replacement description, and this is the premise that the adversary has no access to that party's laboratory. Only then may you declare the source's reduced state τ_A (trusted-source mode), which removes attacks

from the analysis when A has dimension at least two. If you cannot vouch for this, use untrusted-source mode: the key rate can only drop, and the result stays valid.

Uses Def. 1.1, Def. 3.3 · *Category* source · *Affects* secrecy · *Assumed by* Thm. 4.1, Lem. 4.5, Cor. 9.2, meat-core/ex:fss, meat-core/lem:product-marginal, meat-core/lem:range, meat-core/rem:cgate, meat-core/rem:modes-da1, meat-core/rem:not-licensed, meat-core/thm:accumulated-entropy · *Machine key* `source_replacement_license` · newly derived · accepted

Part II
Engine requirements

What an implementation of this module must do, record or refuse; each requirement names the result that motivates it.

6 Certificate and licence requirements

These requirements bind the engine, here the implementation of this module, which answers a request, a submitted source specification whose marginal slot may be empty, with an artefact or a refusal. The licence is what a verified certificate with its retention attestation gives a consumer, the right to impose the declared marginal on the states it optimises over, and the licence record is its data form. The requirements fix when a certificate is demanded, how the one-dimensional case is served, when a certificate is refused, how a declared marginal is decided, and how the retention attestation is recorded.

A trusted-source request must carry the certificate, and with it the retention attestation, only when Alice's retained system has dimension at least two. At dimension one the licensed marginal excludes no state, so demanding them there would refuse data for nothing.

Requirement 6.1 (Certificate and attestation demanded only at $d_A \geq 2$). The engine demands the source-replacement certificate of Definition 3.3 of a request in trusted-source mode only at $d_A \geq 2$, and there refuses such a request that carries none (`trusted_mode_without_certificate`). At $d_A = 1$, where the licensed condition $\omega_A = \tau_A$ excludes no state (Remark 4.7), neither the engine nor a consumer of its artefact demands the certificate or the attestation of Assumption 5.3.

Motivated by Def. 3.3, Rem. 4.7, Ass. 5.3 · *Used by* Req. 6.5, protocol-compiler/req: gate-outcomes · accepted · *Code* certiqs_key.transmitter.artifact:emit_artifact

With a one-dimensional retained system a trusted-source request may leave the marginal empty; the engine fills it with the only state there and records the fill. The record keeps a filled value distinguishable from a declared one, and no other slot is ever filled.

Requirement 6.2 (The unique marginal served at $d_A = 1$). At $d_A = 1$ a request in trusted-source mode may leave the marginal slot empty. The engine then fills the slot with $\tau_A = [1]$, the unique element of $S_=(A)$ (Remark 4.7); the emitted specification is in trusted-source mode, trusted-source and untrusted-source mode imposing the same condition there; and the emitted artefact records that the slot was filled. This is the only slot the engine ever fills.

Motivated by Rem. 4.7 · accepted · *Code* certiqs_key.transmitter.artifact:emit_artifact

A request whose certificate is present and fails a condition decided of its form, pure, block or ensemble presentation, is refused at every dimension, including dimension one, where none is demanded. A certificate that does not verify is a defect in the supplied data, not a weaker claim.

REQUIREMENT 6.3 (Refusal of a certificate that fails a decided condition). The engine refuses a request whose certificate is present and fails one of the conditions Definition 3.3 decides of its form — in pure form the three on Ψ, (SR1) and (SR2); in block form Hermiticity and positive semidefiniteness of every block, the trace, the block diagonality of every weighted virtual member along the matched sectors, and (3.1); in an ensemble presentation (E1) to (E3) — at every d_A, including $d_A = 1$, where no certificate is demanded (Requirement 6.1): a certificate that does not verify is a defect in the supplied data, not a weaker claim.

Motivated by Def. 3.3, Rem. 4.7, Req. 6.1 · accepted · *Code* certiqs_key.transmitter.artifact:emit_artifact

A declared marginal is checked to be a state on the declared matrix itself, even where a certificate satisfies (SR1). An implication proved of the certificate's objects is no check of the datum, so it is never used.

Theorem 4.1 (i) shows that (SR1) implies that τ_A is a state. That implication is proved of the objects the declared data describe; a check of the declared matrix is a check of the datum itself, and it is the check the engine makes.

REQUIREMENT 6.4 (Direct decision of the declared marginal). Where τ_A is declared, the engine decides $\tau_A \in S_=(A)$ on the declared matrix directly, also where a certificate is present and satisfies (SR1), and never infers it from the implication (SR1) $\Rightarrow \tau_A \in S_=(A)$ of Theorem 4.1 (i); it refuses a request whose declared τ_A is not a state (`alice_marginal_not_admissible`). Consumers of the declared data decide it in the same way under their own rules; this requirement binds the engine.

Motivated by Thm. 4.1 · implemented (effective: accepted) · *Code* certiqs_key.transmitter.artifact:emit_artifact

Where a licence record is emitted, in trusted-source mode at dimension at least two, the retention attestation (SR3) is recorded in it exactly as supplied, never verified and never asserted of a physical apparatus. No function of the declared data decides it, so the engine can only record it.

REQUIREMENT 6.5 (Pass-through of the retention attestation). Where the engine emits the licence record `source_replacement_license`, which it does only in trusted-source mode at $d_A \geq 2$, the scope in which the certificate is demanded (Requirement 6.1), it records the (SR3) attestation it receives (Definition 3.3) in that record, and passes it through: it never verifies it and never asserts it of any physical apparatus (Assumption 5.3; Remark 4.6). No consumer can verify it either, since no function of the declared data decides it (Theorem 4.1 (i)).

Motivated by Def. 3.3, Thm. 4.1, Rem. 4.6, Ass. 5.3, Req. 6.1 · implemented (effective: accepted) · *Code* certiqs_key.transmitter.certificate:SourceReplacementLicense

Part III
Decoy-state sources and the tagging channel

The decoy submodule declares a phase-randomised weak-coherent-pulse source and turns it into a finite virtual source that the engine can read. The physical states live on an infinite-dimensional two-mode Fock space, so they are declared as a source model, with the attestation A-WCP in the place of A-ENS. Up to an entropy cutoff the photon-number sectors stay quantum; above it and up to a source cutoff each photon number and polarisation carries one flag shared by every intensity, which ties a photon number's weight across intensities; the remaining weight sits on one tail flag per polarisation and intensity. Its main results are Proposition 8.1, one channel carrying this virtual ensemble to the physical states for any sector weights at or below the Poisson weights, and Corollary 9.2, by which the emitted declaration, an ensemble presentation, has an exact source-replacement certificate wherever its declared blocks meet the conditions of that form, and a tagging link whose source identity holds at the emitted descriptors. Every number is used as declared: the Poisson weights are met by a certified inequality and never by an equality, and no square root or quotient is ever declared. The submodule states no key rate and compares none, and its emission, the step that outputs the declaration, refuses entropy cutoffs above two, quantum sectors whose signal dyads it cannot form exactly, flag families finer than the tagged declaration's and formed entries that are not binary64 numbers.

7 The source model and the tagged declaration

The weak-coherent-pulse model fixes the declared labels, intensities and phase-averaged coherent states on the two-mode Fock space, which no finite matrix can carry. The tagged declaration Definition 7.2 replaces it by a finite virtual source in three bands: quantum photon-number sectors up to the entropy cutoff; above it and up to the source cutoff, one one-dimensional polarisation flag per photon number and polarisation, shared by all intensities; and one tail flag per polarisation and intensity for the rest. Sharing a flag across intensities is what relates a photon number's weight at different intensities, the decoy content of the declaration. Its sector weights are declared numbers bounded by the Poisson weights, and its sector list is read off them; Lemma 7.3 encloses each Poisson weight between two rationals formed from the declared intensity, which is how that bound is decided. Remark 7.4 separates its two cutoffs from the receiver's.

A weak-coherent-pulse model declares polarisations, intensities, weighted labels and the phase-averaged coherent states on the two-mode Fock space, as mathematical objects rather than as matrices. It is the specified ensemble over which the tagged declaration is built.

A phase-randomised weak-coherent-pulse source prepares, for a polarisation a and an intensity μ, the coherent state of amplitude $\sqrt{\mu}$ in the mode $|a\rangle$ averaged over its global phase. That average is diagonal in the photon number, which is why full phase randomisation is one of the conditions Assumption 10.1 attests; without it the emitted state is not the one below. The state is [2, Eq. (144)], whose $\sigma_{(a,\mu,z)}$, $|N\rangle_a$ and setting choice (a,μ,z) are ρ_x, $|\mathsf{n},a\rangle$ and the letter x here. Its transmitted system is an infinite-dimensional Fock space, so the model fixes the states as mathematical objects and not as matrices. It fixes only the specified ensemble $\{p_x,\rho_x\}$, the weighted states the source is declared to prepare: Alice's retained system, her measurement, her marginal and the certificate come from the tagged declaration built over it (Definition 7.2). We write n for a photon number, n being the number of rounds.

Definition 7.1 (Phase-randomised weak-coherent-pulse source model)**.** A *weak-coherent-pulse model* is a source model, the case (b) of Definition 1.1 in which the specified ensemble $\{p_x,\rho_x\}$ is fixed as mathematical objects rather than as matrices, on a transmitted system whose Hilbert space is separable. It consists of the following declared data.

(W1) *(modes and polarisations)* two declared optical modes with a fixed orthonormal basis of the one-photon space $\mathbb{C}^2$, and a finite non-empty set Pol of *polarisations*, declared labels each standing for a unit vector $|a\rangle \in \mathbb{C}^2$, $a \in \mathrm{Pol}$, with creation operator $c_a^\dagger$. The vector $|a\rangle$ is a mathematical object of the model, as the states of (W4) are, and not an array of declared numbers: a declaration names the vector a label stands for and carries none of its coordinates, which need not be binary64 numbers ($(|H\rangle+|V\rangle)/\sqrt{2}$ has the coordinates $1/\sqrt{2}$). The transmitted system A' is the Fock space over the two modes, with vacuum $|\mathrm{vac}\rangle$; its Hilbert space is separable and infinite dimensional.

(W2) *(intensities and labels)* a finite non-empty set $\mathrm{Int} \subset [0,\infty)$ of *intensities*; the signal alphabet $\mathcal{X} \subseteq \mathrm{Pol}\times\mathrm{Int}\times\{\mathrm{test},\mathrm{gen}\}$ with its declared order, an element of $\mathrm{Pol}\times\mathrm{Int}\times\{\mathrm{test},\mathrm{gen}\}$ being a label and a label in $\mathcal{X}$ a letter, written $x=(a_x,\mu_x,z_x)$, whose third component is the *round type*; and weights $p_x>0$ for every $x\in\mathcal{X}$ with $\sum_{x\in\mathcal{X}} p_x = 1$ in the declared order, rounding nowhere. A label of weight zero is not a letter of $\mathcal{X}$.

(W3) *(number states)* for $a \in \mathrm{Pol}$ and an integer $\mathsf{n} \geq 0$, the unit vector

$$|\mathsf{n},a\rangle \;:=\; \frac{(c_a^\dagger)^{\mathsf{n}}}{\sqrt{\mathsf{n}!}}\,|\mathrm{vac}\rangle\,, \tag{7.1}$$

and the subspace $S_{\mathsf{n}} \coloneqq \operatorname{span}\{|\mathsf{n}, a\rangle : a \in \mathrm{Pol}\}$ of the n-photon subspace. Since the n-photon subspace of a two-mode Fock space has dimension $\mathsf{n}+1$ and S_{n} is spanned by $|\mathrm{Pol}|$ vectors, $\dim S_{\mathsf{n}} \leq \min(\mathsf{n}+1, |\mathrm{Pol}|)$; and $S_{\mathsf{n}} \perp S_{\mathsf{n}'}$ for $\mathsf{n} \neq \mathsf{n}'$, the two subspaces lying in different photon-number eigenspaces. $S_0 = \mathbb{C}\,|\mathrm{vac}\rangle$ is one dimensional, every polarisation sharing the vacuum.

(W4) *(the specified states)* with the *Poisson weight* $P_\mu(\mathsf{n}) \coloneqq e^{-\mu}\mu^{\mathsf{n}}/\mathsf{n}!$, the state prepared at the letter $x = (a, \mu, z)$ is

$$\rho_x \;=\; \sum_{\mathsf{n}\geq 0} P_\mu(\mathsf{n})\; |\mathsf{n}, a\rangle\langle \mathsf{n}, a|\,, \tag{7.2}$$

which does not depend on the round type z. It is a state: $\sum_{\mathsf{n}\geq 0} P_\mu(\mathsf{n}) = 1$ and each $|\mathsf{n}, a\rangle\langle \mathsf{n}, a|$ is a rank-one projector.

The physical-source assumption of the model, which asserts that the device emits (7.2), is Assumption 10.1; the model declares it in the place of Assumption 5.1.

Uses Def. 1.1 · *Used by* Def. 7.2, Lem. 7.3, Prop. 8.1, Prop. 9.1, Cor. 9.2, Ass. 10.1, Rem. 10.2, Req. 11.4 · newly derived · implemented · *Code* certiqs_key.transmitter.decoy.model:WcpModel

A tagged declaration is a finite virtual source on the sectors a letter weights: quantum to the entropy cutoff, then to the source cutoff a flag per photon number and polarisation shared by all intensities, then a tail flag per polarisation and intensity. Its weights are at most the Poisson ones.

The model's states live on an infinite-dimensional Fock space, which no declaration can carry as matrices. The tagged declaration replaces them by a finite virtual source with three bands of photon numbers: up to the entropy cutoff each photon-number sector stays quantum; above it, up to the source cutoff, each photon number and polarisation keeps a one-dimensional flag; and the remaining weight goes to one tail flag per polarisation and intensity. Only the sectors that some letter weights are declared. A polarisation flag is shared by every intensity, so that the weight of one photon number at different intensities sits on one sector; that relation is the decoy content of the declaration (Remark 8.2 gives the detail). The Poisson weights $P_\mu(\mathsf{n})$ the sector weights are bounded by are not declared numbers, so the inequality of (T2) is not decided against them: the emission decides it against a rational lower bound on $P_\mu(\mathsf{n})$ that it forms from the declared intensity (Lemma 7.3, Requirement 11.5). The polarisation vectors, the number states and the basis of each quantum sector are objects of the model as well, so the entries of the signal dyads in that basis are not declared either: the emission forms them by a construction (Requirement 11.4). The construction generalises the virtual source of [2, Eq. (145)], which is the case $N_{\mathrm{src}} = N_{\mathrm{ent}}$ with the

exact Poisson weights (Proposition 8.1 (iv)): here the declared weights need only lie at or below the Poisson weights, and the polarisation flags between the two cutoffs and the sector map are new. Remark 7.4 compares the two cutoffs with the receiver's.

Definition 7.2 (Tagged virtual source: cutoffs, flags and sector map)**.** Let a weak-coherent-pulse model be declared (Definition 7.1). A *tagged source declaration* over it consists of the following.

(T1) *(the two cutoffs)* integers N_{ent} and N_{src} with $0 \le N_{\mathrm{ent}} \le N_{\mathrm{src}}$: the *entropy cutoff*, the largest photon number whose sector stays quantum, and the *source cutoff*, the largest photon number that keeps a sector of its own.

(T2) *(the sector weights)* declared numbers $\tilde{p}(\mathsf{n} \mid \mu)$ for $0 \le \mathsf{n} \le N_{\mathrm{src}}$ and $\mu \in \mathrm{Int}$, depending on the intensity alone and not on the polarisation or the round type, satisfying

$$0 \;\le\; \tilde{p}(\mathsf{n} \mid \mu) \;\le\; P_\mu(\mathsf{n}) \qquad \text{for every } \mathsf{n} \le N_{\mathrm{src}},\ \mu \in \mathrm{Int}, \tag{7.3}$$

and the *tail weight*

$$\tilde{w}_\mu \;:=\; 1 - \sum_{\mathsf{n}=0}^{N_{\mathrm{src}}} \tilde{p}(\mathsf{n} \mid \mu), \tag{7.4}$$

the sum formed in increasing n and rounding nowhere. Then $\tilde{w}_\mu \ge 0$, because $\sum_{\mathsf{n} \le N_{\mathrm{src}}} \tilde{p}(\mathsf{n} \mid \mu) \le \sum_{\mathsf{n} \le N_{\mathrm{src}}} P_\mu(\mathsf{n}) \le 1$ by (7.3).

(T3) *(the sector index set and the virtual transmitted system)* a sector that no letter of $\mathcal{X}$ puts positive weight on is not declared: the *sector index set* carries the sectors the declaration uses and is read off the weights,

$$\begin{aligned}\mathbb{T} \;&= \{\mathsf{n} \le N_{\mathrm{ent}} :\ \tilde{p}(\mathsf{n} \mid \mu_x) > 0 \text{ for some } x \in \mathcal{X}\} \\ &\quad \cup \{(\mathsf{n}, a) : N_{\mathrm{ent}} < \mathsf{n} \le N_{\mathrm{src}},\ \tilde{p}(\mathsf{n} \mid \mu_x) > 0 \text{ for some } x \in \mathcal{X} \text{ with } a_x = a\} \\ &\quad \cup \{(a, \mu) : \tilde{w}_\mu > 0 \text{ and } (a, \mu) = (a_x, \mu_x) \text{ for some } x \in \mathcal{X}\},\end{aligned} \tag{7.5}$$

in that declared order: the quantum sectors in increasing photon number, then the polarisation flags in the declared order of (n, a), then the tail flags in the declared order of (a, μ). The finite-dimensional virtual transmitted system is the orthogonal direct sum over it,

$$A'' \;=\; \bigoplus_{t \in \mathbb{T}} S_t, \qquad S_t = \begin{cases} S_{\mathsf{n}} & t = \mathsf{n} \le N_{\mathrm{ent}}, \\ \mathbb{C} f_{\mathsf{n},a} & t = (\mathsf{n}, a), \\ \mathbb{C} f_{a,\mu} & t = (a, \mu), \end{cases} \tag{7.6}$$

with the basis of each summand described next; it is the declared basis of A'' in the sense of Definition 3.3, the basis in which the blocks of

the declaration are written. At a quantum sector $t = \mathsf{n}$ the summand is the space S_{n} of Definition 7.1 (W3), taken into the direct sum with its vectors $|\mathsf{n}, a\rangle$ and an orthonormal basis of S_{n}; like those vectors, that basis is a mathematical object of the model and not an array of declared numbers (Proposition 8.1 returns S_{n} to the Fock space by an isometry V_{n}). There is one *polarisation flag* $f_{\mathsf{n},a}$ per photon number above the entropy cutoff and per polarisation of a letter that weights it, shared across intensities and round types; and one *tail flag* $f_{a,\mu}$ per pair a letter carries with $\tilde{w}_\mu > 0$, shared across round types. The flag vectors are declared orthonormal, the summands of (7.6) are mutually orthogonal in A'', and Π_t^{v} is the projector onto S_t. Every flag sector is one dimensional, and so is the quantum sector $\mathsf{n} = 0$. The index set is a function of the declared weights.

(T4) *(the virtual ensemble)* for $x = (a, \mu, z) \in \mathcal{X}$,

$$\rho_x^{\mathrm{v}} = \sum_{\mathsf{n} \le N_{\mathrm{ent}}} \tilde{p}(\mathsf{n} \mid \mu)\, |\mathsf{n}, a\rangle\langle \mathsf{n}, a| + \sum_{N_{\mathrm{ent}} < \mathsf{n} \le N_{\mathrm{src}}} \tilde{p}(\mathsf{n} \mid \mu)\, |f_{\mathsf{n},a}\rangle\langle f_{\mathsf{n},a}| + \tilde{w}_\mu\, |f_{a,\mu}\rangle\langle f_{a,\mu}|\,, \tag{7.7}$$

every sum running over the sectors of the index set (7.5) only: a quantum sector or a polarisation flag that no letter puts positive weight on is not a sector of A'' by (T3) and contributes no term, as the tail flag contributes none where $\tilde{w}_\mu = 0$. Each ρ_x^{v} is a state on A'' that is block diagonal along $\mathbb{T}$. Its component in a quantum sector $\mathsf{n} \le N_{\mathrm{ent}}$ is $\tilde{p}(\mathsf{n} \mid \mu)\, |\mathsf{n}, a\rangle\langle \mathsf{n}, a|$; in the polarisation flag sector (n, a') it is $\tilde{p}(\mathsf{n} \mid \mu)$ if $a' = a$ and zero otherwise; in the tail flag sector (a', μ') it is $\tilde{w}_\mu$ if $(a', \mu') = (a, \mu)$ and zero otherwise. It is positive semidefinite, being a sum of non-negative multiples of rank-one projectors, and its trace is $\sum_{\mathsf{n} \le N_{\mathrm{src}}} \tilde{p}(\mathsf{n} \mid \mu) + \tilde{w}_\mu = 1$ by (7.4), the absent terms contributing nothing to it: a sector that (7.5) leaves out is one on which $\tilde{p}(\mathsf{n} \mid \mu)$ vanishes for every letter, this one included, or a tail flag with $\tilde{w}_\mu = 0$. Two labels of the same polarisation and different intensities put weight on the same polarisation flag, and two labels of the same polarisation and intensity and different round types put weight on the same tail flag.

(T5) *(the matched sector map)* the retained system and its sector structure are not declared: a tagged source is presented as an ensemble presentation (Definition 3.3: a certificate defined from declared weighted blocks rather than declared), in which A, its structure, τ_A and $\{M_x\}$ are the objects (3.2) defines, A_t being a copy of S_t sector by sector and in the same order. In particular $\dim \mathcal{H}(A_t) = \dim \mathcal{H}(S_t)$ whatever the rank of the sector's marginal (Proposition 9.1 (iii)). The matched pair $(\{\Pi_t^A\}, \{\Pi_t^{\mathrm{v}}\})$, with Π_t^A the projectors onto Alice's sectors (Definition 1.1), is the structure of (T3) read on both sides. It

is the retained-register sector map of the link of (T6), the slot (e) of Definition 2.1, which fixes how the certificate, τ_A and $\{M_x\}$ are presented, sector by sector. It is a statement about those three objects only: it does not assert that any joint state of A and another system is block diagonal (Definition 1.1).

(T6) *(the link)* the channel Ξ_{tag} that Proposition 8.1 constructs and proves to carry each ρ_x^{v} to ρ_x, recorded as one source link (Definition 2.1) with the following slots: the kind tagging (slot (a)); the source descriptor $(A'', \{p_x, \rho_x^{\mathrm{v}}\})$ of (T3) and (T4) and the target descriptor $(A', \{p_x, \rho_x\})$ of the model (slot (b)); the channel witness supplied as a theorem constructor, namely Proposition 8.1, with an obligation record that names Assumption 10.1 (slot (c2)); the declared descriptor family, every tagged source declaration over every weak-coherent-pulse model (slot (d)); the retained-register sector map, the matched pair of (T5) (slot (e)); an empty list of transported constraint entries, so that no constraint is carried across the link, and hence no licence record, a licence being present only beside a non-empty list (slot (f)); and the status exact, the identity holding with no error term (slot (g)).

The cutoffs of (T1), the weights $\tilde{p}(\mathsf{n} \mid \mu)$ of (T2), and the model's alphabet, polarisation labels, intensities and label weights are *declared*. The polarisation vectors $|a\rangle$, the number states $|\mathsf{n}, a\rangle$ and the orthonormal basis of each quantum sector S_{n} of (T3) are objects of the model and are not declared numbers. The tail weight (7.4), the index set (7.5) and the weighted members $p_x \rho_x^{\mathrm{v}}$ of (7.7) are *formed*: the tail weight and the index set from the declared numbers in exact arithmetic, and each entry of a weighted member, in the basis of its sector, as the product of declared numbers with an entry of the dyad of the sector's signal vector in that basis, a number fixed by the model and the basis. They are the declaration's own data: read sector by sector, the weighted members are the blocks that the ensemble presentation of (T5) takes as its data. Everything on Alice's side — A, τ_A, $\{M_x\}$ and the certificate — is defined from those blocks and declared by no one.

Uses Def. 2.1, Def. 2.2, Def. 7.1 · *Used by* Rem. 7.4, Prop. 8.1, Rem. 8.2, Prop. 9.1, Cor. 9.2, Ass. 10.1, Rem. 10.2, Req. 11.1, Req. 11.2, Req. 11.4, Req. 11.5, Req. 11.6 · newly derived · implemented · *Code* certiqs_key.transmitter.decoy.tagged: TaggedSourceDeclaration

For a rational intensity $\mu \geq 0$ and an even integer $K \geq \mu$, two consecutive partial sums of the exponential series, scaled by $\mu\hat{}n/n!$, enclose the Poisson weight between rationals. The enclosure narrows monotonically to zero width as K grows, so the bound on the sector weights is decided exactly.

The inequality (7.3) of Definition 7.2 (T2) compares a declared number with $P_\mu(\mathsf{n})$, which at a positive intensity is transcendental. A declared intensity is a binary64 number, hence a dyadic rational, and the exponential series at a negative rational argument is alternating with terms that decrease from some index on, so two consecutive partial sums bracket $e^{-\mu}$. Multiplied by the rational factor $\mu^{\mathsf{n}}/\mathsf{n}!$ they bracket the Poisson weight, and every number involved is a rational formed from μ in finitely many steps.

Lemma 7.3 (Rational enclosure of the Poisson weights). Let $\mu \geq 0$ be a rational number. For integers $k \geq 0$ put $t_k := \mu^k/k!$, with $\mu^0 := 1$, and for integers $K \geq 0$

$$S_K \;:=\; \sum_{k=0}^{K} (-1)^k \, t_k. \tag{7.8}$$

Let K be an even integer with $K \geq \mu$, and let $\mathsf{n} \geq 0$ be an integer. Then:

(i) *(the enclosure)* with the Poisson weight $P_\mu(\mathsf{n}) = e^{-\mu}\mu^{\mathsf{n}}/\mathsf{n}!$ of Definition 7.1 (W4),

$$\max\left(0, S_{K+1}\right) t_{\mathsf{n}} \;\leq\; P_\mu(\mathsf{n}) \;\leq\; S_K \, t_{\mathsf{n}}; \tag{7.9}$$

(ii) *(rational ends)* both ends of (7.9) are rational numbers, formed from μ by finitely many additions, subtractions, multiplications and divisions by positive integers, and a comparison of either end with a binary64 number is an exact comparison of two rationals;

(iii) *(the width)* $S_K - S_{K+1} = t_{K+1}$, so the two ends of (7.9) differ by at most $t_{K+1}t_{\mathsf{n}}$, and $t_{K+1} \to 0$ as $K \to \infty$ through the even integers. At $\mu = 0$ both ends equal $P_0(\mathsf{n})$, which is 1 for $\mathsf{n} = 0$ and 0 for $\mathsf{n} \geq 1$.

(iv) *(monotone in K)* over the even integers $K \geq \mu$ the lower end $\max(0, S_{K+1})\, t_{\mathsf{n}}$ is non-decreasing and the upper end $S_K\, t_{\mathsf{n}}$ is non-increasing: for even $K \leq K'$ with $K \geq \mu$, the enclosure at K' lies inside the enclosure at K.

Uses Def. 7.1 · *Used by* Req. 11.5 · newly derived · implemented · *Code* certiqs_key.transmitter.decoy.enclosure:poisson_enclosures

Proof. Idea. The tail of the exponential series beyond K is alternating with non-increasing terms, so grouping it in consecutive pairs fixes its sign.

Step 1 (the series). The exponential series $e^{-\mu} = \sum_{k\geq 0}(-\mu)^k/k! = \sum_{k\geq 0}(-1)^k t_k$ converges absolutely, $\sum_{k\geq 0} t_k = e^{\mu}$ being finite. Hence for

every $K' \geq 0$

$$e^{-\mu} - S_{K'} = \sum_{k>K'} (-1)^k t_k,$$

and this tail may be summed in consecutive pairs of terms: the partial sums of the paired series are partial sums of the tail, which converge to its value.

Step 2 (the tail decreases). For $k \geq K+1$, $t_{k+1} = t_k\,\mu/(k+1)$ with $0 \leq \mu \leq K < k+1$, so $0 \leq t_{k+1} \leq t_k$.

Step 3 (the upper end). K is even, so the first term of the tail of S_K is $(-1)^{K+1}t_{K+1} = -t_{K+1}$, and pairing gives

$$e^{-\mu} - S_K = -\sum_{j\geq 0} \big(t_{K+1+2j} - t_{K+2+2j}\big) \leq 0,$$

every bracket being non-negative by Step 2. So $e^{-\mu} \leq S_K$.

Step 4 (the lower end). $K+2$ is even, so the first term of the tail of S_{K+1} is t_{K+2}, and pairing gives

$$e^{-\mu} - S_{K+1} = \sum_{j\geq 0} \big(t_{K+2+2j} - t_{K+3+2j}\big) \geq 0,$$

again by Step 2. So $S_{K+1} \leq e^{-\mu}$, and since $e^{-\mu} > 0$ also $\max(0, S_{K+1}) \leq e^{-\mu}$.

Step 5 (the Poisson weight). $P_\mu(\mathsf{n}) = e^{-\mu}t_{\mathsf{n}}$ with $t_{\mathsf{n}} \geq 0$, and multiplying the inequalities of Steps 3 and 4 by the non-negative number t_{n} preserves them. This is (i).

(ii) $t_0 = 1$ and $t_k = t_{k-1}\,\mu/k$ for $k \geq 1$, so every t_k is a rational formed from μ by multiplications and divisions by positive integers; S_K and S_{K+1} are finite sums and differences of them, and the maximum of two rationals is one of them. A binary64 number is a dyadic rational, so the comparison is one of two rationals.

(iii) $S_K - S_{K+1} = -(-1)^{K+1}t_{K+1} = t_{K+1}$, $K+1$ being odd. The upper end minus the lower end is $\big(S_K - \max(0, S_{K+1})\big)t_{\mathsf{n}} \leq (S_K - S_{K+1})t_{\mathsf{n}} = t_{K+1}t_{\mathsf{n}}$. The t_k are the terms of the convergent series of Step 1, so $t_k \to 0$. At $\mu = 0$, $t_k = 0$ for $k \geq 1$, so the width $t_{K+1}t_{\mathsf{n}}$ is zero and both ends equal $P_0(\mathsf{n})$; and $S_K = S_{K+1} = 1$, so the common value is t_{n}, which is 1 for $\mathsf{n} = 0$ and 0 for $\mathsf{n} \geq 1$.

(iv) It suffices to compare K with $K+2$, both even and at least μ. By (7.8), $S_{K+3} - S_{K+1} = t_{K+2} - t_{K+3} \geq 0$ and $S_{K+2} - S_K = -t_{K+1} + t_{K+2} \leq 0$, both by Step 2, which applies from the index $K+1$ on. So S_{K+1} does not decrease and S_K does not increase from K to $K+2$; $\max(0, \cdot)$ preserves the first, and multiplying by $t_{\mathsf{n}} \geq 0$ preserves both. □

The entropy cutoff bounds the quantum sectors and the source cutoff the flagged ones, and raising the second never requires raising the first. The receiver's cutoff is a different object, which no condition of this submodule reads.

Remark 7.4 (Entropy, source and receiver cutoffs). A reader of a protocol with a decoy source meets three cutoffs that are easily conflated, and only two of them are data of this submodule.

The entropy cutoff N_{ent} is the largest photon number whose sector of A'' stays quantum: at $\mathsf{n} \le N_{\mathrm{ent}}$ the sector is S_{n} and carries the signal states $|\mathsf{n}, a\rangle$ themselves, one sector for all polarisations of that photon number. Above it the polarisation is tagged: the sector is one dimensional and there is one per polarisation, so the declaration separates what a quantum sector keeps together. Whether keeping a photon number quantum or tagging it per polarisation gives the higher key rate is a question for the consuming program, settled by computation and not here.

The source cutoff N_{src} is the largest photon number that keeps a sector of its own at all. Raising it by one adds at most $|\mathrm{Pol}|$ one-dimensional sectors, one per polarisation some letter carries at a positive weight, the index set of Definition 7.2 (T3) being read off the weights; and it moves the weight $\tilde{p}(\mathsf{n} \mid \mu)$ out of the tail flag into them, so that the weight of that photon number sits on a flag shared by every intensity instead of on a flag carried by one intensity alone. Raising N_{src} never requires raising N_{ent}: (T1) demands only $N_{\mathrm{ent}} \le N_{\mathrm{src}}$, and the sectors between the two cutoffs are one dimensional whatever N_{src} is.

The receiver cutoff, the photon number at which a receiver-side reduction stops retaining blocks and starts flagging, is a datum of the receiver and of the reduction that consumes it, and not of this submodule. No condition of this submodule mentions it, no claim here reads it, and nothing ties it to the entropy or the source cutoff: the transmitter's data and the receiver's meet only where the protocol compiler composes them. A declaration with $N_{\mathrm{ent}} = 1$ and a receiver cutoff of two is as well formed as any other.

Every claim of this submodule is stated for arbitrary $N_{\mathrm{ent}} \le N_{\mathrm{src}}$. The bound $N_{\mathrm{ent}} \le 2$ is a requirement of the emission, the step that outputs the declaration (Requirement 11.3), and concerns which numbers a declaration can carry exactly, not any of the mathematics here: for the four polarisations $\{H, V, +, -\}$, at one and at two photons, Corollary 9.2 (iv)(b) exhibits a declared basis in which the signal dyads have the entries 0 and $\pm\frac{1}{2}$ (and at one photon also 1); at three photons and above no such basis is given here.

Uses Def. 7.2 · newly derived · accepted

8 The tagging channel

The virtual and the physical source are linked by one channel, the tagging channel, which is fixed before a letter is drawn. The main result of the section, Proposition 8.1, constructs it for any sector weights at or below the Poisson weights: it keeps each quantum sector, prepares from each polarisation flag the number state it stands for, and prepares from each tail flag the normalised remainder of the Poisson distribution. So a declared weight need only satisfy a certified inequality, never equal a Poisson weight. Remark 8.2 explains why a polarisation flag is shared across intensities: a finer

family would still admit such a channel but would lose the relation across intensities that is the decoy content of the declaration.

One channel, independent of the label, maps each tagged virtual state to its phase-randomised weak coherent state whenever the declared sector weights lie at or below the Poisson weights. The weights therefore need only a certified inequality, never an exact Poisson weight.

Recall the virtual source of Definition 7.2 (T3) and (T4): A'' is the direct sum, over the sectors that some letter weights, of the quantum sectors S_{n} for $\mathsf{n} \le N_{\mathrm{ent}}$, the one-dimensional polarisation flags $\mathbb{C}f_{\mathsf{n},a}$ for $N_{\mathrm{ent}} < \mathsf{n} \le N_{\mathrm{src}}$ and the one-dimensional tail flags $\mathbb{C}f_{a,\mu}$; the virtual state ρ_x^{v} of a letter $x = (a,\mu,z)$ puts $\tilde{p}(\mathsf{n} \mid \mu)\,|\mathsf{n},a\rangle\langle\mathsf{n},a|$ on each quantum sector, the weight $\tilde{p}(\mathsf{n} \mid \mu)$ on the polarisation flag $f_{\mathsf{n},a}$ of its own polarisation, and the tail weight $\tilde{w}_\mu$ on its tail flag $f_{a,\mu}$. The tail flag absorbs whatever weight the declared sectors leave out, so the construction needs the sector weights only to lie at or below the Poisson weights: a declared weight may be any number certified to lie in $[0, P_\mu(\mathsf{n})]$, and none need equal a Poisson weight. Lowering $\tilde{p}(\mathsf{n} \mid \mu)$ moves the difference into $\tilde{w}_\mu$, and so into the tail flag of every polarisation at that intensity; Corollary 9.2 (v) says what else changes. Each flag, polarisation or tail, is sent to a state that depends on the flag alone, so one channel, fixed before a letter is drawn, serves every letter, which is what a source link requires of its channel witness (Definition 2.1). That holds for any flag family, the tail flags indexed by μ included; Remark 8.2 says why the polarisation flags are nonetheless shared across intensities.

Proposition 8.1 (Tagging channel for weights at or below the Poisson weights). Let a weak-coherent-pulse model be given (Definition 7.1): labels $x = (a,\mu,z)$ with z the round type, unit polarisation vectors $|a\rangle$, Poisson weights $P_\mu(\mathsf{n}) = e^{-\mu}\mu^{\mathsf{n}}/\mathsf{n}!$ and model states $\rho_x = \sum_{\mathsf{n}\ge 0} P_\mu(\mathsf{n})\ |\mathsf{n},a\rangle\langle\mathsf{n},a|$ on the two-mode Fock space. Let a tagged source be declared on it (Definition 7.2) with cutoffs $0 \le N_{\mathrm{ent}} \le N_{\mathrm{src}}$, virtual states ρ_x^{v} of (7.7) and declared weights $\tilde{p}(\mathsf{n}\,|\,\mu)$, $\mathsf{n} \le N_{\mathrm{src}}$, satisfying the inequality (7.3) of Definition 7.2 (T2), which reads

$$0 \le \tilde{p}(\mathsf{n}\,|\,\mu) \le P_\mu(\mathsf{n}) \qquad \text{for every intensity } \mu \text{ and every } \mathsf{n} \le N_{\mathrm{src}}, \tag{8.1}$$

with tail weights $\tilde{w}_\mu = 1 - \sum_{\mathsf{n}\le N_{\mathrm{src}}} \tilde{p}(\mathsf{n}\,|\,\mu)$. Then:

(i) *(the tail states)* for every polarisation a and every intensity μ with $\tilde{w}_\mu > 0$,

$$\vartheta_{a,\mu} \coloneqq \frac{1}{\tilde{w}_\mu}\Big[\sum_{\mathsf{n}\le N_{\mathrm{src}}} \big(P_\mu(\mathsf{n}) - \tilde{p}(\mathsf{n}\,|\,\mu)\big)\,|\mathsf{n},a\rangle\langle\mathsf{n},a| + \sum_{\mathsf{n}>N_{\mathrm{src}}} P_\mu(\mathsf{n})\ |\mathsf{n},a\rangle\langle\mathsf{n},a|\Big] \tag{8.2}$$

is a state on the Fock space. Where $\tilde{w}_\mu = 0$ the declaration carries no tail flag for (a,μ) (Definition 7.2) and no tail state is defined.

(ii) *(one channel)* the map

$$\Xi_{\mathrm{tag}}(\zeta) \coloneqq \sum_{\mathsf{n}\le N_{\mathrm{ent}}} V_{\mathsf{n}}\,\Pi^{\mathrm{v}}_{\mathsf{n}}\,\zeta\,\Pi^{\mathrm{v}}_{\mathsf{n}}V_{\mathsf{n}}^{\dagger} + \sum_{N_{\mathrm{ent}}<\mathsf{n}\le N_{\mathrm{src}}}\ \sum_{a} \langle f_{\mathsf{n},a}|\,\zeta\,|f_{\mathsf{n},a}\rangle\ |\mathsf{n},a\rangle\langle\mathsf{n},a| + \sum_{(a,\mu)\colon \tilde{w}_\mu>0} \langle f_{a,\mu}|\,\zeta\,|f_{a,\mu}\rangle\ \vartheta_{a,\mu}, \tag{8.3}$$

in which $\Pi^{\mathrm{v}}_{\mathsf{n}}$ projects $\mathcal{H}(A'')$ onto the quantum sector S_{n}, V_{n} is the isometry that embeds that sector into the Fock space as $\mathrm{span}\{|\mathsf{n},a\rangle\}_a$, and every sum runs over the sectors of the declared index set $\mathbb{T}$ only, is a channel from A'' to the Fock space, and

$$\Xi_{\mathrm{tag}}(\rho^{\mathrm{v}}_x) = \rho_x \qquad \text{for every label } x. \tag{8.4}$$

A quantum sector or a polarisation flag that no letter puts weight on is not a sector of A'' by Definition 7.2 (T3) and contributes no term to (8.3), exactly as for the tail flags. The channel Ξ_{tag} depends on the declared weights and the cutoffs and on no label.

(iii) *(the role of the inequality)* (8.4) holds for every family of weights satisfying (8.1). Where (8.1) fails at an intensity μ, its two halves fail differently.

(a) If $\tilde{p}(\mathsf{n}\,|\,\mu) > P_\mu(\mathsf{n})$ for some $\mathsf{n} \le N_{\mathrm{src}}$ and $\tilde{w}_\mu > 0$, the operator (8.2) is not positive semidefinite, for every polarisation a, and the construction is not available: where a letter x has intensity μ, the map (8.3) sends the state $|f_{a_x,\mu}\rangle\langle f_{a_x,\mu}|$ to $\vartheta_{a_x,\mu}$ and is not positive.

(b) If $\tilde{p}(\mathsf{n}\,|\,\mu) < 0$ for some $\mathsf{n} \le N_{\mathrm{src}}$ while $\tilde{p}(\mathsf{n}'\,|\,\mu) \le P_\mu(\mathsf{n}')$ for every $\mathsf{n}' \le N_{\mathrm{src}}$, then $\tilde{w}_\mu > 0$ and the operator (8.2) is a state for every polarisation a. What fails is the virtual state: for every letter x of intensity μ the operator ρ^{v}_x of (7.7) is not positive semidefinite or has trace greater than one, so it is not a state.

(iv) *(the published case)* for $\tilde{p}(\mathsf{n}\,|\,\mu) = P_\mu(\mathsf{n})$ and $N_{\mathrm{ent}} = N_{\mathrm{src}}$ the virtual states are those of [2, Eq. (145)] and (8.4) is [2, Lemma 9.2].

Uses Def. 2.1, Def. 7.1, Def. 7.2 · *Used by* Cor. 9.2, Req. 11.5 · newly derived · accepted

Proof. Idea. The channel keeps each quantum sector, prepares from each polarisation flag the number state it stands for, and prepares from each tail flag the normalised remainder of the Poisson distribution, which the inequality makes a state.

(i) Every coefficient of (8.2) is non-negative, the first family by (8.1), and the vectors $|\mathsf{n},a\rangle$, $\mathsf{n}\ge 0$, are orthonormal for a fixed a, being number states of the one mode of polarisation a with the orthogonal mode in its vacuum.

So the bracket is a positive trace-class operator with trace $\sum_{\mathsf{n}\geq 0} P_\mu(\mathsf{n}) - \sum_{\mathsf{n}\leq N_{\mathrm{src}}} \tilde{p}(\mathsf{n}\,|\,\mu) = 1-(1-\tilde{w}_\mu) = \tilde{w}_\mu$, and $\vartheta_{a,\mu}$ has unit trace.

(ii) *A channel.* The sector projectors $\Pi^{\mathrm{v}}_{\mathsf{n}}$, $|f_{\mathsf{n},a}\rangle\langle f_{\mathsf{n},a}|$ and $|f_{a,\mu}\rangle\langle f_{a,\mu}|$ are mutually orthogonal and sum to $\mathbb{1}_{A''}$ (Definition 7.2). Each of the three sums of (8.3) is completely positive: the first is a sum of conjugations by $V_{\mathsf{n}}\Pi^{\mathrm{v}}_{\mathsf{n}}$; the second and third are measure-and-prepare maps, a rank-one projective measurement followed by the preparation of a fixed state, the states $\vartheta_{a,\mu}$ being states by (i). For the trace, V_{n} is an isometry on S_{n} and every prepared operator has unit trace, so $\operatorname{tr}\Xi_{\mathrm{tag}}(\zeta) = \sum_{\mathsf{n}\leq N_{\mathrm{ent}}} \operatorname{tr}[\Pi^{\mathrm{v}}_{\mathsf{n}}\zeta] + \sum_{\mathsf{n},a} \langle f_{\mathsf{n},a}|\,\zeta\,|f_{\mathsf{n},a}\rangle + \sum_{(a,\mu):\,\tilde{w}_\mu>0} \langle f_{a,\mu}|\,\zeta\,|f_{a,\mu}\rangle = \operatorname{tr}\zeta$, the projectors summed being all the sector projectors of the declaration. The input is finite dimensional and every output operator is trace class, so Ξ_{tag} is a completely positive trace-preserving map into the trace class of the Fock space.

The identity. Fix $x=(a,\mu,z)$. By Definition 7.2,

$$\rho^{\mathrm{v}}_x = \sum_{\mathsf{n}\leq N_{\mathrm{ent}}} \tilde{p}(\mathsf{n}\,|\,\mu)\;|\mathsf{n},a\rangle\langle\mathsf{n},a|_{S_{\mathsf{n}}} + \sum_{N_{\mathrm{ent}}<\mathsf{n}\leq N_{\mathrm{src}}} \tilde{p}(\mathsf{n}\,|\,\mu)\;|f_{\mathsf{n},a}\rangle\langle f_{\mathsf{n},a}| + \tilde{w}_\mu\;|f_{a,\mu}\rangle\langle f_{a,\mu}|\,,$$

which is block diagonal along the sectors. The first sum of (8.3) maps its first sum to $\sum_{\mathsf{n}\leq N_{\mathrm{ent}}} \tilde{p}(\mathsf{n}\,|\,\mu)\,|\mathsf{n},a\rangle\langle\mathsf{n},a|$, the embedding carrying the vector $|\mathsf{n},a\rangle$ of S_{n} to the Fock vector of the same name. The second maps the flag $f_{\mathsf{n},a}$, which carries weight $\tilde{p}(\mathsf{n}\,|\,\mu)$ in ρ^{v}_x while the flags $f_{\mathsf{n},a'}$ of the other polarisations $a'\neq a$ carry none, to $\tilde{p}(\mathsf{n}\,|\,\mu)\,|\mathsf{n},a\rangle\langle\mathsf{n},a|$. The third maps the one tail flag of ρ^{v}_x, where $\tilde{w}_\mu>0$, to $\tilde{w}_\mu\vartheta_{a,\mu}$, which by (8.2) is the bracket there. When $\tilde{w}_\mu=0$ there is no tail flag and ρ^{v}_x has no third term; then (8.1) and $\sum_{\mathsf{n}} P_\mu(\mathsf{n})=1$ force $\tilde{p}(\mathsf{n}\,|\,\mu)=P_\mu(\mathsf{n})$ for $\mathsf{n}\leq N_{\mathrm{src}}$ and $P_\mu(\mathsf{n})=0$ beyond, so the bracket vanishes as well. Adding the three,

$$\Xi_{\mathrm{tag}}(\rho^{\mathrm{v}}_x) = \sum_{\mathsf{n}\leq N_{\mathrm{src}}} \tilde{p}(\mathsf{n}\,|\,\mu)\,|\mathsf{n},a\rangle\langle\mathsf{n},a| + \sum_{\mathsf{n}\leq N_{\mathrm{src}}} \big(P_\mu(\mathsf{n})-\tilde{p}(\mathsf{n}\,|\,\mu)\big)\,|\mathsf{n},a\rangle\langle\mathsf{n},a| + \sum_{\mathsf{n}>N_{\mathrm{src}}} P_\mu(\mathsf{n})\,|\mathsf{n},a\rangle\langle\mathsf{n},a| = \rho_x.$$

Nothing in (8.3) refers to x: the state prepared from a flag, a polarisation flag indexed by (n,a) or a tail flag indexed by (a,μ), is a function of the flag alone.

(iii) The first sentence is (ii), whose only hypothesis on the weights is (8.1).

(a) For every a the vectors $|\mathsf{n}',a\rangle$, $\mathsf{n}'\geq 0$, are orthonormal (proof of (i)), so $\langle\mathsf{n},a|\,\vartheta_{a,\mu}\,|\mathsf{n},a\rangle = \big(P_\mu(\mathsf{n})-\tilde{p}(\mathsf{n}\,|\,\mu)\big)/\tilde{w}_\mu < 0$ and (8.2) is not positive semidefinite. A letter x of intensity μ carries the tail flag $f_{a_x,\mu}$, which is a sector of $\mathbb{T}$ by (7.5) because $\tilde{w}_\mu>0$. On $|f_{a_x,\mu}\rangle\langle f_{a_x,\mu}|$ the first two sums of (8.3) vanish, the sectors being orthogonal, and the third gives $\vartheta_{a_x,\mu}$; so (8.3) maps a positive operator to one that is not positive semidefinite.

(b) Here $\sum_{\mathsf{n}'\leq N_{\mathrm{src}}} \tilde{p}(\mathsf{n}'\,|\,\mu) < \sum_{\mathsf{n}'\leq N_{\mathrm{src}}} P_\mu(\mathsf{n}') \leq 1$, the inequality being strict at $\mathsf{n}'=\mathsf{n}$, where $\tilde{p}(\mathsf{n}\,|\,\mu)<0\leq P_\mu(\mathsf{n})$, so $\tilde{w}_\mu>0$; and the proof of (i),

which reads only the upper half of (8.1), makes $\vartheta_{a,\mu}$ a state for every a. Let x be a letter with $\mu_x = \mu$, write $a = a_x$, and let t be the sector that carries the term of $\tilde{p}(\mathsf{n}\,|\,\mu)$ in (7.7): $t = \mathsf{n}$ if $\mathsf{n} \le N_{\mathrm{ent}}$ and $t = (\mathsf{n}, a)$ otherwise. If $t \in \mathbb{T}$, the component of ρ^{v}_x in S_t is $\tilde{p}(\mathsf{n}\,|\,\mu)$ times the dyad of the unit vector $|\mathsf{n}, a\rangle$, resp. $f_{\mathsf{n},a}$, so the expectation of ρ^{v}_x in that vector is $\tilde{p}(\mathsf{n}\,|\,\mu) < 0$ and ρ^{v}_x is not positive semidefinite. If $t \notin \mathbb{T}$, (7.7) has no term in t. By (7.5) every sector on which x puts a positive weight lies in $\mathbb{T}$, the tail flag $f_{a,\mu}$ among them since $\tilde{w}_\mu > 0$; so every term that (7.7) omits for x has a weight $\tilde{p}(\mathsf{n}'\,|\,\mu) \le 0$, the term at n among them, and by (7.4)

$$\operatorname{tr} \rho^{\mathrm{v}}_x = 1 - \sum_{\mathsf{n}' \text{ omitted}} \tilde{p}(\mathsf{n}'\,|\,\mu) \;\ge\; 1 - \tilde{p}(\mathsf{n}\,|\,\mu) \;>\; 1.$$

(iv) With $\tilde{p} = P_\mu$ and $N_{\mathrm{ent}} = N_{\mathrm{src}}$ the middle sum of ρ^{v}_x is empty, the low sectors carry the exact Poisson weights on $|\mathsf{n}, a\rangle\langle\mathsf{n}, a|$, and the tail flag carries $1 - \sum_{\mathsf{n} \le N_{\mathrm{src}}} P_\mu(\mathsf{n})$. This is the displayed state of the cited equation with its photon cutoff equal to N_{src} and its flag $|a, \mu\rangle\langle a, \mu|$ equal to $|f_{a,\mu}\rangle\langle f_{a,\mu}|$; the cited lemma asserts a channel with (8.4) for every choice of a, μ and z. □

A flag family split by intensity would still admit a label-independent tagging channel, but it would drop the relation between one photon number's behaviour at different intensities, which is the decoy content of the declaration. The emission therefore refuses it with a named refusal.

Remark 8.2 (Polarisation flags shared across intensities). Definition 7.2 (T3) declares *one* polarisation flag $f_{\mathsf{n},a}$ per photon number $N_{\mathrm{ent}} < \mathsf{n} \le N_{\mathrm{src}}$ and per polarisation, shared by every intensity and both round types. A declarer might instead declare a finer family $f_{\mathsf{n},a,\mu}$, one flag per intensity as well; this remark says what that would and would not be.

The source identity would still hold. The finer declaration admits a channel of the shape Proposition 8.1 constructs, and with it the source identity, for this link (8.4): send $f_{\mathsf{n},a,\mu}$ to $|\mathsf{n}, a\rangle\langle\mathsf{n}, a|$, which is independent of μ, and leave the rest as it is. So splitting is not excluded by the source identity and is not a security error of the kind this submodule is about.

The decoy content would be lost. With one shared flag, the letters (a, μ, z) and (a, μ', z') of the same polarisation put their weight on the same one-dimensional sector of A'', so one sector of the declaration carries the whole of that photon number's weight across intensities and round types. With one flag per intensity they are different sectors, and nothing in the declaration relates what happens in the sector of μ to what happens in the sector of μ': a tagged photon number of one intensity and the same photon number of another intensity become unrelated parts of the declaration. That relation is the decoy content of the declaration, the reason for declaring intensities at all being that the same photon number behaves the same way whichever intensity produced it; so a split declaration is a declaration of a different protocol, one with intensities and without decoy. The same applies to the round type: the shared flag is why a tagged photon number of a test round and of a generation round are one sector, which is not an assumption added here but a consequence of declaring one flag.

The direction of the loss. Splitting removes a relation from the declaration and adds none, so a consumer of a split declaration is told less about the source than a consumer of the shared one. The loss would not show as an error but as a rate that does not improve when an intensity is added, which is why the emission, the step that outputs the declaration, refuses a split family with a named refusal (Requirement 11.6). No claim of this repository compares the key lengths of two declarations, and this remark asserts no such comparison: it asserts which relation is present in which declaration. The reverse, a coarser flag than Definition 7.2 (T3) declares (for instance one flag for two polarisations of the same photon number), is excluded for a stricter reason: it is not a tagged source declaration at all. Which vectors the flags are and which letters put weight on them is fixed by Definition 7.2 (T4), and Proposition 8.1 proves the source identity for those declarations only.

Uses Def. 7.2 · newly derived · accepted

9 The certificate and the emitted declaration

The emission, the step of this submodule that outputs a tagged declaration as a decoy declaration, declares the weighted virtual blocks sector by sector and nothing on Alice's side. Proposition 9.1 shows that these blocks satisfy the positivity and weight conditions (E1) and (E2) of an ensemble presentation, which gives an exact certificate, and computes the marginal and the numbers it forms; whether those numbers are binary64, as (E3) requires, is decided on each declaration and not there. The main result of the section, Corollary 9.2, states what the emitted declaration satisfies: an exact certificate against the tagged ensemble where its blocks meet (E1) to (E3), the source identity at the emitted descriptors as a conclusion of Proposition 8.1, block-diagonal states after any attack, the entries on which exactness is decided, and that lowering a weight gives another tagged declaration. Remark 9.3 says how the consuming modules receive it.

Read sector by sector, the tagged virtual ensemble meets (E1) and (E2) of an ensemble presentation, giving an exact certificate with pure blocks whose sector marginals are transposed sums of signal dyads. Whether the formed numbers are binary64, as (E3) requires, is not decided here.

The certificate of a tagged source is never declared: by the ensemble presentation of Definition 3.3 it suffices to declare the weighted virtual ensemble sector by sector. A presentation has three conditions: (E1) every block is Hermitian and positive semidefinite; (E2) the traces of a letter's blocks sum to p_x, and the p_x to one; (E3) for every sector the sum of the blocks over the letters, accumulated entry by entry in the declared order of $\mathcal{X}$, rounds nowhere, so that its entries are binary64 numbers. The certificate it defines satisfies (SR1), its partial trace over the virtual system is Alice's marginal τ_A, and (SR2), Alice's outcome x leaves $p_x\rho_x^{\mathrm{v}}$ on the virtual system; (SR3),

that Alice retains her system, is an attestation. The sectors are those of Definition 7.2 (T3), the ones that some letter weights: a quantum sector $t = \mathsf{n}$, with $S_{\mathsf{n}} = \mathrm{span}\{|\mathsf{n}, a\rangle : a \in \mathrm{Pol}\}$, for each $\mathsf{n} \le N_{\mathrm{ent}}$ (the entropy cutoff); a polarisation flag $t = (\mathsf{n}, a)$ for $N_{\mathrm{ent}} < \mathsf{n} \le N_{\mathrm{src}}$ (the source cutoff); and a tail flag $t = (a, \mu)$ for each pair a letter carries with $\tilde{w}_\mu > 0$. The proposition checks (E1) and (E2), which every tagged declaration meets, computes the sector marginals and their ranks (a rank below $\dim S_t$ is where the completion term $\mathbb{1}_{A_t} - P_t$ of (3.2) is non-zero), and records which numbers the presentation forms. Whether those products and sums are binary64 numbers, as the number model and (E3) require, is a property of a particular declaration, decided on it and not asserted here (Corollary 9.2 (iv)).

Proposition 9.1 (Ensemble presentation of the tagged source). Let a tagged source be declared on a weak-coherent-pulse source model (Definitions 7.1 and 7.2), with labels $x = (a_x, \mu_x, z_x)$, z_x the round type, of weights $p_x > 0$, sector index set $\mathbb{T}$ and virtual sectors $S_t \subseteq \mathcal{H}(A'')$. For $t \in \mathbb{T}$ and a label x write $w_{x,t} \ge 0$ and $|\varphi_{x,t}\rangle \in \mathcal{H}(S_t)$ for the weight and the unit vector that ρ^{v}_x carries in sector t: $w_{x,t} = \tilde{p}(\mathsf{n} \,|\, \mu_x)$ and $|\varphi_{x,t}\rangle = |\mathsf{n}, a_x\rangle$ on a quantum sector $t = \mathsf{n}$; $w_{x,t} = \tilde{p}(\mathsf{n} \,|\, \mu_x)$ if $a_x = a$ and 0 otherwise, with $|\varphi_{x,t}\rangle = |f_{\mathsf{n},a}\rangle$, on a polarisation flag $t = (\mathsf{n}, a)$; and $w_{x,t} = \tilde{w}_\mu$ if $(a_x, \mu_x) = (a, \mu)$ and 0 otherwise, with $|\varphi_{x,t}\rangle = |f_{a,\mu}\rangle$, on a tail flag $t = (a, \mu)$. So $\rho^{\mathrm{v}}_x = \sum_t w_{x,t} \, |\varphi_{x,t}\rangle\langle\varphi_{x,t}|$ and $\sum_t w_{x,t} = 1$. Put $\mathcal{X}_t := \{x : w_{x,t} > 0\}$, let the sector structure carry exactly the sectors with $\mathcal{X}_t \neq \varnothing$, and let

$$\sigma^{\mathrm{v},(t)}_x \;:=\; p_x \, w_{x,t} \, |\varphi_{x,t}\rangle\langle\varphi_{x,t}| \;\in\; \mathrm{Pos}(S_t), \qquad x \in \mathcal{X},\ t \in \mathbb{T}, \tag{9.1}$$

the block of the weighted member $p_x \rho^{\mathrm{v}}_x$ in sector t.

(i) *(an ensemble presentation)* the block lists (9.1) satisfy conditions (E1) and (E2) of Definition 3.3 as identities of real numbers, with the weights p_x of Definition 7.1. Hence, by Lemma 3.4, the objects (3.2) formed from them — a retained system $A = \bigoplus_t A_t$ with A_t a copy of S_t, the marginal τ_A, a POVM $\{M_x\}$ and blocks $(\Psi_t)_t$ — are a block-form source-replacement certificate, (SR1) and (SR2) holding exactly against the tagged virtual ensemble $\{p_x, \rho^{\mathrm{v}}_x\}$ of Definition 7.2 (T4) itself, and every non-zero Ψ_t is a multiple of a rank-one projector.

(ii) *(the marginal)* in the declared basis of S_t, copied to A_t,

$$\tau^{(t)}_A = \Big(\sum_{x \in \mathcal{X}_t} p_x w_{x,t} \, |\varphi_{x,t}\rangle\langle\varphi_{x,t}| \Big)^{\mathsf{T}}, \qquad g_t := \mathrm{tr}\, \tau^{(t)}_A = \sum_{x \in \mathcal{X}_t} p_x w_{x,t}, \tag{9.2}$$

g_t being the probability of sector t and $\sum_t g_t = 1$. On the vacuum sector and on every flag sector A_t is one dimensional and $\tau^{(t)}_A$ is the number g_t.

(iii) *(ranks)* $r_t \coloneqq \operatorname{rank} \tau_A^{(t)} = \dim \operatorname{span}\{|\varphi_{x,t}\rangle : x \in \mathcal{X}_t\}$. Hence $r_t = 1$ on every polarisation flag, on every tail flag and on the vacuum sector $\mathsf{n} = 0$, and on a quantum sector $\mathsf{n} \geq 1$

$$r_{\mathsf{n}} = \dim \operatorname{span}\{|\mathsf{n}, a_x\rangle : x \in \mathcal{X}_{\mathsf{n}}\} \leq \min\big(\dim \mathcal{H}(S_{\mathsf{n}}),\ |\{a_x : x \in \mathcal{X}_{\mathsf{n}}\}|\big), \tag{9.3}$$

while $\dim \mathcal{H}(A_{\mathsf{n}}) = \dim \mathcal{H}(S_{\mathsf{n}})$ whatever the rank.

(iv) *(the formed numbers)* every entry of (9.1) is a product of the declared numbers p_x and $w_{x,t}$ with an entry of $|\varphi_{x,t}\rangle\langle\varphi_{x,t}|$ in the declared basis, and every entry of $\tau_A^{(t)}$ is a sum of such products: no square root and no quotient of a declared number occurs in anything a declaration of this presentation carries.

Uses Def. 3.3, Lem. 3.4, Def. 7.1, Def. 7.2 · *Used by* Cor. 9.2 · newly derived · implemented · *Code* certiqs_key.transmitter.decoy.emit:DecoySourceDeclaration

Proof. (i) (E1): $\sigma_x^{\mathrm{v},(t)}$ is a non-negative multiple of a dyad, hence Hermitian and positive semidefinite. (E2): $\sum_t \operatorname{tr} \sigma_x^{\mathrm{v},(t)} = p_x \sum_t w_{x,t} = p_x$, the $|\varphi_{x,t}\rangle$ being unit vectors, and $\sum_x p_x = 1$ by Definition 7.1; a sector with $\mathcal{X}_t = \varnothing$ carries the zero block for every letter, so leaving it out of the structure changes neither sum. That the blocks are those of $p_x \rho_x^{\mathrm{v}}$ is the decomposition $\rho_x^{\mathrm{v}} = \sum_t w_{x,t} |\varphi_{x,t}\rangle\langle\varphi_{x,t}|$, which is (7.7) read sector by sector, the sectors being mutually orthogonal. The hypotheses of Lemma 3.4 are (E1) and the total trace $\sum_x \sum_t \operatorname{tr} \sigma_x^{\mathrm{v},(t)} = \sum_x p_x = 1$, so its parts (i) and (ii) give the statement; the virtual ensemble of its part (i) has the weighted members $\bigoplus_t \sigma_x^{\mathrm{v},(t)} = p_x \rho_x^{\mathrm{v}}$.

(ii) is (3.2) with the blocks (9.1), the terms with $w_{x,t} = 0$ vanishing; the trace of a transpose is the trace, and $\sum_t g_t = \sum_x p_x \sum_t w_{x,t} = 1$. The vacuum sector is spanned by the one vector $|0, a\rangle$, which does not depend on a, and a flag sector by its flag vector, so there $\dim \mathcal{H}(S_t) = 1$.

(iii) Transposition preserves the rank, and the range of $\sum_{x \in \mathcal{X}_t} p_x w_{x,t} |\varphi_{x,t}\rangle\langle\varphi_{x,t}|$, a sum of positive multiples of dyads, is the span of the vectors $|\varphi_{x,t}\rangle$, $x \in \mathcal{X}_t$. On a flag sector and on the vacuum that span is one dimensional. On a quantum sector it is spanned by the vectors $|\mathsf{n}, a\rangle$ with a ranging over the polarisations of $\mathcal{X}_{\mathsf{n}}$ and lies in $\mathcal{H}(S_{\mathsf{n}})$, which gives both bounds. A_{n} is a copy of S_{n} by construction.

(iv) is read off (9.1) and (9.2). □

The emitted decoy declaration has an exact certificate against the tagged ensemble, its source identity by the tagging channel, and block-diagonal attacked states. Exactness is decided on products of declared weights and formed dyad entries; lowering a weight gives another tagged declaration.

The emission is the step of this submodule that outputs a tagged source declaration as a decoy declaration; the emitted declaration is that output, a record and not a pulse. It declares the tagged virtual ensemble itself, sector by sector, and defines everything on Alice's side from it. The corollary collects what such a declaration satisfies: a certificate and a source specification (i), the source identity at the emitted descriptors and the admission of its segment (ii), what the certificate reaches (iii), the entries on which exactness is decided for a particular declaration (iv), and the effect of lowering a weight (v), which gives a different declaration and not a perturbation of the first one. The inequality on the weights is decided by the emission against the rational lower bound on $P_\mu(\mathsf{n})$ of Lemma 7.3 (Requirement 11.5), and the entries of the signal dyads are formed by the construction of (iv)(b) (Requirement 11.4). A-SEQ and A-SRC enter through Theorem 4.1 and Lemma 4.5, in (i) and (iii). A-WCP is the physical-source assumption that a specification declared by this source model carries (Definition 1.1, case (b)) and that the obligation record of (ii) names; it is no hypothesis of Proposition 8.1. The artefact the emission outputs lists A-WCP in its assumption ledger (Requirement 11.1), so the admission of its segment in (ii) applies there. Part (iv) matters because the emission requires every formed entry to be a binary64 number and refuses one that is not (Requirement 11.7). Remark 9.3 says what the consuming modules do with the declaration.

Corollary 9.2 (The emitted decoy declaration). Let a weak-coherent-pulse model (Definition 7.1) and a tagged source declaration over it (Definition 7.2) be presented, with the certified inequality (7.3), $0 \le \tilde{p}(\mathsf{n} \mid \mu) \le P_\mu(\mathsf{n})$, discharged for every $\mathsf{n} \le N_{\mathrm{src}}$ and every intensity $\mu \in \mathrm{Int}$. Its *decoy declaration* is the ensemble presentation, in the sense of Definition 3.3, whose declared data are exactly

- the alphabet $\mathcal{X}$ with its declared order and the weights p_x;
- the matched sector structure, matched meaning that Alice's system and A'' carry one ordered index set: the sectors S_t of Definition 7.2 (T3) that some letter puts positive weight on, in the declared order of $\mathbb{T}$, each with its orthonormal basis of (T3);
- the weighted blocks $\sigma_x^{\mathrm{v},(t)} = p_x w_{x,t} \, |\varphi_{x,t}\rangle\langle\varphi_{x,t}|$ of (9.1), as block lists;
- the specified ensemble $\{p_x, \rho_x\}$ of the model as the source model, the source segment with the one tagging link of Definition 7.2 (T6) and its obligation record, and the assumption ledger;

so that no retained system, no measurement, no marginal and no certificate

is among them. Below, *emitted* refers to this decoy declaration: the emitted source descriptor, segment and assumption ledger are the ones it declares. Then the following hold.

(i) *(a certificate, and an admissible specification)* where (E1) to (E3) of Definition 3.3 hold of the declared blocks, the objects (3.2) — the retained system $A = \bigoplus_t A_t$ with A_t a copy of S_t, the marginal τ_A, the POVM $\{M_x\}$ and the blocks $(\Psi_t)_t$ — are a source-replacement certificate in block form with every non-zero block a multiple of a rank-one projector, satisfying (SR1) and (SR2) exactly against the tagged virtual ensemble of Definition 7.2 (T4); the data are a source specification in the sense of Definition 1.1 in trusted-source mode; and Theorem 4.1 holds of it, its part (i) reading (E1) to (E3) as the decided conditions.

(ii) *(the segment, and the source identity at the emitted descriptors)*

(a) *(the source identity)* The emitted source descriptor is $(A'', \{\sigma_x^{\mathrm{v}}\})$ with $\sigma_x^{\mathrm{v}} = p_x \rho_x^{\mathrm{v}}$ for the tagged ensemble of Definition 7.2 (T4) with the declared weights, hence a member of the declared descriptor family of Proposition 8.1, every tagged source declaration over every weak-coherent-pulse model; so the source identity (2.3), here $\Xi_{\mathrm{tag}}(\rho_x^{\mathrm{v}}) = \rho_x$, holds exactly at the emitted descriptors, as that proposition's conclusion.

(b) *(the segment record)* The emitted segment satisfies the interface conditions (S1) (descriptor match), (S2) (index preservation) and (S4) (slots) of Definition 2.2, and its (S3) (source identity) is the conclusion of (a). Its list of transported constraint entries is empty and it carries no licence record. Its obligation record, the triple a consumer compares with the soundness registry to admit the link, is $(\mathsf{tagging},$ the id of Proposition 8.1, $\{\text{A-WCP}\})$, together with the member of the declared descriptor family that its descriptors are (Definition 2.1 (c2)). The third component is the union of the assumption closure of Proposition 8.1, which is empty, and the physical-source assumption A-WCP that this submodule declares for the model that proposition is stated over (Definition 2.2 (A1)). Inside an artefact that this submodule emits and whose assumption ledger lists A-WCP, the segment meets the admission conditions (A1) to (A3) of Definition 2.2: the registry entry, the decided triple and the provenance.

(iii) *(what the certificate reaches)* every state reached from the certificate of (i) by an attack on the transmitted system and the adversary's registers is block diagonal along $\mathbb{T}$ with A-marginal τ_A, and pinching

A along $\mathbb{T}$ after the attack is the same operation as pinching the virtual signal before it (Lemma 4.5); Theorem 4.1 (iii) holds against the tagged ensemble of Definition 7.2 (T4) itself; and a positive operator constrained only to have the marginal τ_A need not be block diagonal.

(iv) *(the entries of the declared blocks)* (E1) to (E3) are conditions on the declared blocks, and every entry of a block is a product $p_x\, w_{x,t}\, \big(\, |\varphi_{x,t}\rangle\langle\varphi_{x,t}| \,\big)_{ij}$ of the declared weight p_x, the weight $w_{x,t}$, which is a declared sector weight or a tail weight formed from them, and an entry of the dyad of $|\varphi_{x,t}\rangle$ in the orthonormal basis of S_t of Definition 7.2 (T3), a number fixed by the model and that basis and not a declared number.

(a) *(the weights)* The products $p_x w_{x,t}$ are formed from the declared p_x, the declared $\tilde{p}(\mathsf{n} \mid \mu)$ and the tail weight (7.4), which is itself formed exactly and rounds nowhere. Choosing the p_x and the $\tilde{p}(\mathsf{n} \mid \mu)$ with few enough significant bits makes every product exact, and lowering a $\tilde{p}$ to a value that is still non-negative is permitted by (7.3) and yields again a tagged source declaration by (v).

(b) *(the signal dyads)* The entries of $|\varphi_{x,t}\rangle\langle\varphi_{x,t}|$ are the number 1 on every flag sector and on the vacuum, and on a quantum sector $\mathsf{n} \geq 1$ the entries of $|\mathsf{n}, a\rangle\langle\mathsf{n}, a|$ in the basis of S_{n}. For every unit vector $|b\rangle \in \mathbb{C}^2$, in Pol or not, write $|\mathsf{n}, b\rangle := (c_b^\dagger)^{\mathsf{n}}\, |\mathrm{vac}\rangle\, /\sqrt{\mathsf{n}!}$, which for $b \in \mathrm{Pol}$ is the vector of (7.1); then $\big|\, \langle \mathsf{n}, a | \mathsf{n}, b\rangle \,\big|^2 = |\, \langle a|b\rangle \,|^{2\mathsf{n}}$ for all unit vectors a, b. Let every polarisation be one of the four vectors of $\{H, V, +, -\}$, with $|H\rangle, |V\rangle$ the fixed orthonormal basis of the one-photon space of Definition 7.1 (W1) and $|\pm\rangle = (|H\rangle \pm |V\rangle)/\sqrt{2}$, distinct polarisations being distinct vectors, and put $v_H = (1, 1, 0)$, $v_V = (1, -1, 0)$ and $v_\pm = (1, 0, \pm 1)$. If $|\mathrm{Pol}| \geq 2$, then S_1 is the one-photon space, $\{|1, H\rangle, |1, V\rangle\}$ is an orthonormal basis of it, and in that basis $|1, H\rangle\langle 1, H|$, $|1, V\rangle\langle 1, V|$ and $|1, \pm\rangle\langle 1, \pm|$ are $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $\frac{1}{2}\left(\begin{smallmatrix} 1 & \pm 1 \\ \pm 1 & 1 \end{smallmatrix}\right)$. If $|\mathrm{Pol}| \geq 3$, then S_2 is the two-photon space, and it has an orthonormal basis in which $|2, a\rangle\langle 2, a| = \frac{1}{2}\, v_a v_a^{\mathsf{T}}$ for every $a \in \{H, V, +, -\}$. So every entry of these dyads is 0, 1 or $\pm\frac{1}{2}$ at one photon and 0 or $\pm\frac{1}{2}$ at two.

(c) *(the sector sums)* With the entries of (a) and (b) representable, (E3) is a condition on the sums $\sum_x \sigma_x^{\mathrm{v},(t)}$, accumulated entry by entry in the declared order, and on nothing else.

(v) *(lowered weights)* if $0 \leq \tilde{p}'(\mathsf{n} \mid \mu) \leq \tilde{p}(\mathsf{n} \mid \mu)$ for every $\mathsf{n} \leq N_{\mathrm{src}}$ and every μ, then the weights $\tilde{p}'$ are again a tagged source declaration over the same model and Proposition 8.1 holds of it, the difference passing into the tail weight. Its weights, its virtual ensemble, its tail

states and its sector list are those formed from $\tilde{p}'$: the sector list may gain a tail flag, where $\tilde{w}_\mu$ was zero and $\tilde{w}'_\mu$ is not, and it may lose a polarisation flag or a quantum sector, where a weight is lowered to zero at every letter that carried it, so that neither sector list need contain the other.

Uses Def. 1.1, Def. 2.1, Def. 2.2, Def. 3.3, Lem. 3.4, Thm. 4.1, Lem. 4.5, Def. 7.1, Def. 7.2, Prop. 8.1, Prop. 9.1 · *Assumes* A-WCP, A-SRC, A-SEQ · *Used by* Rem. 9.3, Req. 11.1, Req. 11.3, Req. 11.4, Req. 11.7, Req. 11.8 · newly derived · implemented (effective: accepted) · *Code* certiqs_key.transmitter.decoy.emit:emit_decoy_declaration

Proof. (i) Proposition 9.1 (i) is exactly this statement for the blocks (9.1): (E1) and (E2) hold as identities of real numbers, and Lemma 3.4 makes the objects (3.2) a block-form certificate with pure blocks against the weighted members $\bigoplus_t \sigma_x^{\mathrm{v},(t)} = p_x \rho_x^{\mathrm{v}}$, which are those of Definition 7.2 (T4). (E3) is the one condition of the presentation that is not an identity of real numbers but a property of the declared numbers, and (iv) says what it is a condition on. That the data are a source specification is Definition 1.1: the alphabet and its order are declared, the specified ensemble is declared by a source model, which is that definition's case (b) and comes with a non-empty source segment, and in trusted-source mode with more than one sector an ensemble presentation is one of the two forms it admits, under which A, $\{M_x\}$ and τ_A are not declared but defined, the conditions on them holding by Lemma 3.4 (i). The dimension of Alice's system is $d_A = \sum_t \dim \mathcal{H}(S_t) \geq 2$ whenever the structure carries two sectors, and Definition 1.1 demands a certificate in trusted-source mode at $d_A \geq 2$; so the certificate is demanded, and it is supplied. Theorem 4.1 applies to a certificate in either form, and its part (i) says what an ensemble presentation decides.

(ii)(a) By Definition 7.2 (T4) the emitted weighted members are $p_x \rho_x^{\mathrm{v}}$ with the declared $\tilde{p}(\mathsf{n} \mid \mu)$ and $\tilde{w}_\mu$, which is a tagged source declaration over the given model: it is the one presented, no number having been changed. The declared descriptor family of the link is every tagged source declaration over every weak-coherent-pulse model (Definition 7.2 (T6)), so the emitted descriptor is a member of it and Proposition 8.1 (ii) concludes $\Xi_{\mathrm{tag}}(\rho_x^{\mathrm{v}}) = \rho_x$ for every letter, which is (2.3) for the one-link composite.

(ii)(b) (S1) holds by Definition 7.2 (T6): the source descriptor is the virtual pair and the target descriptor is the specified pair of the model. (S2) holds because both are indexed by $\mathcal{X}$ and $\operatorname{tr} \sigma_x^{\mathrm{v}} = p_x \sum_t w_{x,t} = p_x$ by Proposition 9.1 (i), the same p_x the model declares. (S4) holds because the link carries its empty entry list, its empty licence slot, its status and its obligation record. The triple of that record is the registry entry of the kind tagging (Definition 2.2 (A1)), which is what a consumer decides against. The third component is as that definition specifies it. Its first half, the assumption closure of Proposition 8.1, is empty by the criterion of (A1)(a), no claim of that closure listing an assumption among its dependencies: the

proposition lists none and uses Definitions 7.1, 7.2 and 2.1; what these use, directly or in turn, lies among them and Definitions 2.2 and 1.1; and none of these definitions lists an assumption. Its second half is the physical-source assumption this submodule declares for the model, A-WCP; so the union is {A-WCP}. A-WCP is not a hypothesis of Proposition 8.1, and nothing here makes it one. For (A1) to (A3), let the segment be carried by an artefact that this submodule emits and whose assumption ledger lists A-WCP. The registry entry of the kind tagging is the triple above, which gives (A1) and the comparison of triples of (A2); the one assumption of that entry, A-WCP, is in the artefact's ledger by that hypothesis, which is the rest of (A2); and the artefact is emitted by this submodule, the module that admits the kind, which is (A3).

(iii) The certificate of (i) is in block form against the matched pair of Definition 7.2 (T5), which is the hypothesis of Lemma 4.5, and the attacks quantified over are those of Theorem 4.1 (iii), whose conclusion is stated against the virtual ensemble the certificate is written against: the tagged one, by (i). The operator of the last clause is the one exhibited in the proof of that lemma.

(iv) That every entry of a block is such a product is Proposition 9.1 (iv). (a) is the definition of $w_{x,t}$ in that proposition and (7.4); that lowering a $\tilde{p}$ leaves a tagged declaration is (v). (b) On a flag sector and on the vacuum S_t is one dimensional (Proposition 9.1 (ii)) and the entry is $\langle\varphi_{x,t}|\varphi_{x,t}\rangle = 1$.

The overlaps. For unit vectors $|a\rangle, |b\rangle \in \mathbb{C}^2$, in Pol or not, write $|a\rangle = \sum_i a_i |i\rangle$ in the fixed basis of (W1) and $c_a^\dagger = \sum_i a_i c_i^\dagger$ for the mode operators c_i of that basis, so that $[c_a, c_b^\dagger] = \sum_i \bar{a}_i b_i = \langle a|b\rangle$ and $c_a|\mathrm{vac}\rangle = 0$. The commutator is a number, so $[c_a, (c_b^\dagger)^{\mathsf{n}}] = \mathsf{n}\,\langle a|b\rangle\,(c_b^\dagger)^{\mathsf{n}-1}$ by induction on n, and hence $c_a(c_b^\dagger)^{\mathsf{n}}|\mathrm{vac}\rangle = \mathsf{n}\,\langle a|b\rangle\,(c_b^\dagger)^{\mathsf{n}-1}|\mathrm{vac}\rangle$. Applying this n times, $\langle\mathrm{vac}|\, c_a^{\mathsf{n}}(c_b^\dagger)^{\mathsf{n}}|\mathrm{vac}\rangle = \mathsf{n}!\,\langle a|b\rangle^{\mathsf{n}}$, and the definition of $|\mathsf{n}, b\rangle$ in (b), which is (7.1) for $b \in$ Pol, gives $\langle \mathsf{n}, a|\mathsf{n}, b\rangle = \langle a|b\rangle^{\mathsf{n}}$, whence $\big|\langle \mathsf{n}, a|\mathsf{n}, b\rangle\big|^2 = |\langle a|b\rangle|^{2\mathsf{n}}$.

One photon. The map $|a\rangle \mapsto c_a^\dagger|\mathrm{vac}\rangle = |1, a\rangle$ is linear and isometric from $\mathbb{C}^2$ onto the one-photon space, by the overlaps at $\mathsf{n} = 1$. If $|\mathrm{Pol}| \geq 2$, Pol holds two distinct vectors of $\{H, V, +, -\}$, and no vector of $\{H, V, +, -\}$ is a multiple of another, their overlaps having modulus 0 or $1/\sqrt{2}$; so two vectors of Pol span $\mathbb{C}^2$, and S_1, the span of the $|1, a\rangle$, $a \in$ Pol, is the whole one-photon space. $|H\rangle$ and $|V\rangle$ are the fixed basis of (W1), whether or not they lie in Pol, so $\{|1, H\rangle, |1, V\rangle\}$ is an orthonormal basis of it, and $|1, \pm\rangle = (|1, H\rangle \pm |1, V\rangle)/\sqrt{2}$. The four dyads are therefore $|1, H\rangle\langle 1, H|$, $|1, V\rangle\langle 1, V|$ and $\frac{1}{2}\big(|1, H\rangle\langle 1, H| + |1, V\rangle\langle 1, V| \pm |1, H\rangle\langle 1, V| \pm |1, V\rangle\langle 1, H|\big)$, which are the stated matrices.

Two photons. Write $|2, H\rangle, |1_H 1_V\rangle, |2, V\rangle$ for the orthonormal number states of the two modes, which span the two-photon space, three dimensional. Expanding $(c_\pm^\dagger)^2 = \frac{1}{2}\big((c_H^\dagger)^2 \pm 2c_H^\dagger c_V^\dagger + (c_V^\dagger)^2\big)$ gives $|2, \pm\rangle = \frac{1}{2}|2, H\rangle \pm \frac{1}{\sqrt{2}}|1_H 1_V\rangle + \frac{1}{2}|2, V\rangle$, so the four vectors $|2, a\rangle$, $a \in \{H, V, +, -\}$,

have the Gram matrix with diagonal 1, $\langle 2,H|2,V\rangle = 0$, $\langle 2,+|2,-\rangle = 0$ and $\langle 2,H|2,\pm\rangle = \langle 2,V|2,\pm\rangle = \frac{1}{2}$. The four vectors $r_a \coloneqq v_a/\sqrt{2}$,

$$r_H = \tfrac{1}{\sqrt{2}}(1,1,0), \quad r_V = \tfrac{1}{\sqrt{2}}(1,-1,0), \quad r_+ = \tfrac{1}{\sqrt{2}}(1,0,1), \quad r_- = \tfrac{1}{\sqrt{2}}(1,0,-1)$$

of $\mathbb{C}^3$ are unit vectors with the same Gram matrix — each inner product is a sum of three products of coordinates, and $r_H^\dagger r_V = \frac{1}{2}(1-1+0) = 0$, $r_+^\dagger r_- = \frac{1}{2}(1+0-1) = 0$, $r_H^\dagger r_\pm = r_V^\dagger r_\pm = \frac{1}{2}$ — and any three of them are linearly independent, the determinant of any three of v_H, v_V, v_+, v_- being ± 2. Equal Gram matrices force the same linear relations, a vector c of coefficients annihilating one family exactly when $c^\dagger G c = 0$ for the common Gram matrix G, so the map $|2,a\rangle \mapsto r_a$ extends to a well-defined linear map U from the span of the $|2,a\rangle$, $a \in \{H,V,+,-\}$, onto $\mathbb{C}^3$ that preserves inner products; by the same relations any three of the $|2,a\rangle$ are linearly independent, so that span is the whole two-photon space and U is unitary. If $|\mathrm{Pol}| \geq 3$, S_2 contains three of the $|2,a\rangle$ and is therefore the whole two-photon space. The inverse image under U of the standard basis is then an orthonormal basis of S_2, in which $|2,a\rangle\langle 2,a|$ has the entries $(r_a)_i\overline{(r_a)_j} = \frac{1}{2}(v_a)_i(v_a)_j \in \{0, \pm\frac{1}{2}\}$, the v_a being real. (c) is (E3) read on the entries of (a) and (b).

(v) Let $0 \leq \tilde{p}' \leq \tilde{p}$ pointwise. Then $0 \leq \tilde{p}'(\mathsf{n} \mid \mu) \leq \tilde{p}(\mathsf{n} \mid \mu) \leq P_\mu(\mathsf{n})$, so (7.3) holds of $\tilde{p}'$, and $\tilde{w}'_\mu = 1 - \sum_{\mathsf{n} \leq N_{\mathrm{src}}} \tilde{p}'(\mathsf{n} \mid \mu) \geq \tilde{w}_\mu \geq 0$ by (7.4). Every item of Definition 7.2 is then satisfied by the primed data, the sector structure (T3) included, which is formed from the weights: a tail flag exists exactly where $\tilde{w}'_\mu > 0$, a polarisation flag or a quantum sector exactly where some letter puts positive weight on it, and both sets are read off $\tilde{p}'$. So the primed declaration is a tagged source declaration and Proposition 8.1 holds of it. Neither inclusion between the two sector lists holds in general: raising a tail weight from zero adds a tail flag, and lowering $\tilde{p}(\mathsf{n} \mid \mu)$ to zero for every intensity that a polarisation a occurs with removes the sector (n, a). □

The ensemble form of the consuming declarations reads the decoy declaration, and exactly trusted-source mode, the rule against general constraint entries, which refuses nothing it emits, and the emission's own refusals stand in between. It licenses no restriction of a consumer's feasible set.

Remark 9.3 (Consumers of the emitted decoy declaration). By Corollary 9.2 (i) and (ii) the decoy declaration is a source specification in trusted-source mode given by an ensemble presentation, and its source segment meets the admission conditions of the transmitter inside the artefact the emission outputs (Requirement 11.1). Item (a) says which consumer reads it; what stands between the decoy declaration and a consumer of that form is exactly (b) to (d), each a named refusal.

(a) *The consumer's form.* The protocol compiler's declaration (Definition "Protocol declaration" (protocol-compiler/def:declaration)) and the MEAT core's (Definition "The protocol declaration as the MEAT core reads it" (meat-core/def:declaration)) each have an ensemble form, which carries $\{N_x\}$ in the place

of Alice's POVM, and Definition "Key-length eligibility" (protocol-compiler/def:eligible) admits an ensemble presentation beside a declaration of that form, so the decoy declaration has a consumer. A consumer whose declaration carries $\{M_x\}$ refuses it (Definition 3.3), which is a refusal of that consumer's form and not a condition on the decoy declaration.

(b) *Trusted-source mode only.* A non-empty source segment is admitted in trusted-source mode alone, by the admission rule of Definition "Reduction chain" (reductions/def:chain), and the two-sided reduction theorem (Theorem "The two-sided reduction in trusted-source mode" (reductions/thm:two-sided)) is stated for that mode alone; an ensemble form is refused in untrusted-source mode, the mode without a marginal or a certificate (Definition 1.1), by the consuming module.

(c) *Scalar constraint entries only.* Beside a non-empty source segment the protocol compiler requires every entry of a consumer's constraint list $\mathfrak{L}$ to have a real scalar as its right-hand side, and refuses an entry declared general (the licensed-only rule of Definition "Interface conditions on a chain" (protocol-compiler/def:chaincond)). The decoy declaration carries no constraint entry (Corollary 9.2 (ii)), so the rule refuses nothing it emits.

(d) *The emission's own refusals.* The requirements of this submodule refuse a declaration before any consumer reads it; the bound $N_{\mathrm{ent}} \leq 2$ on the entropy cutoff (Requirement 11.3) and the exactness of the formed entries (Requirement 11.7) are the two of them a declarer of a physically interesting source meets first.

One further point is not a refusal but the absence of a licence: the emission licenses no restriction of any consumer's feasible set. The states reached from its certificate are block diagonal, but a positive operator constrained only to have the marginal τ_A need not be (Corollary 9.2 (iii)), and a consumer that restricts its own optimisation does so under a theorem of its own module (Remark 4.6).

Uses Cor. 9.2, Req. 11.1 · newly derived · accepted

10 The physical-source assumption

One attestation carries the model to the device: A-WCP asserts that the physical source emits the declared phase-randomised weak coherent states, exactly and independently in every round. It refines A-ENS and takes its place for a specification declared by this model, and an artefact emitted from a tagged declaration carries A-WCP and not A-ENS in its assumption ledger (Requirement 11.1). No function of the declared data decides it. Remark 10.2 reads it as making the tagged declaration, with its tagging link and at the declared numbers exactly, a virtual model of the device.

The physical source emits exactly the declared phase-randomised weak coherent state of each round's letter, drawn with its declared weight, in the declared modes only and independently across rounds. It ties the tagged declaration to the device, in the place of A-ENS.

The model declares its states as mathematical objects, and this assumption ties the device to them. It refines Assumption 5.1: for a specification whose specified ensemble is declared by this source model, A-WCP is declared in the place of A-ENS; it asserts the emission A-ENS asserts and adds the conditions this model needs, so it is never the weaker of the two. Full phase randomisation makes the emitted state diagonal in the photon number. That no degree of freedom outside the two declared modes depends on the letter excludes side channels: no timing, spectral or spatial correlation with the label and no leakage of the label beyond the declared modes.

Assumption 10.1 [A-WCP] (Emission of the declared weak-coherent-pulse states)**.** For the tagged source declaration of Definition 7.2, with the cutoffs, the weights and the polarisations it declares, the physical source satisfies: (i) in every round it emits the specified state ρ_x of Definition 7.1 of the letter $x = (a, \mu, z)$ drawn in that round, and nothing else; (ii) the intensity is the declared μ exactly, with no fluctuation and no uncertainty interval; (iii) the phase randomisation is full, so that the emitted state is the phase average (7.2); (iv) the emission is in the two optical modes declared in Definition 7.1 (W1), and the polarisation is the declared $|a\rangle$; (v) no degree of freedom other than those two modes carries any dependence on x; (vi) the round type z changes nothing the source emits; (vii) the letters $x_1, \ldots, x_n$ of the n rounds are drawn independently, each with the weights p_x of Definition 7.1 (W2); (viii) given those letters, the systems the source emits in the n rounds are in the product state $\rho_{x_1} \otimes \cdots \otimes \rho_{x_n}$ of the per-round states; (ix) neither the letter of a round nor the state emitted in it depends on what happens in the other rounds, their outcomes and public messages included.

The engine does not verify this assumption; a run that rests on it records it as unverified.

Reading: Your weak-coherent-pulse source emits, in every round, a fully phase-randomised coherent pulse at exactly the declared intensity μ and polarisation, in the two declared optical modes only, and nothing else; the numbers are those of your tagged declaration, exactly as declared. No other property of the pulse (timing, spectrum, spatial shape) depends on the chosen bit, basis, intensity or round type, and the round type (test or generation) changes nothing emitted. Intensities do not fluctuate. Each round's letter is drawn independently with the declared probabilities, and no round's letter or emitted pulse depends on another round, its detection results or its public messages. Intensity fluctuations, imperfect phase randomisation or pattern effects violate this.

Uses Def. 7.1, Def. 7.2 · *Category* source · *Affects* secrecy · *Refines* A-ENS · *Assumed by* Cor. 9.2 · *Machine key* `wcp_source_model` · newly derived · accepted

Under A-WCP the tagged declaration with its tagging link is a virtual model of the device: the states the tagging channel produces are the ones it emits. The assumption carries no arithmetic identity, so it says nothing of a declaration whose numbers were altered.

Remark 10.2 (The tagged declaration as a virtual model of the device). Assumption 10.1 is asserted of the tagged source declaration of Definition 7.2, with the cutoffs, the weights and the polarisations it declares, which is what the emitted decoy declaration of Corollary 9.2 presents. That declared virtual source, together with its tagging link, is then a virtual model of the physical source: the states $\Xi_{\mathrm{tag}}(\rho_x^{\mathrm{v}}) = \rho_x$ that the link's channel is proved to produce (Proposition 8.1 (ii)) are the states the device emits, the device being the one Assumption 10.1 describes at those intensities and polarisations. The assumption asserts no arithmetic identity: the source identity of the link is a theorem about the declared numbers and holds of them exactly (Corollary 9.2 (ii)), and if a declared number is altered in any way, rounded or shifted included, the assumption says nothing about the result. This contract asserts Assumption 10.1 of no particular apparatus.

Uses Def. 7.1, Def. 7.2 · newly derived · accepted

11 Emission and refusals

These requirements bind the engine, the implementation, where it turns a tagged declaration into a decoy declaration. Recall the three bands: photon numbers up to the entropy cutoff keep quantum sectors, those up to the source cutoff one flag per polarisation, and the rest sits on the tail flags. What the emission outputs is the tagged ensemble as declared, with no number altered and with the entries of its signal dyads formed by the engine. It refuses, each with a named reason, cutoffs that are not integers or are out of order, entropy cutoffs above two, quantum sectors whose signal dyads the construction it carries does not cover, sector weights that are negative or exceed the rational lower bound it forms on their Poisson weights or are presented as equalities, flag families finer than the tagged declaration's, formed entries that are not binary64 numbers, and blocks or weight sums that fail the conditions of an ensemble presentation.

What the engine emits from a tagged declaration is the decoy declaration of Corollary 9.2 and nothing else, with every declared number unchanged. Its ledger carries A-WCP in the place of A-ENS.

The source identity holds at the emitted descriptors because the emitted ensemble is the tagged one with the declared weights (Corollary 9.2 (ii)).

Requirement 11.1 (Emission of the tagged ensemble as declared). When the engine emits an artefact from a tagged source declaration over a weak-coherent-pulse model, the artefact is that declaration's decoy declaration (Corollary 9.2) and nothing else: the engine declares no retained system, no measurement, no marginal and no certificate, and it rounds or otherwise alters no declared number. The matched sector structure among the declared

data of that corollary is emitted as the retained-register sector map of Definition 7.2 (T5), with the block kind `proof_sector`, the tag that records a decomposition licensed by a proof, here source replacement, rather than by the device's own mode structure; it names Alice's sectors only as copies of the declared sectors S_t, so it declares no retained system. An artefact emitted from a tagged declaration carries A-WCP and not A-ENS in its assumption ledger (Assumption 10.1).

Motivated by Def. 7.2, Cor. 9.2, Ass. 10.1 · *Used by* Rem. 9.3 · implemented (effective: accepted) · *Code* certiqs_key.transmitter.decoy.emit:emit_decoy_declaration

The engine refuses cutoffs unless both are integers and the entropy cutoff lies between zero and the source cutoff, the first condition of a tagged source declaration.

REQUIREMENT 11.2 (Order of the entropy and source cutoffs). The engine refuses, with no artefact, a declaration whose cutoffs are not integers with $0 \leq N_{\text{ent}} \leq N_{\text{src}}$, which is (T1) of Definition 7.2 (`cutoffs_out_of_order`, the one refusal for a cutoff that is not an integer and for cutoffs out of order).

Motivated by Def. 7.2 · implemented · *Code* certiqs_key.transmitter.decoy.emit: emit_decoy_declaration

The engine refuses an entropy cutoff of three or more. This is a scope limit and no consequence of the claims, which hold for every entropy cutoff; a basis with exactly declarable dyad entries is exhibited only for the polarisations H, V, $+$ and $-$, at one and at two photons.

REQUIREMENT 11.3 (Entropy cutoff of at most two photons). The engine refuses, with no artefact, a declaration with $N_{\text{ent}} \geq 3$ (`entropy_cutoff_above_two`). The refusal is a scope decision and not a consequence of anything the claims state: every claim of this submodule is stated for arbitrary $N_{\text{ent}} \leq N_{\text{src}}$ and nothing in them fails at $N_{\text{ent}} \geq 3$. What is missing is a construction of the kind Corollary 9.2 (iv)(b) supplies for the four polarisations $\{H, V, +, -\}$ at one and at two photons: a declared basis in which the signal dyads have exactly declarable entries. This repository has no such construction at three photons and above. Admitting such a sector would mean deciding exact conditions on dyad entries that differ from the entries of the signal states, so until the construction exists the sector is refused rather than declared with entries that are not the numbers written.

Motivated by Cor. 9.2 · implemented (effective: accepted) · *Code* certiqs_key.transmitter.decoy.emit:emit_decoy_declaration

The engine forms the entries of the quantum-sector signal dyads itself, by the construction of Corollary 9.2 (iv)(b), and takes no dyad or basis from a request. It refuses a quantum sector above the vacuum that the construction does not cover.

The source identity at the emitted descriptors (Corollary 9.2 (ii)) holds because the emitted blocks are the tagged ensemble of the model: on a quantum sector their dyads are the matrices of the projectors $|\mathsf{n}, a\rangle\langle\mathsf{n}, a|$ in an orthonormal basis of S_{n}. The polarisation vectors and that basis are objects of the model whose coordinates are in general not binary64 numbers, so those entries cannot be declared; they exist as numbers only through a construction, and Corollary 9.2 (iv)(b) gives one for polarisations among the four of $\{H, V, +, -\}$ at one and at two photons.

Requirement 11.4 (Signal dyads formed by the exhibited construction). The engine forms the entries of the signal dyads of a decoy declaration itself, and takes neither a dyad nor a basis of a sector from a request. On the vacuum sector and on every flag sector it forms the number 1. On a quantum sector $\mathsf{n} \in \{1, 2\}$ of the index set it forms them only where every polarisation of Pol is declared as one of the four vectors of $\{H, V, +, -\}$, distinct polarisations as distinct vectors, and $|\mathrm{Pol}| \geq \mathsf{n} + 1$; it then forms the dyad of $|\mathsf{n}, a\rangle$ with the entries Corollary 9.2 (iv)(b) states for it. It decides of the formed dyads, exactly, that each is a rank-one projector on S_{n} (Hermitian, idempotent and of unit trace) and that $\mathrm{tr}\left[\, |\mathsf{n}, a\rangle\langle\mathsf{n}, a| \; |\mathsf{n}, b\rangle\langle\mathsf{n}, b| \,\right] = |\,\langle a|b\rangle\,|^{2\mathsf{n}}$ for every pair of polarisations of Pol, the value that Definition 7.1 (W3) gives by that corollary and that is 0, $2^{-\mathsf{n}}$ or 1 for two vectors of $\{H, V, +, -\}$. It refuses, with no artefact, a declaration whose index set carries a quantum sector $\mathsf{n} \geq 1$ outside this construction, or at which one of these decisions fails (`quantum_sector_outside_construction`). A declaration with $N_{\mathrm{ent}} \geq 3$ is refused under its own reason (Requirement 11.3).

Motivated by Def. 7.1, Def. 7.2, Cor. 9.2, Req. 11.3 · implemented (effective: accepted) · *Code* certiqs_key.transmitter.decoy.emit:emit_decoy_declaration

Each sector weight is decided against a rational lower bound on its Poisson weight that the engine forms from the declared intensity, and a negative weight, one above that bound, or one presented as an equality is refused. No Poisson weight is compared for equality or at a tolerance.

Proposition 8.1 needs the inequality (7.3) and nothing more: its identity holds for every family of weights at or below the Poisson weights, the tail absorbing the difference. The right-hand side is not a number the engine can compare with, so it compares with the lower end of the rational enclosure of Lemma 7.3, which lies at or below the Poisson weight. A weight between that lower end and the Poisson weight is refused as well. The engine fixes the summation index by one rule, so a declarer can compute the same lower end and declare at or below it; by parts (iii) and (iv) of that lemma the rule's index gives a lower end at least as large as any smaller index would,

within $2^{-90}\mu^{\mathsf{n}}/\mathsf{n}!$ of the Poisson weight.

REQUIREMENT 11.5 (Certified bound on the sector weights). The obligation (7.3) of Definition 7.2 (T2), $0 \le \tilde{p}(\mathsf{n} \mid \mu) \le P_\mu(\mathsf{n})$ for every $\mathsf{n} \le N_{\mathrm{src}}$ and every intensity $\mu \in$ Int of Definition 7.1 (W2), is an inequality and never an equality: its right-hand side is not a declared datum, and at a positive intensity, declared as a binary64 number, it is a transcendental number (at $\mu = 0$ it is 1 for $\mathsf{n} = 0$ and 0 for $\mathsf{n} \ge 1$). For every such (n, μ) the engine forms, from the declared intensity μ and in exact rational arithmetic, the lower end $\underline{P}_\mu(\mathsf{n}) := \max(0, S_{K+1})\,\mu^{\mathsf{n}}/\mathsf{n}!$ of the enclosure of Lemma 7.3, with S_{K+1} the partial sum (7.8) at $K = K_\mu$, the least even integer $K \ge \mu$ with $t_{K+1} = \mu^{K+1}/(K+1)! \le 2^{-90}$, which exists because $t_{K+1} \to 0$. It records K_μ for every intensity in the artefact, and it reads no enclosure of $P_\mu(\mathsf{n})$ from the request. By part (iv) of that lemma no even K with $\mu \le K < K_\mu$ gives a larger lower end, so the rule refuses no weight that a smaller admissible index would admit. It refuses, with no artefact, a declaration in which $\tilde{p}(\mathsf{n} \mid \mu) < 0$ or $\tilde{p}(\mathsf{n} \mid \mu) > \underline{P}_\mu(\mathsf{n})$ at some (n, μ) (`sector_weight_above_poisson`). Since $\underline{P}_\mu(\mathsf{n}) \le P_\mu(\mathsf{n})$ by that lemma, every declaration violating (7.3) is refused, and every declaration it admits satisfies (7.3). It refuses, with no artefact, a declaration presenting the obligation as an equality $\tilde{p}(\mathsf{n} \mid \mu) = P_\mu(\mathsf{n})$ to be decided (`sector_weight_as_equality`). No condition of this submodule compares $\tilde{p}(\mathsf{n} \mid \mu)$ with $P_\mu(\mathsf{n})$ for equality and none is decided at a tolerance.

Motivated by Def. 7.1, Def. 7.2, Lem. 7.3, Prop. 8.1 · implemented (effective: accepted) · *Code* certiqs_key.transmitter.decoy.emit:emit_decoy_declaration

The engine refuses a flag family finer than the one the tagged declaration declares, in particular polarisation flags split by intensity. For that split the refusal keeps the loss of the decoy content from being silent.

REQUIREMENT 11.6 (Shared polarisation and tail flags). The engine refuses, with no artefact, a declaration carrying a flag family finer than the one Definition 7.2 (T3) declares (`flag_family_split_by_intensity`). That family is one polarisation flag $f_{\mathsf{n},a}$ for each sector (n, a) of the index set of (T3), $N_{\mathrm{ent}} < \mathsf{n} \le N_{\mathrm{src}}$, shared by every intensity and both round types, and one tail flag $f_{a,\mu}$ for each sector (a, μ) of that index set, shared by both round types; a finer family is one with separate flags where (T3) declares one. In particular the engine refuses a family whose polarisation flags are split by intensity at a photon number $N_{\mathrm{ent}} < \mathsf{n} \le N_{\mathrm{src}}$. A family split by intensity still satisfies the source identity but gives up the relation across intensities that is the decoy content of the declaration, one photon number's weight at every intensity sitting on one flag (Remark 8.2), and without the refusal that loss would be silent.

Motivated by Def. 7.2, Rem. 8.2 · implemented · *Code* certiqs_key.transmitter.decoy.emit:emit_decoy_declaration

Every entry of an emitted block and of each sector sum must be a binary64 number, and the engine refuses rather than rounds one that is not. Rounding would change which ensemble is declared, and so what the source identity is asserted of.

The three conditions of an ensemble presentation (Definition 3.3) are: (E1) every block is Hermitian and positive semidefinite; (E2) the traces of a letter's blocks sum to p_x, and the p_x to one; (E3) for every sector the sum of the blocks over the letters, accumulated entry by entry in the declared order of $\mathcal{X}$, rounds nowhere. A declarer for which a product is not representable removes the refusal by declaring on a coarser dyadic grid (Corollary 9.2 (iv)(a)).

Requirement 11.7 (Binary64 entries of the emitted blocks and sector sums). The engine decides (E3) of Definition 3.3 on the sums of the declared blocks of a decoy declaration, accumulated entry by entry in the declared order. Each entry of a block, a product of declared numbers, the tail weight formed from them and a formed entry of a signal dyad (Corollary 9.2 (iv)), is required to be a binary64 number and is refused, with no artefact, when it is not (`formed_entry_not_declarable`). Those sums are required to round nowhere, so that their entries are binary64 numbers, and a declaration at a sector of which the accumulation rounds fails (E3) and is refused, with no artefact (`ensemble_presentation_not_admissible`). No entry is ever rounded, because rounding an entry of a declared block would change which ensemble is declared and so which ensemble the source identity of Corollary 9.2 (ii) is asserted of.

Motivated by Def. 3.3, Cor. 9.2 · implemented (effective: accepted) · *Code* certiqs_key.transmitter.decoy.emit:emit_decoy_declaration

The engine decides on the declared blocks that each is positive semidefinite and that a letter's block traces sum to its weight, the weights to one. It refuses a declaration in which either fails.

Requirement 11.8 (Positivity and weight sums of the emitted blocks). The engine decides (E1) and (E2) of Definition 3.3 on the declared blocks of a decoy declaration, as it decides (E3) (Requirement 11.7). It refuses, with no artefact, a declared block failing (E1) or a weight sum failing (E2) (`ensemble_presentation_not_admissible`).

Motivated by Def. 3.3, Cor. 9.2, Req. 11.7 · implemented (effective: accepted) · *Code* certiqs_key.transmitter.decoy.emit:emit_decoy_declaration

Interface

The declared inputs and outputs of the module and its submodules: each key with its symbol, when it is present and the claims that fix it. The domains, types and refusal rules are in the interface files themselves.

Inputs: Transmitter source model (transmitter)

From `interface.yaml`.

Key and symbol	Presence	Fixed by
`signal_alphabet` X	always	Def. 1.1
`alice_system` A, d_A = dim H(A)	always, EXCEPT in an ensemble presentation, where A is not declared but defined (the 'ensemble_presentation' row)	Def. 1.1, Def. 3.3
`alice_sector_structure` T, {Pi^A_t}, d_{A_t}	always; THE ONE-SECTOR STRUCTURE IS THE DEFAULT	Def. 1.1, Def. 7.2
`alice_povm_spec` {M_x}_{x in X}	always, EXCEPT in an ensemble presentation, where {M_x} is not declared but defined and the POVM condition is a theorem (the 'ensemble_presentation' row)	Def. 1.1
`specified_ensemble` {p_x, rho_x}_{x in X} on A'	always	Def. 1.1, Def. 7.1
`source_mode_request`	always	Def. 1.1, Rem. 4.7
`alice_marginal` tau_A	trusted mode only; null in untrusted mode; NOT DECLARED in an ensemble presentation, where tau_A is the transpose of the declared sum (the 'ensemble_presentation' row)	Def. 1.1, Rem. 1.2, Rem. 4.7
`source_replacement_certificate` (tau_A, Psi, {p_x, rho^v_x})	trusted mode; demanded only at d_A >= 2	Def. 3.3, Def. 1.1, Thm. 4.1
`ensemble_presentation` (S_t)_t and {sigma^v_x} as block lists	the third form of a certificate; trusted mode	Def. 3.3, Lem. 3.4, Def. 1.1
`source_segment` Ch_src = (Xi_1, ..., Xi_{L_s})	always	Def. 2.2, Def. 2.1

Outputs: Transmitter source model (transmitter)

From `interface.yaml`.

Key and symbol	Presence	Fixed by
`signal_alphabet` X	always	Def. 1.1
`alice_dim` d_A	always	Def. 1.1
`alice_marginal` tau_A, or none	always	Def. 1.1, Rem. 1.2
`alice_marginal_slot_filled`	always	Req. 6.2
`alice_povm` {M_x}_{x in X}	always; an empty mapping with an ensemble presentation, where {M_x} is defined by eq:enspres and formed by no one. A consumer in ensemble form reads ‘alice_ensemble_operators‘ instead, and one whose declaration carries {M_x} refuses the presentation (def:srcert, ’What a consumer reads of a presentation’).	Def. 1.1
`alice_ensemble_operators` {N^(t)_x}	with an ensemble presentation only	Lem. 3.4, Def. 3.3
`virtual_transmitted_system` A^v	always	Def. 2.2
`soundness_registry`	always	Def. 2.2, Def. 2.1
`virtual_ensemble` {p_x, rho^v_x}_{x in X}	always; with an ensemble presentation its ‘states‘ are an empty mapping, rho^v_x = sigma^v_x / p_x being a quotient of declared numbers that need not be binary64 and is formed by no one (‘virtual_ensemble_weighted_members‘).	Def. 2.2, Def. 3.3
`virtual_ensemble_weighted_members` {sigma^v_x}, per virtual sector	with a block-form certificate or an ensemble presentation only	Def. 3.3, Lem. 3.4
`alice_sector_map` T, {Pi^A_t}, and the matched {Pi^v_t} where a link emitted one	always; the one-sector structure by default	Def. 1.1, Def. 2.1, Lem. 4.5
`source_segment` Ch_src	always; EMPTY unless a submodule supplied an admitted link	Def. 2.2, Def. 2.1

Key and symbol	Presence	Fixed by
`source_replacement_license`	trusted mode at d_A >= 2 only	Def. 3.3, Ass. 5.3, Rem. 4.7, Req. 6.1, Req. 6.5, Rem. 4.6
`certificate_form`	on every record but the degenerate one; an absent key means ‘pure‘	Def. 3.3
`assumption_ledger`	always	Ass. 5.1, Ass. 5.2, Ass. 5.3, Rem. 4.7, Def. 1.1, Def. 2.1, Def. 2.2
`assumption_bindings`	always	Ass. 5.1, Ass. 5.2, Ass. 5.3, Ass. 10.1
`mathematical_spec_version`	always	—

Inputs: Decoy-state sources and the tagging channel

From `decoy/interface.yaml`.

Key and symbol	Presence	Fixed by
`polarisation_set` Pol, with |a> in C^2 per a	always	Def. 7.1, Cor. 9.2, Req. 11.4
`intensity_set` Int	always	Def. 7.1
`signal_alphabet` X, x = (a, mu, z)	always	Def. 7.1, Def. 1.1
`entropy_cutoff` N_ent	always	Def. 7.2, Req. 11.3, Rem. 7.4
`source_cutoff` N_src	always	Def. 7.2, Rem. 7.4
`sector_weights` ptilde(n | mu), 0 <= n <= N_src	always	Def. 7.2, Lem. 7.3, Req. 11.5
`weighted_blocks` sigma^{v,(t)}_x = p_x w_{x,t} |phi_{x,t}><phi_{x,t}|	always	Prop. 9.1, Cor. 9.2, Req. 11.4, Def. 3.3
`sector_structure` T, {Pi^v_t}, the declared basis of each S_t	always	Def. 7.2, Prop. 9.1, Req. 11.1

Outputs: Decoy-state sources and the tagging channel

From `decoy/interface.yaml`.

Key and symbol	Presence	Fixed by
`source_specification` S_src	always	Cor. 9.2, Def. 1.1, Def. 3.3
`virtual_ensemble` {sigmaˆv_x} on Aˆv, by weighted blocks	always	Def. 7.2, Prop. 9.1, Def. 3.3
`source_segment` Ch_src = (Xi_tag)	always	Def. 7.2, Cor. 9.2
`link_obligation_record`	always	Def. 2.1, Def. 2.2, Cor. 9.2
`alice_sector_map` ({PiˆA_t}, {Piˆv_t})	always	Def. 7.2, Cor. 9.2
`assumption_ledger`	always	Ass. 10.1, Req. 11.1, Def. 1.1, Cor. 9.2
`mathematical_spec_version`	always	—

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References

[1] Lars Kamin, John Burniston, and Ernest Y.-Z. Tan. Rényi security framework against coherent attacks applied to decoy-state QKD, 2025.

[2] Devashish Tupkary, Shlok Nahar, Amir Arqand, Ernest Y.-Z. Tan, and Norbert Lütkenhaus. A rigorous and complete security proof of decoy-state BB84 quantum key distribution, 2026.

[3] Adam Winick, Norbert Lütkenhaus, and Patrick J. Coles. Reliable numerical key rates for quantum key distribution. *Quantum 2, 77 (2018)*, 2018.